Lecture Notes of the course:

Networked Control

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CONTENTS

PART I: BASICS

THEORETICAL BACKGROUND

A. LINEAR TIME-INVARIANT (LTI) CONTINUOUS-TIME SYSTEMS

A linear time-invariant continuous-time system is described by the following model

$$\mathcal{S}: \begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \quad (i) \\ y(t) &= Cx(t) + Du(t) \quad (ii) \end{cases}$$

where $t \in \mathbb{R}$, $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^m$ is the input vector, $y \in \mathbb{R}^p$ is the output vector. Equation (i) is denoted state equation, while (ii) is the output equation. Consistently with the vector dimensions, $A \in \mathbb{R}^{n\times n}$, $B \in \mathbb{R}^{n\times m}$, $C \in \mathbb{R}^{p\times n}$, and $D \in \mathbb{R}^{p\times m}$. For simplicity and without lack of generality (the system is time invariant) we will set the initial time instant $t_0 = 0$ s.

1) Lagrange equation

Given the initial condition x_0 (i.e., such that $x(0) = x_0$) and the input trajectory $u(t)$, $t \in \mathbb{R}$, the state and output motions are provided by the Lagrange equation

$$x(t) = \underbrace{e^{At}x_0}_{x_{\mathrm{FR}}(t):\ \text{free motion}} + \underbrace{\int_0^t e^{A(t-\nu)}Bu(\nu)d\nu}_{x_{\mathrm{FO}}(t):\ \text{forced motion}}$$

$$y(t) = \underbrace{Ce^{At}x_0}_{y_{\mathrm{FR}}(t):\ \text{free motion}} + \underbrace{C\int_0^t e^{A(t-\nu)}Bu(\nu)d\nu + Du(t)}_{y_{\mathrm{FO}}(t):\ \text{forced motion}}$$

2) Equilibria

Given a constant input trajectory $u(t) = \bar{u}$ for all $t \in \mathbb{R}$, the vector $\bar{x} \in \mathbb{R}^n$ is an equilibrium for system $\mathcal{S}$ iff $A\bar{x} + B\bar{u} = 0$. The following remarks are in order.

- If $\bar{u} = 0$, $\bar{x} = 0$ is an equilibrium condition for any possible matrix A.
- If $\bar{u} \neq 0$
  - if A is invertible (i.e., if $\det(A) \neq 0$), then $\bar{x} = -A^{-1}B\bar{u}$;
  - if A is not invertible (i.e., if $\det(A) = 0$) the equation $A\bar{x} + B\bar{u} = 0$ may have no or infinitely many solutions.

3) Stability

We define

- a nominal state trajectory $x(t)$ as the solution to the Lagrange equation with initial condition $x(0) = x_0$ and input signal $u(t)$;
- a perturbed trajectory $\tilde{x}(t)$ with $x(0) = \tilde{x}_0 \neq x_0$ and input signal $u(t)$.

The nominal trajectory is

- *stable* if, for all $\varepsilon > 0$, $\exists \rho > 0$ such that, for all $\tilde{x}_0 \neq x_0$ such that $||\tilde{x}_0 - x_0|| < \rho$, then $||x(t) - \tilde{x}(t)|| < \varepsilon$ for all $t > 0$;
- (globally) *asymptotically stable* (AS) if it is stable and attractive, i.e., if $||x(t) - \tilde{x}(t)|| \to 0$ as $t \to +\infty$ for all $\tilde{x}_0 \in \mathbb{R}^n$;
- *unstable* if it is not stable.

Remarks:

- For linear time-invariant systems, stability and asymptotic stability are *system properties*. Indeed, if a motion is asymptotically stable (respectively, stable or unstable), all system motions are asymptotically stable (respectively, stable or unstable). For this reason, for simplicity, we say that a linear time-invariant *system* is asymptotically stable/stable/unstable to characterize the stability properties of all its motions. Note that this is not possible for other classes of systems (e.g., nonlinear).
- If a linear time-invariant systems is asymptotically stable (AS) it is also *exponentially stable* (ES), i.e., $\exists \alpha > 0, \lambda < 0$ such that $||x(t) - \tilde{x}(t)|| \leq \alpha e^{\lambda t} ||x_0 - \tilde{x}_0||$. The scalar λ is denoted convergence rate.

Theorem 1

An LTI system is

- asymptotically stable if and only if all the eigenvalues of A have strictly negative real part (a matrix with this property is denoted Hurwitz stable);
- unstable if at least one eigenvalue of A has positive real part;
- stable if and only if all the eigenvalues of A satisfy $Re(\lambda) \leq 0$ and those verifying $Re(\lambda) = 0$ have algebraic multiplicity equal to the geometric one.

The following definitions and properties are useful for stating the next result.

- A symmetric matrix $P \in \mathbb{R}^{n \times n}$ is
  - *positive definite* if, for all $x \neq 0$, $x^T P x > 0$. In this case we write that $P > 0$;
  - *positive semidefinite* if, for all $x \neq 0$, $x^T P x \geq 0$. In this case we write that $P \geq 0$;
  - *negative definite* if, $-P > 0$;
  - *negative semidefinite* if $-P \geq 0$.
- If matrix $P > 0$ is symmetric, letting $\lambda_{\min}(P)$ and $\lambda_{\max}\ (P)$ be the minimum and maximum, respectively, eigenvalue of P, it holds that
  - $\lambda_{\max}(P) \geq \lambda_{\min}\ (P)\ > 0$;
  - $\lambda_{\min}(P)||x||^2 \leq x^T P x \leq \lambda_{\max}(P)||x||^2$ for all x;
  - $||Px|| \leq ||P|| \cdot ||x||$

Theorem 2 (Lyapunov inequality)

Matrix A is Hurwitz stable if and only if there exists a symmetric positive definite matrix P such that

$$A^T P + PA < 0 \quad (iii)$$

Sketch of the proof

I) A is Hurwitz stable ⇒ there exists a symmetric positive definite matrix P such that (*iii*) holds.

If A is Hurwitz stable then, for any symmetric matrix $Q > 0$, we can define

$$P = \int_0^{+\infty} e^{A^T t} Q e^{At} dt$$

Note that P exists and is well defined thanks to the fact that A is Hurwitz stable. Also, it is positive definite. We can write

$$A^T P + PA = \int_0^{+\infty} \underbrace{A^T e^{A^T t} Q e^{At} A}_{\frac{d}{dt}(e^{A^T t} Q e^{At})} dt = \int_0^{+\infty} d(e^{A^T t} Q e^{At}) = \left[e^{A^T t} Q e^{At}\right]_0^{+\infty} = -Q < 0$$

II) There exists a symmetric positive definite matrix P such that (*iii*) holds ⇒ A is Hurwitz stable.

For any eigenvector v ($\neq 0$) of A

$$0 > v^T (A^T P + PA) v = 2Re(\lambda) v^T P v$$

Since $P > 0$, the latter inequality is verifiable only if $Re(\lambda) < 0$. This proves that every eigenvalue of A has negative real part, i.e., is Hurwitz stable.

Note that v^T is the Hermite transpose of vector v (i.e., transpose + complex conjugate).

Remark

In view of (*iii*), the quadratic function $V(x) = x^T P x$ is a Lyapunov function for the autonomous system $\dot{x} = Ax$. In fact

$$\dot{V}(x) = x^T A^T P x + x^T P A x = x^T (A^T P + PA) x < 0$$

4) Modes

The modes of an LTI system are, for $i = 1, \dots, n$, $t^j e^{\lambda_i t}$, where

- λ_i is the i-th eigenvalue of matrix A;
- j spans the discrete set $\{0, \dots, n_i - g_i\}$, where n_i and g_i are the algebraic and the geometric, respectively, multiplicity of eigenvalue λ_i.

B. LINEAR TIME-INVARIANT (LTI) DISCRETE-TIME SYSTEMS

1) Discretization

Consider a continuous-time system of the type (i)- (ii). We define the sampling time with h and we make the assumption that we use zero-order hold actuators, i.e., that the input signal $u(t)$ is piecewise constant. Specifically, $u(t) = u_k$ for all $t \in [kh, (k+1)h)$. We define $x_k = x(kh)$ and $y_k = y(kh)$. The evolution of x_k can be obtained resorting to the Lagrange equation, i.e.,

$$x_{k+1} = x\big((k+1)h\big) = e^{A\left((k+1)h-kh\right)}x(kh) + \int_{kh}^{(k+1)h} e^{A\left((k+1)h-v\right)}dvBu_k = e^{Ah}x_k + \int_0^h e^{Av}dvBu_k$$

Overall, the obtained discrete-time linear time invariant system is characterized by the following equations

$$\mathcal{S}_{DT}\colon \begin{cases} x_{k+1} & = & Fx_k + Gu_k & \quad (iv) \\ y_k & = & Hx_k + Lu_k & \quad (v) \end{cases}$$

where $k \in \mathbb{N}$, $F = e^{Ah}$, $G = \int_0^h e^{Av}dvB$, $H = C$, $L = D$.

2) Solution

Given the initial condition x_0 and the input trajectory u_k, $k \in \mathbb{N}$, the state and output motions are provided by the following equation

$$x_k = \underbrace{F^k x_0}_{x_{\mathrm{FR}_k}\text{: free motion}} + \underbrace{\sum_{i=1}^{k} F^{k-i}Gu_{i-1}}_{x_{\mathrm{FO}_k}\text{: forced motion}} = F^k x_0 + \sum_{i=1}^{k} F^{i-1}Gu_{k-i}$$

$$y_k = \underbrace{HF^k x_0}_{y_{\mathrm{FR}_k}\text{: free motion}} + \underbrace{Hx_{\mathrm{FO}_k} + Lu_k}_{y_{\mathrm{FO}_k}\text{: forced motion}}$$

3) Equilibria

Given a constant input trajectory $u_k = \bar{u}$ for all $k \in \mathbb{N}$, the vector $\bar{x} \in \mathbb{R}^n$ is an equilibrium for system $\mathcal{S}_{DT}$ iff $\bar{x} = F\bar{x} + G\bar{u}$. The following remarks are in order.

- If $\bar{u} = 0$, $\bar{x} = 0$ is an equilibrium condition for any possible matrix F.
- If $\bar{u} \neq 0$
  - if $(I - F)$ is invertible (i.e., if $\det(I - F) \neq 0$, which implies that also matrix A is invertible), then $\bar{x} = (I - F)^{-1}G\bar{u} = \left(I - e^{Ah}\right)^{-1}A^{-1}\left(e^{Ah} - I\right)B\bar{u} = -A^{-1}B\bar{u}$;
  - if $(I - F)$ is not invertible (i.e., if $\det(I - F) = 0$) the equation $\bar{x} = F\bar{x} + G\bar{u}$ may have no or infinitely many solutions.

4) Stability

The following considerations hold.

- The definition of stability, asymptotic stability, and instability, in case of discrete-time systems, is similar to the one of continuous-time systems and will not be repeated here for conciseness.
- As in the continuous-time case, stability, asymptotic stability, and instability are *system properties*.
- If a discrete-time LTI system is asymptotically stable (AS) it is also *exponentially stable* (ES), i.e., $\exists \alpha > 0, \lambda \in [0,1)$ such that $||x_k|| \leq \alpha\lambda^k ||x_0||$ if $u_k = 0$.

Theorem 3

A discrete-time LTI system is

- asymptotically stable if and only if all the eigenvalues of F have modulus strictly smaller than 1 (a matrix with this property is denoted Schur stable);
- unstable if at least one eigenvalue of F has modulus greater than 1;
- stable if and only if all the eigenvalues of F satisfy $|\lambda| \leq 1$ and those verifying $|\lambda| = 1$ have algebraic multiplicity equal to the geometric one.

Remark the following: letting $\lambda_i(A)$ denote the *i*-th eigenvalue of the continuous-time system and with $\lambda_i(F)$ the *i*-th eigenvalue of the corresponding discrete-time system matrix, it holds that $\lambda_i(F) = e^{\lambda_i(A)h}$. Therefore

- if $Re(\lambda_i(A)) < 0$ then $|\lambda_i(F)| < 1$;
- if $Re(\lambda_i(A)) = 0$ then $|\lambda_i(F)| = 1$;
- if $Re(\lambda_i(A)) > 0$ then $|\lambda_i(F)| > 1$.

Therefore, the (continuous-time) stability properties of A and those (in discrete-time) of F are strictly correlated.

Theorem 4 (Lyapunov inequality)

Matrix F is Schur stable if and only if there exists a symmetric positive definite matrix P such that

$$F^T PF - P < 0 \quad (vi)$$

Sketch of the proof

<u>I) F is Schur stable $\Rightarrow$ there exists a symmetric positive definite matrix P such that (*vi*) holds.</u>

If F is Schur stable then, for any symmetric matrix $Q > 0$, we can define

$$P = \sum_{i=0}^{+\infty} (F^T)^i Q F^i$$

Note that P exists and is well defined thanks to the fact that F is Schur stable. Also, it is positive definite. We can write

$$F^T P F = \sum_{i=0}^{+\infty} (F^T)^{i+1} Q F^{i+1} = \sum_{i=1}^{+\infty} (F^T)^i Q F^i = \sum_{i=0}^{+\infty} (F^T)^i Q F^i - Q = P - Q$$

Therefore

$$F^T P F - P = -Q < 0$$

<u>II) There exists a symmetric positive definite matrix P such that (*vi*) holds $\Rightarrow$ F is Schur stable.</u>

For any eigenvector v ($\neq 0$) of F

$$0 > v^T (F^T P F - P) v = (|\lambda|^2 - 1) v^T P v$$

Since $P > 0$, the latter inequality is verifiable only if $|\lambda|^2 - 1 < 0$. This proves that every eigenvalue of F has modulus strictly smaller than 1, i.e., *F* is Schur stable.

Remark

In view of (*vi*), the quadratic function $V(x) = x^T P x$ is a Lyapunov function for the autonomous system $x_{k+1} = F x_k$. In fact

$$V(x_{k+1}) = x_k^T F^T P F x_k < x_k^T P x_k = V(x_k)$$

5) Modes

The modes of an LTI system are, for $i = 1, \dots, n$, $k^j \lambda_i^k$, where

- λ_i is the i-th eigenvalue of matrix F;
- j spans the discrete set $\{0, n_i - g_i\}$, where n_i and g_i are the algebraic and the geometric, respectively, multiplicity of eigenvalue λ_i.

Note that, the modes of the corresponding continuous-time system, computed in $t = kh$, are $h^j \left(k^j (e^{\lambda_i(A)h})^k\right)$, where $\lambda_i(A)$ is the i-th eigenvalue of A. Recalling that the i-th eigenvalue of matrix F verifies $\lambda_i(F) = e^{\lambda_i(A)h}$, it results clear that the modes of the discrete-time system $\mathcal{S}_{DT}$ (i.e., $k^j \lambda_i(F)^k$) are, as a matter of fact, a sampled version of the modes of the continuous-time system $\mathcal{S}$.

6) The Jury criterion

To analyze the Schur stability property of matrix F from its characteristic polynomial we can rely on the Jury criterion. It consists of the following steps.

- <u>Construction of the Jury table</u>
  Consider the characteristic polynomial of F, i.e.,
  $$p_F(\lambda) = \det(\lambda I - F) = \varphi_0\lambda^n + \varphi_1\lambda^{n-1} + \cdots + \varphi_n$$
  The Jury table is triangular and is constructed in a recursive way, i.e., thanks to the following rule
  $$\begin{array}{cccccc} \varphi_0 & \varphi_1 & \varphi_2 & \cdots & \varphi_{n-1} & \varphi_n \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ h_1 & h_2 & \dots & h_{\nu-1} & h_\nu & \\ l_1 & l_2 & \dots & l_{\nu-1} & & \\ \vdots & \vdots & \ddots & & & \end{array}$$
  The i-th element of the r-th line is defined based on the coefficients of the $(r-1)$-th according to the following equation.
  $$l_i = \frac{1}{h_1}\det\left(\begin{bmatrix} h_1 & h_{\nu-i+1} \\ h_\nu & h_i \end{bmatrix}\right)$$
  If $h_1 = 0$ we say that the Jury table is not well-defined.

- <u>Application of the Jury stability criterion</u>
  F is Schur stable if and only if the Jury table is well defined and all the elements of its first column are concordant (i.e., have the same sign).

EXAMPLE: 2x2 system

The characteristic polynomial of a 2x2 matrix can be written in the following form

$$p_F(\lambda) = \lambda^2 + a\lambda + b$$

The corresponding Jury table is

$$\begin{array}{ccc} 1 & a & b \\ 1-b^2 & a(1-b) & \\ \frac{(1-b)^2}{1-b^2}((1+b)^2 - a\text{^}2) & & \end{array}$$

According to the Jury criterion, the matrix F is Scur stable if and only if the following conditions hold at the same time

$$\begin{cases} -1 < b < 1 \\ -(1+b) < a < 1+b \end{cases}$$

C. Linear matrix inequalities

When checking the stability properties of LTI systems, we have encountered some inequalities involving matrices, i.e.,

$$P > 0$$

$$A^T P + PA < 0$$

$$F^T PF - P < 0$$

They are typical examples of *linear matrix inequalities* (LMI).

Definition

An LMI is an inequality $f(X) > 0$ where $f: V \to S^n$. In this definition

- V is a finite-dimensional vector space,
- S^n is the set of symmetric $n \times n$ matrices,
- f is an affine function.

$f(X) > 0$ means that $f(X)$ is definite positive, while $f(X)$ is affine if $f(X) = f_0 + t(X)$ where $f_0 \in S^n$ and $t(X): V \to S^n$ is linear.

Typical *LMI optimization problems* are of the type

$$\min_X c(X)$$

subject to

$$\begin{cases} f_1(X) > 0 \\ \vdots \\ f_p(X) > 0 \end{cases}$$

where $c(X)$ is a linear function and $f_i(X) > 0$ are LMIs.

On the other hand, solving an *LMI feasibility problem* consists of checking if there exists X such that

$$\begin{cases} f_1(X) > 0 \\ \vdots \\ f_p(X) > 0 \end{cases}$$

Some final remarks are due. A number of interesting control problems can be formulated as LMIs.

LMI feasibility and optimization problems are convex programs, for which there are efficient (polynomial-time) algorithms. Available softwares that can be used with MATLAB are Yalmip, CVX, LMI Matlab toolbox, and many others. Available LMI solvers are, e.g., SDPT3 and SEDUMI.

RELATED REFERENCES

- S. Boyd and L.V. Vanderberghe. *Convex Optimization*. Cambridge University press, 2004, available online.
- S. Boyd *et al*. *Linear matrix inequalities in systems and control theory*. SIAM, Volume 15 of studies in applied mathematics, 1994, available online.

EXERCISES: THEORETICAL BACKGROUND

EXERCISE 1

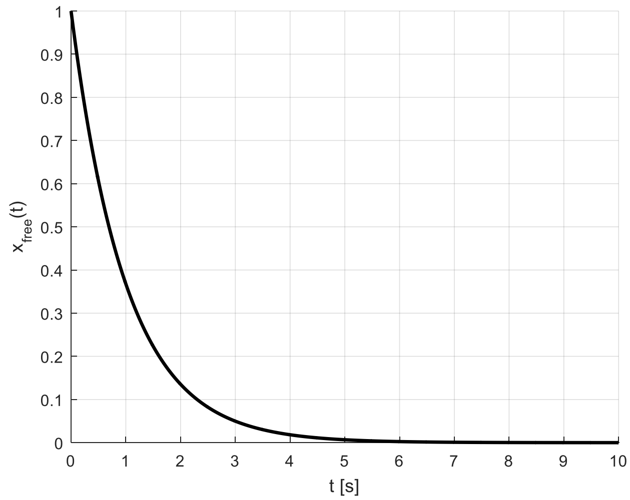
Consider the continuous-time system

$$\dot{x} = -x + u$$

A) Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
B) Compute the free motion of the system state starting from the initial condition $x(0) = 1$.
C) Considering the corresponding open-loop dynamical equation $\dot{x} = -x$, compute a value of $P > 0$ that satisfies the corresponding Lyapunov inequality. Based on P, define a quadratic Lyapunov function of the system and verify that it decreases in time.
D) Compute the forced motion of the state of the system in the following cases:
   a. $u(t) = 5$ for all $t \geq 0$;
   b. $u(t) = e^{-t}, t \geq 0$;
   c. $u(t) = e^{-2t}, t \geq 0$.
E) Assume that the system is controlled using a control law of the type $u(t) = Kx(t)$.
   a. Define the range of values of K that make the closed-loop system asymptotically stable.
   b. Define the value of K that makes the settling time of the closed-loop system smaller or equal to 0.5 s.

SOLUTION

A) The eigenvalue of the system is $\lambda = a = -1$ and the corresponding mode is e^{-t}. The system is asymptotically stable.
B) The free motion of the system, starting from the initial condition $x(0) = 1$, is $x_{free}(t) = e^{-t}$, and is depicted I the following plot.



C) The corresponding Lyapunov inequality is

$$a^T P + Pa = -2P < 0$$

which holds for any $P > 0$. The corresponding quadratic Lyapunov function is

$$V(x) = Px^2$$

We compute that, for all $P > 0$,

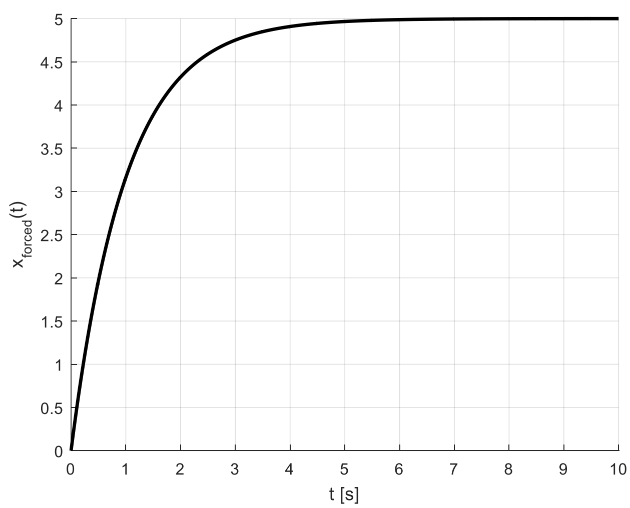
$$\dot{V}(x) = 2Px(ax) = -2Px^2 < 0$$

D) The forced motion is computed as

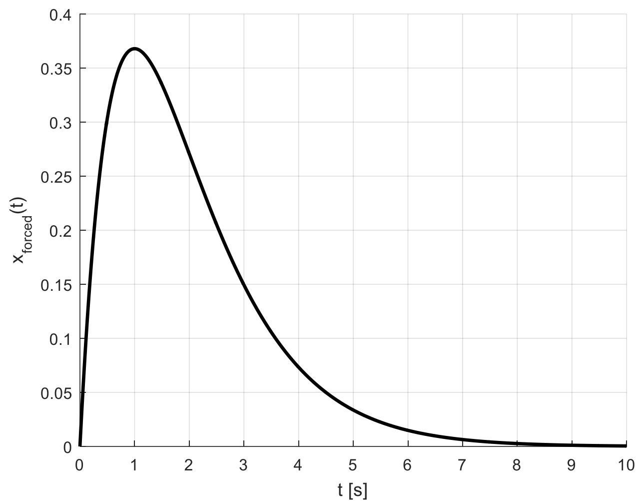
$$x_{forced}(t) = \int_0^t e^{a(t-\tau)} b u(\tau) d\tau$$

Consider now the different cases.

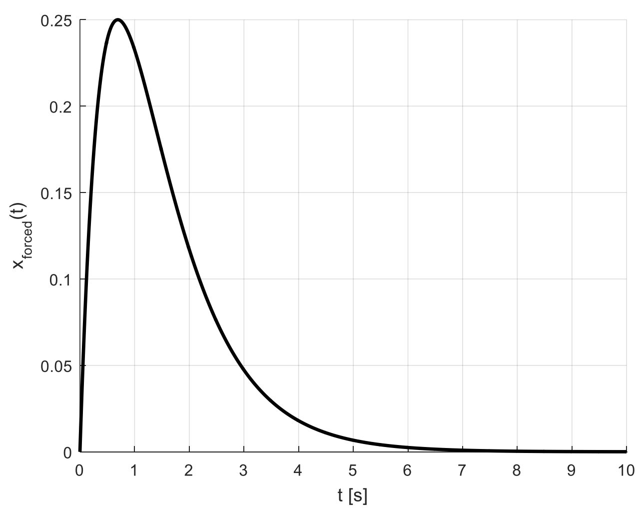
a. For $u(t) = 5$ for all $t \geq 0$, $x_{forced}(t) = 5(1 - e^{-t})$, depicted in the following plot.



b. For $u(t) = e^{-t}, t \geq 0$, $x_{forced}(t) = te^{-t}$, depicted in the following plot.



c. For $u(t) = e^{-2t}, t \geq 0$, $x_{forced}(t) = e^{-t} - e^{-2t}$, depicted in the following plot.



E) Setting $u(t) = Kx(t)$, the closed loop dynamics is described by the equation

$$\dot{x} = (-1 + K)x$$

whose eigenvalue is $\lambda = -1 + K$.

a. The system is closed-loop asymptotically stable for $K < 1$.
b. The settling time is given by

$$T_{set} \cong \frac{5}{|Re(\lambda)|}$$

Therefore, to have $T_{set} < 0.5$ s, we must set $K < -9$.

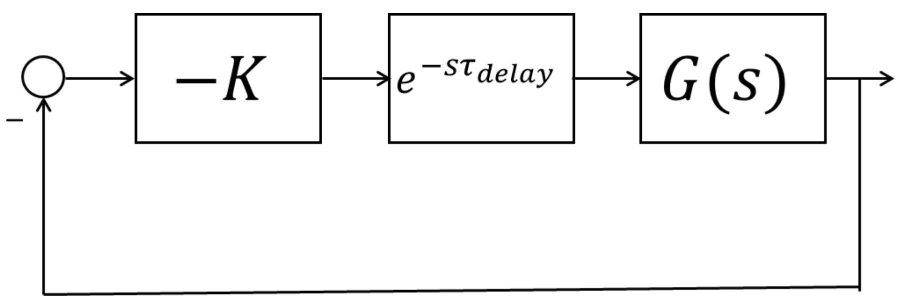
EXERCISE 2

Consider the continuous-time system

$$\dot{x} = -x + u$$

First of all, devise the corresponding discrete-time dynamical equation, with sampling time $h = 2$ s.

A) Compute the eigenvalue and the mode of the discretized system. Is the system open-loop asymptotically stable?
B) Compute, using the discrete-time system equation, the free motion of the system state starting from the initial condition $x_0 = 1$.
C) Considering the corresponding open-loop discrete-time dynamical system, compute a value of $P > 0$ that satisfies the corresponding (discrete-time) Lyapunov inequality. Based on P, define a quadratic Lyapunov function of the discrete-time system and verify that it decreases in time.
D) Compute the forced motion of the state of the system in the following cases:
    a. $u_k = 5$ for all $k \in \mathbb{N}$;
    b. $u_k = e^{-2k}, k \in \mathbb{N}$;
    c. $u_k = \left(\frac{1}{2}\right)^k, k \in \mathbb{N}$.
E) Assume that the system is controlled using a discrete-time control law of the type $u_k = Kx_k$.
    a. Define the range of values of K that make the closed-loop system asymptotically stable.
    b. Is the value of the gain K computed in EXERCISE 1.E.b suitable to stabilize the system under the assumption of sample-and-hold actuation?
F) For comparison, repeat the analysis required at point E) using frequency-domain (Bode-type) arguments, considering that the closed loop system, controlled with a sampled control law $u_k = Kx_k$, can be approximated by the scheme



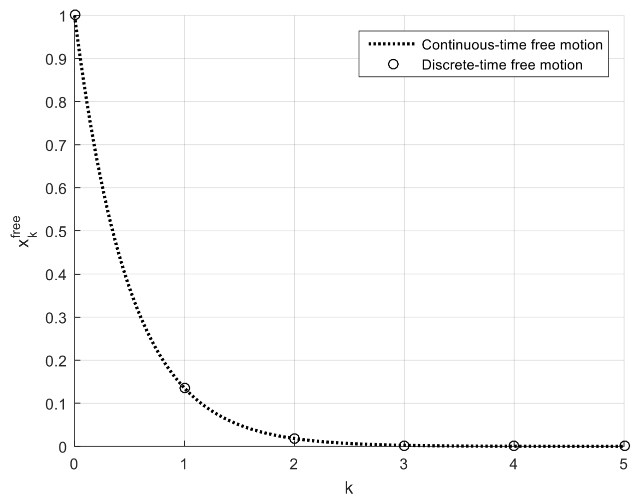
Here $G(s) = \frac{1}{s+1}$ is the transfer function of the original continuous-time system and, as usually done, the effect of sampling is assumed to be equivalent to a delay of magnitude $\tau_{delay} = \frac{h}{2} = 1$ s.

SOLUTION

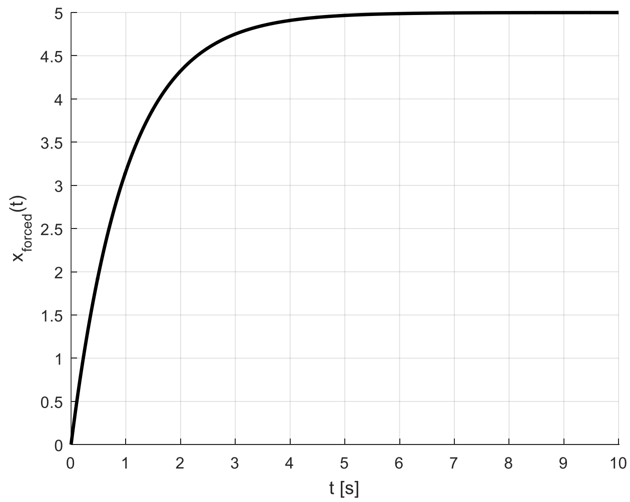
The corresponding discrete-time system is

$$x_{k+1} = fx_k + gu_k = e^{-h}x_k + (1 - e^{-h})u_k = 0.1353x_k + 0.8647u_k$$

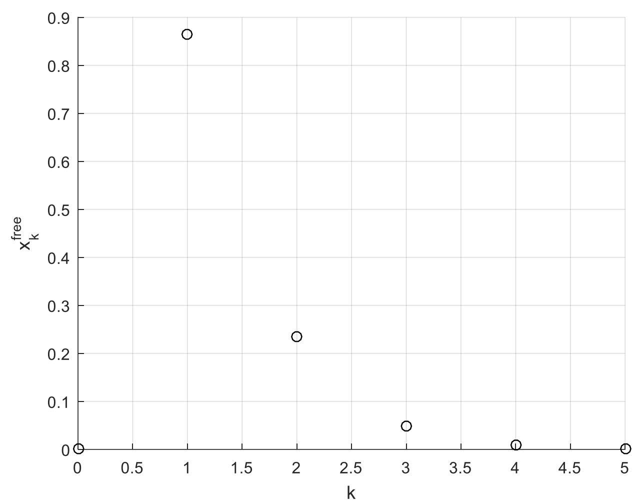
A) The eigenvalue of the system is $\lambda = f = 0.1353$ and the corresponding mode is $(0.1353)^k$. The system is asymptotically stable.
B) The free motion of the system, starting from the initial condition $x(0) = 1$, is $x_{free}(t) = (0.1353)^k = (e^{-h})^k = e^{-kh}$. Its plot is depicted in the following figure, and compared with the corresponding plot of the free motion in continuous time.



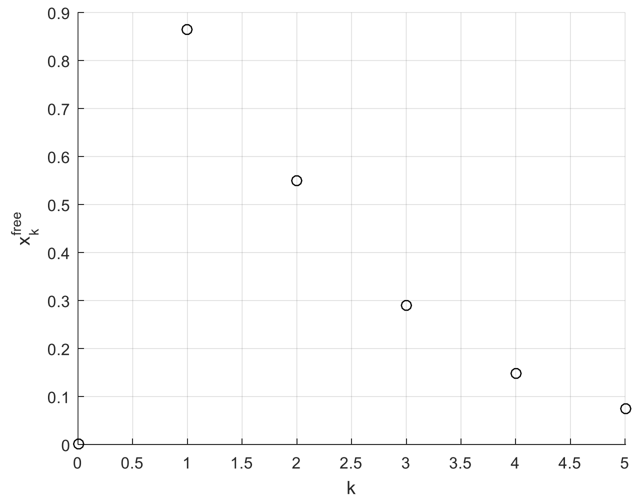
C) The corresponding Lyapunov inequality is
$$f^T Pf - P = -0.9817P < 0$$
which holds for any $P > 0$. The corresponding quadratic Lyapunov function is
$$V(x) = Px^2$$
We compute that, for all $P > 0$,
$$V(x_{k+1}) - V(x_k) = V(fx_k) - V(x_k) = 0.0183Px_k^2 - Px_x^2 = -0.9817Px_k^2 < 0$$
D) The forced motion is computed as
$$x_k^{forced} = \sum_{i=1}^{k} f^{i-1} g\, u_{k-i}$$
Consider now the different cases.
   a. For $u_k = 5$, $x_k^{forced} = 5g\frac{1-f^k}{1-f} = \frac{5(1-e^{-h})}{1-e^{-h}}(1 - e^{-kh}) = 5(1 - e^{-kh})$, depicted in the following plot.



b. For $u_k = e^{-2k} = e^{-hk}$, $x_k^{forced} = \sum_{i=1}^{k} e^{-h(i-1)}(1 - e^{-h})\, e^{-h(k-i)} = ke^{-h(k-1)}(1 - e^{-h})$, depicted in the following plot.



c. For $u_k = \left(\frac{1}{2}\right)^k, t \geq 0$, $x_k^{forced} = 2\frac{1-e^{-h}}{1-2e^{-h}}(\frac{1}{2}^k - e^{-kh})$, depicted in the following plot.

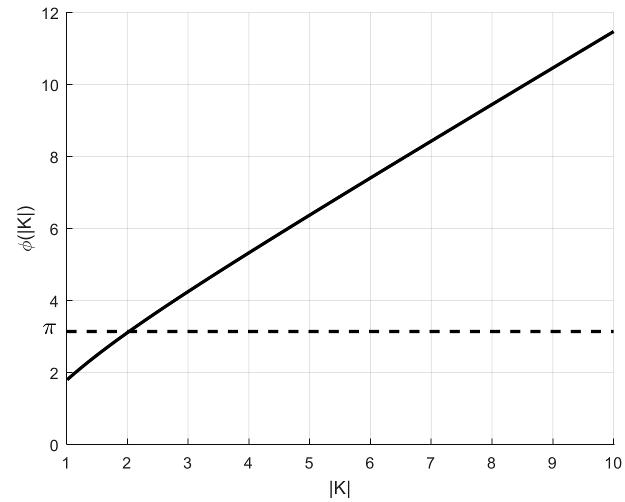


E) Setting $u_k = Kx_k$, the closed loop dynamics is described by the equation
$$x_{k+1} = (e^{-h} + (1 - e^{-h})K)x_k$$
whose eigenvalue is $\lambda = e^{-h} + (1 - e^{-h})K$.

a. The system is closed-loop asymptotically stable iff $|e^{-h} + (1 - e^{-h})K| = |1 + (e^{-h} - 1) + (1 - e^{-h})K| = |1 + (K - 1)(1 - e^{-h})| < 1$, i.e., iff $-1 - \frac{2e^{-h}}{1-e^{-h}} < K < 1$, i.e., $-1.3130 < K < 1$.

b. The value of the gain $K = -9$ computed in EXERCISE 1.E.b is not suitable to stabilize the system under the assumption of sample-and-hold actuation.

F) First note that, $-1 \le K < 1$, asymptotic stability is guaranteed by the Nyquist criterion. Secondly, to verify the applicability assumptions of the Bode criterion, $|K| > 1$, since otherwise the Bode diagram of the modulus of $L^o(s) = -KG(s)$ would not cross the zero dB axis. Also, to keep the gain (i.e., $-K$) of $L^o(s)$ positive we set $K < 0$. It is easy to see that, for $|K| > 1$, the approximated critical angular frequency (at least if we consider the asymptotic Bode diagrams) is equal to $\omega_c = |K|$. The corresponding critical phase deviation (considering the transfer function $L(s) = L^o(s)e^{-s\tau_{delay}}$) is equal to $\measuredangle L(\omega_c) = -\operatorname{atan}(\omega_c)\frac{180^\circ}{\pi} - \omega_c \tau_{delay}\frac{180^\circ}{\pi}$. According to the Bode criterion, stability is preserved as long as $\measuredangle L(\omega_c) > -180^\circ$, i.e., as long as
$$\phi(|K|) = \operatorname{atan}(|K|) + |K|\frac{h}{2} < \pi$$
The plot of $\phi(|K|)$ is shown below.



As shown in the plot, for $1 < |K| < 2$, $\phi(|K|) < \pi$, which should guarantee asymptotic stability of the scheme for $-2 < K < -1$ (in addition to $-1 \leq K < 1$). However, this result is prone to approximations, and therefore not extremely precise.

EXERCISE 3

Consider the continuous-time system

$$\dot{x} = x + u$$

A) Compute the eigenvalue of the system. Is it open-loop asymptotically stable?
B) Assuming that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.

Now, devise the corresponding discrete-time equation, with generic sampling time h.

C) Compute the eigenvalue of the discretized system. Is it open-loop asymptotically stable?
D) Assuming that the system is controlled using a discrete-time control law of the type $u_k = Kx_k$, compute the range of values of K that make the closed-loop system asymptotically stable as a function of h.

SOLUTION

A) The eigenvalue of the system is $\lambda = a = 1 > 0$. The system is unstable.
B) Setting $u(t) = Kx(t)$, the closed loop dynamics is described by the equation
$$\dot{x} = (1+K)x$$
whose eigenvalue is $\lambda = 1 + K$. The system is closed-loop asymptotically stable for $K < -1$.

The corresponding discrete-time system is

$$x_{k+1} = fx_k + gu_k = e^h x_k + (e^h - 1)u_k$$

C) The eigenvalue of the system is $\lambda = f = e^h$. Since $|e^h| > 1$, the system is unstable.
D) Setting $u_k = Kx_k$, the closed loop dynamics is described by the equation
$$x_{k+1} = (e^h + (e^h - 1)K)x_k$$
whose eigenvalue is $\lambda = e^h + (e^h - 1)K$. The system is closed-loop asymptotically stable iff $|e^h + (e^h - 1)K| = |1 + (e^h - 1) + (e^h - 1)K| = |1 + (K+1)(e^h - 1)| < 1$, i.e., iff $-1 - \frac{2}{e^h - 1} < K < -1$.

EXERCISE 4

Consider the continuous-time system

$$\dot{x} = Ax + Bu$$
$$y = Cx$$

where

$$A = \begin{bmatrix} -1 & 0 \\ 0.2 & -0.2 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C = [0\ 1]$$

A) Compute the eigenvalues and the modes of the system. Is the system open-loop asymptotically stable?
B) Compute the free motion of the output of the system starting from the initial condition $x(0) = [1,0]^T$.
C) Compute the equilibrium point of the system output when $u(t) = 1$ for all $t \geq 0$.

SOLUTION

A) The eigenvalues of the system are $\lambda_1 = -1 < 0$ and $\lambda_2 = -0.2 < 0$. The system is asymptotically stable.
B) The modes of the system are e^{-t} and $e^{-0.2t}$. The free motion of the output of the system is a linear combination of the system modes, i.e.,
$$y(t) = \gamma_1 e^{-t} + \gamma_2 e^{-0.2t}$$
We compute that $\dot{y}(t) = -\gamma_1 e^{-t} - 0.2\gamma_2 e^{-0.2t}$. The parameters γ_1 and γ_2 can be derived by solving the following system.
$$y(0) = \gamma_1 + \gamma_2 = \quad Cx(0) = 0$$
$$\dot{y}(0) = -\gamma_1 - 0.2\gamma_2 = \quad CAx(0) = 0.2$$
The solution is $\gamma_1 = -0.25$, $\gamma_2 = 0.25$.
C) The equilibrium point of the system output when $u(t) = 1$ is
$$\bar{y} = -CA^{-1}B\bar{u} = 1.5$$

EXERCISE 5

Consider the discrete-time system obtained by discretization of the model in Exercise 4 with $h = 1$ s.

$$x_{k+1} = Fx_k + Gu_k$$

$$y_k = Hx_k$$

where

$$F = \begin{bmatrix} 0.3679 & 0 \\ 0.1127 & 0.8187 \end{bmatrix}, G = \begin{bmatrix} 0.6321 \\ 0.1592 \end{bmatrix}, H = [0\ 1]$$

A) Compute the eigenvalues and the mode of the system. Is the system open-loop asymptotically stable?
B) Compute the free motion of the system state starting from the initial condition $x_0 = [1,0]^T$.
C) Compute the equilibrium point of the system output when $u_k = 1$ for all $k \in \mathbb{R}$.

SOLUTION

A) The eigenvalues of the system are $\lambda_1 = e^{-1} = 0.3679 < 0$ and $\lambda_2 = e^{-0.2} = 0.8187$. Since $|\lambda_1|$<1 and $|\lambda_2| < 1$, the system is asymptotically stable.
B) The modes of the system are e^{-k} and $e^{-0.2k}$. The free motion of the output of the system is a linear combination of the system modes, i.e.,
$$y_k = \gamma_1 e^{-k} + \gamma_2 e^{-0.2k}$$
We compute that $y_{k+1} = -\gamma_1 e^{-(k+1)} - 0.2\gamma_2 e^{-0.2(k+1)}$. The parameters γ_1 and γ_2 can be derived by solving the following system.
$$\begin{aligned} y_0 = \gamma_1 + \gamma_2 = &\quad Hx(0) = 0 \\ y_1 = \gamma_1 e^{-1} + \gamma_2 e^{-0.2} = &\quad HFx(0) = 0.1127 \end{aligned}$$
The solution is $\gamma_1 = -0.25$, $\gamma_2 = 0.25$, consistently with the solution to the previous exercise.
C) The equilibrium point of the system output when $u_k = 1$ is
$$\bar{y} = H(I - F)^{-1}G\bar{u} = 1.5$$

EXERCISE 6

Consider the continuous-time system

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

where

$$A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, C = [1\ 1]$$

A) Compute the eigenvalues and the modes of the system. Is the system open-loop asymptotically stable?
B) Assume that the system is controlled using a static output feedback control law of the type $u(t) = Ky(t)$. Define the range of values of K that make the closed-loop system asymptotically stable.

SOLUTION

A) The eigenvalues of the system are $\lambda_1 = 0$ and $\lambda_2 = -1 < 0$. The system is stable but not asymptotically. The modes of the system are $e^{0t} = 1$ and e^{-t}.
B) Setting $u(t) = Ky(t) = KCx(t)$, the closed loop dynamical system is
$$\dot{x} = (A + BKC)x(t)$$
where
$$A + BKC = \begin{bmatrix} K & K \\ K & K-1 \end{bmatrix}$$
The characteristic polynomial of $A + BKC$ is $p_{A+BKC}(\lambda) = \det(\lambda I - (A + BKC)) = \lambda^2 + (1 - 2K)\lambda - K$. The necessary and sufficient condition for asymptotical stability is that
$$\begin{cases} 1 - 2K > 0 \\ -K > 0 \end{cases}$$
This is verified if and only if $K < 0$.

EXERCISE 7

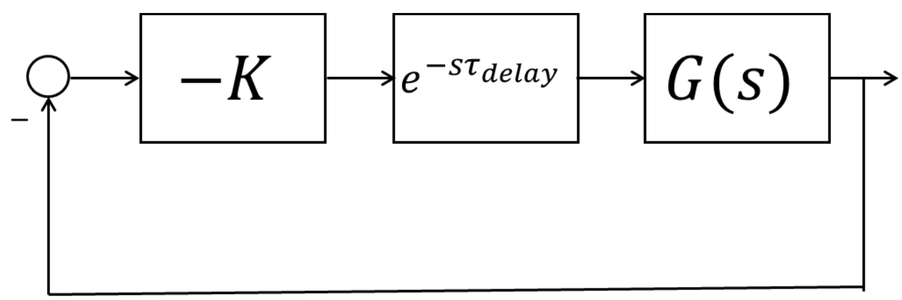
Consider the discrete-time system obtained by discretization of the model in Exercise 6 with $h = 1$ s.

$$x_{k+1} = Fx_k + Gu_k$$
$$y_k = Hx_k$$

where

$$F = \begin{bmatrix} 1 & 0 \\ 0 & 0.3679 \end{bmatrix}, G = \begin{bmatrix} 1 \\ 0.6321 \end{bmatrix}, H = [1\ 1]$$

A) Compute the eigenvalues and the mode of the system. Is the system open-loop asymptotically stable?
B) Assuming that the system is controlled using a static output feedback discrete-time control law of the type $u_k = Ky_k$, compute the range of values of K that make the closed-loop system asymptotically stable as a function of h.
C) For comparison, repeat the analysis required at point B) using frequency-domain (Bode-type) arguments, considering that the closed loop system, controlled with a sampled control law $u_k = Kx_k$, can be approximated by the scheme



Here $G(s) = C(sI - A)^{-1}B = \frac{1+2s}{(s+1)s}$ is the transfer function of the original continuous-time system and, as usually done, the effect of sampling is assumed to be equivalent to a delay of magnitude $\tau_{delay} = \frac{h}{2} = 0.5$ s.

SOLUTION

A) The eigenvalues of the system are $\lambda_1 = e^0 = 1$ and $\lambda_2 = e^{-1}$. The system is stable but not asymptotically. The modes of the system are $(e^0)^k = 1$ and $(e^{-1})^k$.
B) Setting $u_k = Ky_k = KHx_k$, the closed loop dynamical system is

$$x_{k+1} = (F + GKH)x_k$$

where

$$F + GKH = \begin{bmatrix} 1 + K & K \\ 0.6321K & 0.6321K + e^{-1} \end{bmatrix}$$

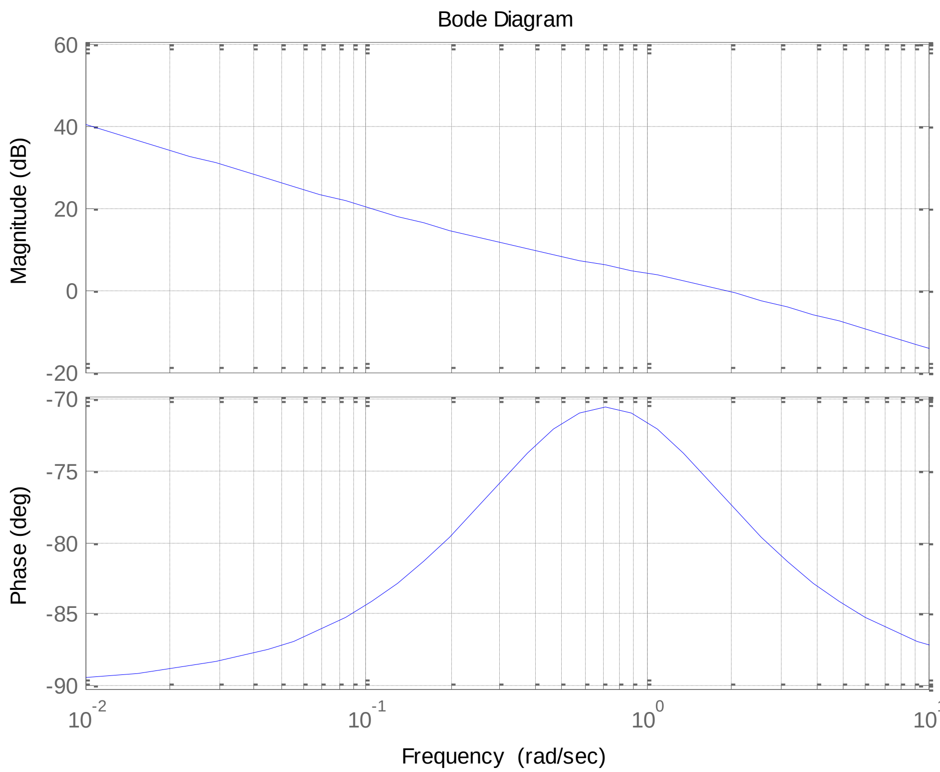
The characteristic polynomial of $F + GKH$ is $p_{F+GKH}(\lambda) = \det(\lambda I - (F + GKH)) = \lambda^2 - (1 + 1.6321K + e^{-1})\lambda + (K + e^{-1})$. According to the Jury criterion, a necessary and sufficient condition for $F + GKH$ to be Schur stable is that

$$\begin{cases} -1 < K + e^{-1} < 1 \\ -1 - K - e^{-1} < -(1 + 1.6321K + e^{-1}) < 1 + K + e^{-1} \end{cases}$$

This occurs if and only if

$$-\frac{2 + 2e^{-1}}{2.6321} = -1.0394 < K < 0$$

G) The Bode diagrams of G(s) are shown below



Note that the Bode criterion can be applied for all values of $K \neq 0$. In particular, when $K > 0$ the closed loop system is not asymptotically stable since the loop transfer function has negative gain. So, we restrict our further analysis to the case $K < 0$. Under this condition, asymptotic stability is guaranteed if and only if

$$-90 - (\text{atan}(\omega_c) - \text{atan}(2\omega_c))\frac{180}{\pi} - \frac{h}{2}\omega_c\frac{180}{\pi} > -180$$

i.e., iff

$$\text{atan}(\omega_c) - \text{atan}(2\omega_c) + \frac{\omega_c}{2} < \frac{\pi}{2}$$

This occurs as $\omega_c < 3.42$. This occurs if, circa, $|K| < 1.8$ (the latter results have been obtained using MATLAB).

PART II: DECENTRALIZED AND DISTRIBUTED CONTROL STRUCTURES

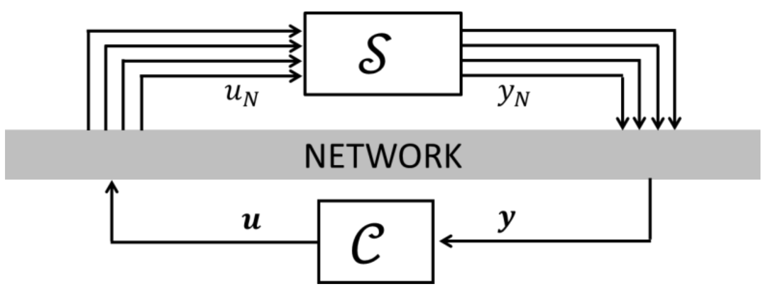
II.1: MODELS FOR DECENTRALIZED AND DISTRIBUTED CONTROL

A. INTRODUCTION

Decentralized and distributed control structures are often necessary when there is a possibly large number of geographically distributed inputs (i.e., actuators) and outputs (i.e., sensors).

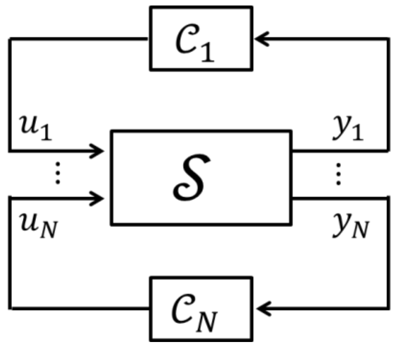
In this case, the *system* to be controlled can be regarded as a *network of subsystems* (each having its "dedicated" input and output), interacting with each other in various ways. In fact, the input of subsystem i may affect the output of a different subsystem j, for $i \neq j$. In view of this, these subsystems cannot be regarded as isolated ones, but must be considered as parts of a *comprehensive system*, to be controlled in an integrated fashion.

The most common (classic) control scheme is the *centralized* one. For the implementation of a centralized controller, a transmission link must exist between each actuator/sensor pair and a monolithic controller (possibly through a *control network*), which is in charge of computing the control action to be applied by each actuator (see the following figure).



However, especially for large-scale systems (i.e., systems with a large number of inputs, states and outputs), centralized control solutions may be impractical, for at least a twofold reason. First, the control of systems with a large number of free manipulable and state variables may become extremely burdensome and even infeasible from the *computational* point of view. Secondly, the implementation of such a control scheme requires a very dense star-like transmission pattern and transmissions to take place instantaneously and synchronously: this may become infeasible as the number of inputs and outputs and the distance between them grow (*information structure constraints*).

Decentralized and distributed control schemes can be adopted as alternatives with respect to the centralized one. They are characterized by the fact that local controllers are embedded at each subsystem/device (i.e., *local*) level. In practice, control loops are closed locally, as in figure.



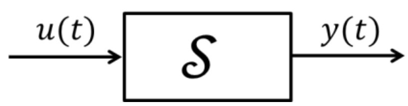
If no communication is possible among the local control stations $\mathcal{C}_1$, …, $\mathcal{C}_N$ the resulting scheme is said to be *decentralized*. One the other hand, if communication is allowed between local controllers $\mathcal{C}_1$, …, $\mathcal{C}_N$, then the resulting scheme is said to be *distributed*.

B. MATHEMATICAL REPRESENTATION: SYSTEM DECOMPOSITION AND CHANNELS

In this section we consider the system under control. The "centralized" system model is the usual continuous-time one.

$$\mathcal{S}: \begin{cases} \dot{x} = & Ax + Bu \\ y = & Cx \end{cases}$$

where $x \in \mathbb{R}^n$, $u \in \mathbb{R}^m$, $y \in \mathbb{R}^p$. Graphically, we can represent it as in the following figure.



Decomposing the model into N submodels consists of decomposing the input and output vectors in N non-overlapping subvectors each, i.e., u_1, …, u_N and y_1, …, y_N, respectively. More specifically, possibly up to a suitable permutation of the elements of u and y, we can write

$$u = \begin{bmatrix} u_1 \\ \vdots \\ u_N \end{bmatrix}, y = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix}$$

where $u_i \in \mathbb{R}^{m_i}$, $y_i \in \mathbb{R}^{p_i}$ and, for consistency, $\sum_{i=1}^{N} m_i = m$ and $\sum_{i=1}^{N} p_i = p$. The input/output pair (u_i, y_i) is denoted i-th *channel*.

We can decompose the system matrices B and C consistently with the decomposition of vectors u and y. More specifically, possibly up to a suitable permutation of the columns of B and of the rows of C, we can define matrices $B_i \in \mathbb{R}^{n \times m_i}$ and $C_i \in \mathbb{R}^{p_i \times n}$ for all $i = 1, \dots, N$ such that

$$B = [B_1 \quad \cdots \quad B_N], C = \begin{bmatrix} C_1 \\ \vdots \\ C_N \end{bmatrix}$$

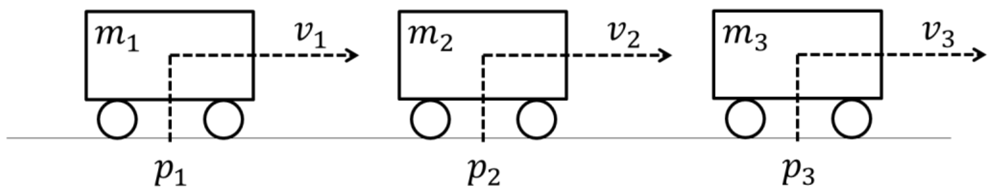
In this way we can rewrite the system model as

$$\begin{cases} \dot{x} = & Ax + B_1 u_1 + \cdots + B_N u_N \\ y_i = & C_i x \qquad\qquad i = 1, \dots, N \end{cases}$$

C. EXAMPLES

1) Detached carts on a railway

We consider the example sketched in the following figure, consisting of $N = 3$ carts moving on a railway.



Each cart (indicated by $i = 1,2,3$) has mass m_i and velocity v_i. Variable p_i indicates the displacement of cart i with respect to a given rest position. Also, we assume that a friction force $-h_i v_i$ acts on each cart, and that each cart has, as a manipulable variable, an input force u_i acting in longitudinal direction.

The dynamics of each cart is described by the following model:

$$m_i \ddot{p}_i = m_i \dot{v}_i = -h_i v_i + u_i$$

In state-space we obtain the following 2nd order model:

$$\begin{bmatrix} \dot{p}_i \\ \dot{v}_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -\dfrac{h_i}{m_i} \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix} + \begin{bmatrix} 0 \\ \dfrac{1}{m_i} \end{bmatrix} u_i$$

Also, we assume that only the position variable p_i is measurable, for each subsystem, i.e.,

$$y_i = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} p_i \\ v_i \end{bmatrix}$$

If we need to control the trucks in a coordinated way, a “monolithic” centralized model must be devised. We define

$$x = \begin{bmatrix} p_1 \\ v_1 \\ p_2 \\ v_2 \\ p_3 \\ v_3 \end{bmatrix}, A = \begin{bmatrix} \begin{matrix} 0 & 1 \\ 0 & -\dfrac{h_1}{m_1} \end{matrix} & 0_{2\times 2} & 0_{2\times 2} \\ 0_{2\times 2} & \begin{matrix} 0 & 1 \\ 0 & -\dfrac{h_2}{m_2} \end{matrix} & 0_{2\times 2} \\ 0_{2\times 2} & 0_{2\times 2} & \begin{matrix} 0 & 1 \\ 0 & -\dfrac{h_3}{m_3} \end{matrix} \end{bmatrix}, B = \begin{bmatrix} \begin{matrix} 0 \\ \dfrac{1}{m_1} \end{matrix} & 0_{2\times 1} & 0_{2\times 1} \\ 0_{2\times 1} & \begin{matrix} 0 \\ \dfrac{1}{m_2} \end{matrix} & 0_{2\times 1} \\ 0_{2\times 1} & 0_{2\times 1} & \begin{matrix} 0 \\ \dfrac{1}{m_3} \end{matrix} \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 \quad 0 & 0_{1\times 2} & 0_{1\times 2} \\ 0_{1\times 2} & 1 \quad 0 & 0_{1\times 2} \\ 0_{1\times 2} & 0_{1\times 2} & 1 \quad 0 \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

In this way we can write

$$\begin{cases} \dot{x} = & Ax + Bu \\ y = & Cx \end{cases}$$

Note that, in view of the fact that the dynamics of each cart is independent both of the dynamics and of the inputs of the other carts, matrices A and B are block-diagonal.

The decomposed model

$$\begin{cases} \dot{x} = & Ax + B_1 u_1 + B_2 u_2 + B_3 u_3 \\ y_i = & C_i x \end{cases} \qquad i = 1,2,3$$

can be obtained by simply defining

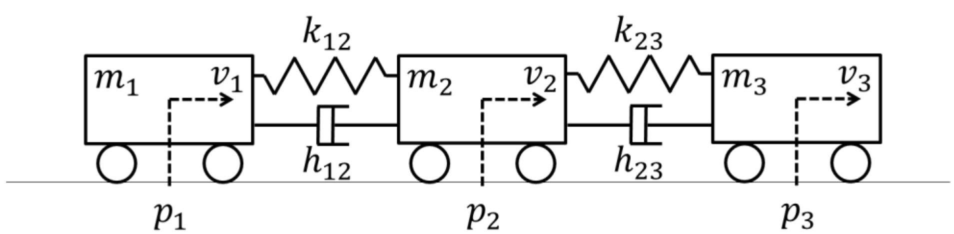
$$B_1 = \begin{bmatrix} 0 \\ \frac{1}{m_1} \\ 0_{2\times1} \\ 0_{2\times1} \end{bmatrix}, B_2 = \begin{bmatrix} 0_{2\times1} \\ 0 \\ \frac{1}{m_2} \\ 0_{2\times1} \end{bmatrix}, B_3 = \begin{bmatrix} 0_{2\times1} \\ 0_{2\times1} \\ 0 \\ \frac{1}{m_3} \end{bmatrix}$$

$$C_1 = [1 \quad 0 \quad 0_{1\times2} \quad 0_{1\times2}], C_2 = [0_{1\times2} \quad 1 \quad 0 \quad 0_{1\times2}],$$

$$C_3 = [0_{1\times2} \quad 0_{1\times2} \quad 1 \quad 0]$$

2) Carts on a railway connected with springs and dampers

We consider the example sketched in the following figure, consisting of $N = 3$ carts moving on a railway, connected together through springs and dampers (with elastic and damping coefficients k_{12}, k_{23} and h_{12}, h_{23}, respectively).



As in the previous example, each cart (indicated by $i = 1,2,3$) has mass m_i and velocity v_i. Variable p_i indicates the displacement of cart i with respect to a given rest position. Also, we assume that a friction force $-h_i v_i$ acts on each cart, and that each cart has, as a manipulable variable, an input force u_i acting in the longitudinal direction. Contrarily to example 1, in this example the dynamics of the carts are coupled. We can write

$$m_1 \ddot{p}_1 = m_1 \dot{v}_1 = -h_1 v_1 + u_1 - k_{12}(p_1 - p_2) - h_{12}(v_1 - v_2)$$

$$m_2 \ddot{p}_2 = m_2 \dot{v}_2 = -h_2 v_2 + u_2 - k_{12}(p_2 - p_1) - h_{12}(v_2 - v_1) - k_{23}(p_2 - p_3) - h_{23}(v_2 - v_3)$$

$$m_3 \ddot{p}_3 = m_3 \dot{v}_3 = -h_3 v_3 + u_3 - k_{23}(p_3 - p_2) - h_{23}(v_3 - v_2)$$

Again, we assume that only the position variable p_i is measurable for each subsystem. The corresponding centralized model is obtained by defining

$$x=\begin{bmatrix}p_1\\v_1\\p_2\\v_2\\p_3\\v_3\end{bmatrix}, A=\begin{bmatrix}\begin{matrix}0 & 1\\ -\dfrac{k_{12}}{m_1} & -\dfrac{(h_1+h_{12})}{m_1}\end{matrix} & \begin{matrix}0 & 0\\ \dfrac{k_{12}}{m_1} & \dfrac{h_{12}}{m_1}\end{matrix} & 0_{2\times 2}\\ \begin{matrix}0 & 0\\ \dfrac{k_{12}}{m_2} & \dfrac{h_{12}}{m_2}\end{matrix} & \begin{matrix}0 & 1\\ -\dfrac{(k_{12}+k_{23})}{m_2} & -\dfrac{(h_2+h_{12}+h_{23})}{m_2}\end{matrix} & \begin{matrix}0 & 0\\ \dfrac{k_{23}}{m_2} & \dfrac{h_{23}}{m_2}\end{matrix}\\ 0_{2\times 2} & \begin{matrix}0 & 0\\ \dfrac{k_{23}}{m_3} & \dfrac{h_{23}}{m_3}\end{matrix} & \begin{matrix}0 & 1\\ -\dfrac{k_{23}}{m_3} & -\dfrac{(h_3+h_{23})}{m_3}\end{matrix}\end{bmatrix},$$

$$B=\begin{bmatrix}\begin{matrix}0\\ \dfrac{1}{m_1}\end{matrix} & 0_{2\times 1} & 0_{2\times 1}\\ 0_{2\times 1} & \begin{matrix}0\\ \dfrac{1}{m_2}\end{matrix} & 0_{2\times 1}\\ 0_{2\times 1} & 0_{2\times 1} & \begin{matrix}0\\ \dfrac{1}{m_3}\end{matrix}\end{bmatrix}, u=\begin{bmatrix}u_1\\u_2\\u_3\end{bmatrix}, C=\begin{bmatrix}1 \quad 0 & 0_{1\times 2} & 0_{1\times 2}\\ 0_{1\times 2} & 1 \quad 0 & 0_{1\times 2}\\ 0_{1\times 2} & 0_{1\times 2} & 1 \quad 0\end{bmatrix}, y=\begin{bmatrix}y_1\\y_2\\y_3\end{bmatrix}$$

Some remarks are in order.

- Matrix A does not display anymore a block-diagonal structure, but a *banded* one (there is a non-zero diagonal band, below and above which there are only zeros). Therefore, we say that the subsystems 1,2,3 are *dynamically coupled*, i.e., since the state variables related to truck i appear in the dynamical model of cart j, with $i \neq j$.
- Matrix B has a block-diagonal structure. In this case, the subsystems are *input-decoupled*, since only the input u_j appears in the dynamical model of cart j.

The decomposed model can be obtained by defining

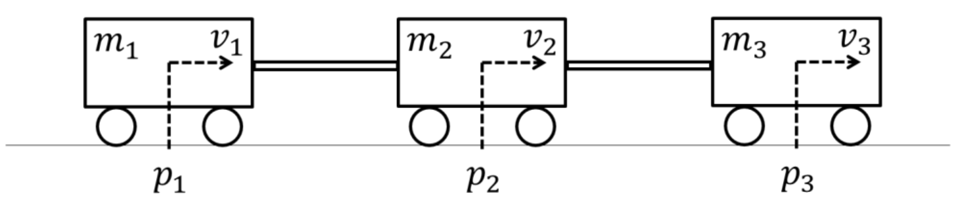
$$B_1=\begin{bmatrix}0\\ \dfrac{1}{m_1}\\ 0_{2\times 1}\\ 0_{2\times 1}\end{bmatrix}, B_2=\begin{bmatrix}0_{2\times 1}\\ 0\\ \dfrac{1}{m_2}\\ 0_{2\times 1}\end{bmatrix}, B_3=\begin{bmatrix}0_{2\times 1}\\ 0_{2\times 1}\\ 0\\ \dfrac{1}{m_3}\end{bmatrix}$$

$$C_1=[1 \quad 0 \quad 0_{1\times 2} \quad 0_{1\times 2}], C_2=[0_{1\times 2} \quad 1 \quad 0 \quad 0_{1\times 2}],$$

$$C_3=[0_{1\times 2} \quad 0_{1\times 2} \quad 1 \quad 0]$$

3) rigidly-connected Carts on a railway

We consider the example sketched in the following figure, consisting of $N=3$ carts moving on a railway, rigidly connected together.



As in the previous example, each cart (indicated by $i = 1,2,3$) has mass m_i and velocity v_i. Variable p_i indicates the displacement of cart i with respect to a given rest position and u_i is the input force acting on it in the longitudinal direction. Since the carts are rigidly connected together, $v_1 = v_2 = v_3 = v_{\text{tot}}$.

A friction force $-h_{\text{tot}} v_{\text{tot}}$ acts on the three-cart system.

The dynamics of each cart is described by the following model:

$$(m_1 + m_2 + m_3)\ddot{p}_i = (m_1 + m_2 + m_3)\dot{v}_{\text{tot}} = -h_{\text{tot}} v_{\text{tot}} + u_1 + u_2 + u_3$$

Again, we assume that only the position variable p_i is measurable for each subsystem.

Depending on how the state vector is defined, we can define different (dynamically equivalent) centralized models. For example, if we define

$$x = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ v_{\text{tot}} \end{bmatrix}, A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -\frac{h_{\text{tot}}}{m_{\text{tot}}} \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

where $m_{\text{tot}} = m_1 + m_2 + m_3$, we obtain the "minimal" representation. The corresponding decomposition is obtained by selecting

$$B_1 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}$$

$$C_1 = [1 \quad 0 \quad 0 \quad 0], C_2 = [0 \quad 1 \quad 0 \quad 0],$$

$$C_3 = [0 \quad 0 \quad 1 \quad 0]$$

An alternative centralized modelling choice consists of defining the state vector as $x = [p_1, v_1, p_2, v_2, p_3, v_3]^T$. However, since $v_1 = v_2 = v_3 = v_{\text{tot}}$, the equations governing the variables v_1, v_2, v_3 and their initial conditions are equal to each other, resulting in an *overlapping model*. In this case

$$A = \begin{bmatrix} \begin{matrix} 0 & 1 \\ 0 & -\frac{h_{\text{tot}}}{m_{\text{tot}}} \end{matrix} & 0_{2\times 2} & 0_{2\times 2} \\ 0_{2\times 2} & \begin{matrix} 0 & 1 \\ 0 & -\frac{h_{\text{tot}}}{m_{\text{tot}}} \end{matrix} & 0_{2\times 2} \\ 0_{2\times 2} & 0_{2\times 2} & \begin{matrix} 0 & 1 \\ 0 & -\frac{h_{\text{tot}}}{m_{\text{tot}}} \end{matrix} \end{bmatrix}, B = \begin{bmatrix} 0 & 0 & 0 \\ \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} \\ 0 & 0 & 0 \\ \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} \\ 0 & 0 & 0 \\ \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} & \frac{1}{m_{\text{tot}}} \end{bmatrix}$$

$$C = \begin{bmatrix} 1 \quad 0 & 0_{1\times 2} & 0_{1\times 2} \\ 0_{1\times 2} & 1 \quad 0 & 0_{1\times 2} \\ 0_{1\times 2} & 0_{1\times 2} & 1 \quad 0 \end{bmatrix}$$

Some remarks are in order.

- Matrix A displays a block-diagonal structure, and therefore this representation regards the systems as "dynamically decoupled".
- Matrix B does not display a block-diagonal structure. Therefore, we say that the subsystems are *input-coupled*.

The corresponding decomposed model can be obtained by defining

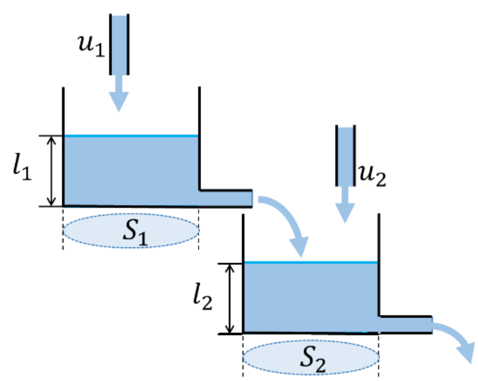
$$B_1 = \begin{bmatrix} 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}, B_3 = \begin{bmatrix} 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \\ 0 \\ \frac{1}{m_{\text{tot}}} \end{bmatrix}$$

$$C_1 = [1 \quad 0 \quad 0_{1\times 2} \quad 0_{1\times 2}], C_2 = [0_{1\times 2} \quad 1 \quad 0 \quad 0_{1\times 2}],$$

$$C_3 = [0_{1\times 2} \quad 0_{1\times 2} \quad 1 \quad 0]$$

4) Cascaded (linearized) water tanks

We consider the example sketched in the following figure, consisting of $N = 2$ connected water tanks.



Each tank (indicated by $i = 1,2$) has basis surface area S_i. Variable h_i indicates the (measurable, i.e., $y_i = h_i$) water level inside tank i. Also, we assume that water with a volumetric flowrate $k_i l_i$ leaves the tank i, and that each tank has, as a manipulable variable, a water input flowrate u_i. The dynamics of the system is described by the following model:

$$S_1 \dot{l}_1 = -k_1 l_1 + u_1$$

$$S_2 \dot{l}_2 = -k_2 l_2 + k_1 l_1 + u_2$$

The corresponding centralized model is obtained by defining

$$x = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}, A = \begin{bmatrix} -\frac{k_1}{S_1} & 0 \\ \frac{k_1}{S_2} & -\frac{k_2}{S_2} \end{bmatrix}, u = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, B = \begin{bmatrix} \frac{1}{S_1} & 0 \\ 0 & \frac{1}{S_2} \end{bmatrix}, y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Some remarks are in order.

- Matrix A displays a block-triangular structure, in view of the cascaded system structure.
- Matrix B is block-diagonal, and therefore the system is *input-decoupled*.

The decomposed model can be obtained by defining

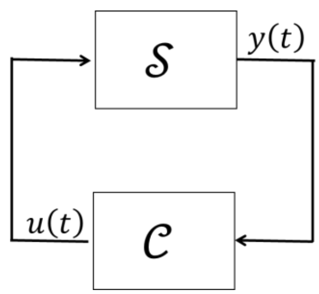
$$B_1 = \begin{bmatrix} \frac{1}{S_1} \\ 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ \frac{1}{S_2} \end{bmatrix}$$
$$C_1 = [1 \quad 0], C_2 = [0 \quad 1].$$

II.2: CONTROL SCHEMES AND CONTROLLABILITY PROPERTIES

In this chapter we define and characterize the *centralized*, *decentralized*, and *distributed control schemes*. Also, we discuss in details the main limitations related to these control schemes, due to the presence of the so-called *fixed modes*.

A. CENTRALIZED CONTROL

We first consider a classic (centralized) control system having the following general structure.



and where the dynamics of system $\mathcal{S}$ is described by the model:

$$\mathcal{S}:\begin{cases}\dot{x}(t) = & Ax(t) + Bu(t) \\ y(t) = & Cx(t)\end{cases} \quad (i)$$

We now describe different control laws applicable, in a centralized fashion, to system $\mathcal{S}$.

1) Static output-feedback control

A static output-feedback control law is of type

$$u(t) = K_y y(t)$$

where $K_y \in \mathbb{R}^{m\times p}$ is the *control gain*. In this way the closed-loop system dynamics is described by the dynamical model

$$\dot{x}(t) = Ax(t) + BK_y y(t) = (A + BK_y C)x(t)$$

As it will be clarified in the following, even in case pairs (A, B) and (A, C) are controllable and observable, respectively, there is no guarantee that, if A is not asymptotically stable, there exists a gain matrix K_y that allows to make $A + BK_y C$ asymptotically (i.e., Hurwitz) stable.

2) *Static state-feedback control*

State-feedback control relies on the assumption that the state is known (e.g., vector $x(t)$ is fully measured, i.e., $y(t) = x(t)$, meaning that $C = I$). In this case a static state-feedback control law can be used, i.e.,

$$u(t) = K_x x(t)$$

where $K_x \in \mathbb{R}^{m\times n}$ is the *control gain.* In this way that the closed-loop system dynamics is described by the dynamical model

$$\dot{x}(t) = (A + BK_x)x(t)$$

A well-known result (see later) states that, if the pair (A, B) is controllable, then we can always design a control gain K_x such that $A + BK_x$ is Hurwitz.

3) *Dynamic output-feedback control*

A generic dynamic output-feedback controller has the following form

$$\mathcal{C}: \begin{cases} \dot{\bar{x}}(t) = & A_C\bar{x}(t) + B_c y(t) \\ u(t) = & K_x\bar{x}(t) + K_y y(t) \end{cases} \quad (ii)$$

The most widespread scheme is the one that relies on a (e.g., Luenberger) state observer. In this case (ii) is replaced by

$$\begin{cases} \dot{\bar{x}}(t) = & A\bar{x}(t) + Bu(t) + L\big(y(t) - C\bar{x}(t)\big) = (A - LC + BK_x)\bar{x}(t) + Ly(t) \\ u(t) = & K_x\bar{x}(t) \end{cases} \quad (iii)$$

In this case, we define the estimation error $e(t) = x(t) - \bar{x}(t)$. Considering (i) and (iii), we write

$$\begin{cases} \dot{e}(t) = & (A - LC)e(t) \\ \dot{x}(t) = & Ax(t) + BK_x\bar{x}(t) = (A + BK_x)x(t) - BK_x e(t) \end{cases}$$

Collectively, we obtain that

$$\begin{bmatrix} \dot{e}(t) \\ \dot{x}(t) \end{bmatrix} = \begin{bmatrix} A - LC & 0 \\ -BK_x & A + BK_x \end{bmatrix} \begin{bmatrix} e(t) \\ x(t) \end{bmatrix}$$

In view of its block-triangular structure, the eigenvalues of

$$A_{cl} = \begin{bmatrix} A - LC & 0 \\ -BK_x & A + BK_x \end{bmatrix}$$

consists of the eigenvalues of $A - LC$ and those of $A + BK_x$.

As discussed in the following section, if the pair (A, B) is controllable, then we can always design a control gain K_x such that the eigenvalues of $A + BK_x$ are placed at will. On the other hand, if the pair (A, C) is observable, then the eigenvalues of $A - LC$ can be placed at will by properly designing the observer gain L. This allows to place at any arbitrary desired positions the eigenvalues of A_{cl} (*separation principle*).

4) Controllability, observability, and centralized fixed modes

We introduce on the following definitions.

- A linear system is *controllable* if, for every initial condition $x(0)$, $\exists T > 0$ and a control signal $u(t)$, $t \in [0, T)$, such that $x(T) = 0$.
- A linear system is *observable* if, for $u(t) = 0$, every initial state $x(0)$ can be reconstructed from the measurement $y(t)$, $t \in [0, T)$, for some $T > 0$.

The following theorem establishes formally the properties introduced in the previous section.

Theorem 1

The system (i)

- is *controllable* if and only if one of the following (equivalent) conditions are satisfied.
  1. $\mathrm{rank}(S_C) = n$, where
  $$S_C = [B \quad \cdots \quad A^{n-1}B]$$
  is denoted *reachability* (*controllability*) *matrix*.
  2. $\mathrm{rank}([\lambda_i I - A \quad B]) = n$ for all the eigenvalues λ_i of matrix A, $i = 1, \dots, n$.
- is *observable* if and only if one of the following (equivalent) conditions are satisfied.
  3. $\mathrm{rank}(S_O) = n$, where
  $$S_O = \begin{bmatrix} C \\ \vdots \\ CA^{n-1} \end{bmatrix}$$
  is denoted *observability matrix*.
  4. $\mathrm{rank}\left(\begin{bmatrix} \lambda_i I - A \\ C \end{bmatrix}\right) = n$ for all the eigenvalues λ_i of matrix A, $i = 1, \dots, n$.

Remark

Consider condition 2. If, for a given eigenvalue λ_i, $\mathrm{rank}([\lambda_i I - A \quad B]) < n$ then the mode associated with the eigenvalue λ_i, cannot be controlled. To see this, assume that (for simplicity of presentation but without lack of generality) A is diagonalizable, i.e., $\exists$ matrix T such that $TAT^{-1} = A_D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$. Defining $\tilde{x} = Tx$ we can write that

$$\dot{\tilde{x}} = T\dot{x} = TAT^{-1}\tilde{x} + TBu = \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} \tilde{x} + \tilde{B}u$$

Where $\tilde{B} = TB$. The dynamics of the i-th element $\tilde{x}_i$ of the state vector $\tilde{x}$ is

$$\dot{\tilde{x}}_i = \lambda_i \tilde{x}_i + \tilde{B}_i u$$

If $\tilde{B}_i = 0$ it is clear that the state $\tilde{x}_i$ cannot be controlled. It can be proved that this happens when $\mathrm{rank}([\lambda_i I - A \quad B]) < n$.

A dual remark applies to condition 4. Indeed, it can be proved that, if

$$\mathrm{rank}\left(\begin{bmatrix} \lambda_i I - A \\ C \end{bmatrix}\right) < n$$

the state $\tilde{x}_i$ cannot be observed.

It can be proved that the *modes* of matrix A (and, with some abuse of notation, its *eigenvalues* and the corresponding *system states*) that cannot be controlled and/or observed correspond with the *centralized fixed modes* (CFM). The CFMs are defined as the eigenvalues of matrix A that do not change position under a *static output feedback control* law $u(t) = K_y y(t)$. We define

$$\mathrm{spec}(A + BK_y C)$$

the spectrum of $A + BK_y C$, i.e., the set of eigenvalues of $A + BK_y C$. The formal definition of CFM is as follows.

Definition 1 (centralized fixed modes)

The elements of the set

$$\Lambda_{CFM} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \mathrm{spec}(A + BK_y C)$$

are the CFMs.

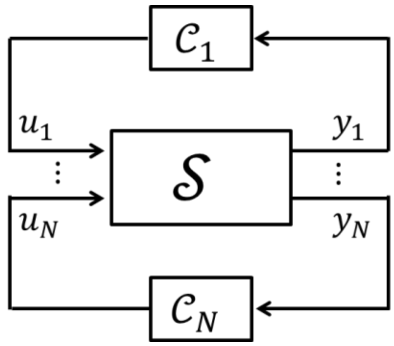
Remark

By definition, the modes that are not CFMs are the modes that change position under *output feedback static control*. It is important to note the following facts.

- The fact that their position "moves" does not mean that, by *output feedback static control*, the system modes can be placed at will. To place them at any desired positions we need either a *state feedback static control law* (under the assumption that the whole state is measured) or a *dynamic output feedback control law*.
- The absence of CFMs (despite their definition is based on the *output feedback static control* law) implies that all the eigenvalues (i.e., all the modes) are both controllable and observable. This implies that *a dynamic output feedback control law* can always be designed, which allows to place all the eigenvalues of A_{cl} at the prescribed positions.
- The absence of CFMs with positive or null real part implies that the system can be made, by a suitable controller, asymptotically stable: otherwise it has *modes which are not asymptotically stable and*, at the same time*, are not stabilizable*.
- When there are CFMs, the closed-loop system eigenvalues cannot be moved in any arbitrary positions with any control law, be it static, dynamic, time-varying, nonlinear. For this reason, we say that CFMs are "*structural*".
- If a continuous-time system has a CFM corresponding to eigenvalue λ_i, the corresponding discrete-time system (obtained, e.g., under the assumption that the actuators are of sample-and-hold type, with sampling time h)
$$\begin{cases} x_{k+1} = & Fx_k + Gu_k \\ y_k = & Hx_k \end{cases}$$
has a corresponding fixed mode (e.g., having the same controllability/observability properties as λ_i) corresponding to eigenvalue $e^{\lambda_i h}$. Note that, similarly to the continuous-time case, the fixed modes for discrete-time models are defined as the elements of the set
$$\Lambda_{CFM}^{\text{discrete-time}} = \bigcap_{K_y \in \mathbb{R}^{m \times p}} \mathrm{spec}(F + GK_y H)$$

B. DECENTRALIZED CONTROL

We consider now a decentralized control system, where a controller $\mathcal{C}_i$ is committed to control output y_i by acting on input u_i, for each $i = 1, \dots, N$, as in figure.



Of course, the design of controllers $\mathcal{C}_1$, …, $\mathcal{C}_N$ must not disregard global properties, such as the *stability of the closed-loop system*.

1) Static output-feedback control

Recalling that each channel i corresponds with the input/output pair (u_i, y_i), a decentralized static output feedback control law is of the type, for all $i = 1, \dots, N$

$$u_i(t) = K_y^i y_i(t)$$

where $K_y^i \in \mathbb{R}^{m_i \times p_i}$. In this way, the whole vector $u(t)$ is defined as follows

$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_N(t) \end{bmatrix} = \begin{bmatrix} K_y^1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & K_y^N \end{bmatrix} \begin{bmatrix} y_1(t) \\ \vdots \\ y_N(t) \end{bmatrix} = K_y^{\mathrm{d}} y(t)$$

where the main peculiarity of the control law above lies in the fact that the overall gain matrix K_y^{d} is block-diagonal. In this way the closed-loop system is described by the dynamical model

$$\dot{x}(t) = Ax(t) + BK_y^{\mathrm{d}} y(t) = (A + BK_y^{\mathrm{d}} C)x(t)$$

Importantly, it is worth noting that, similarly to the centralized case, there is never a guarantee that a gain K_y^{d} exists, that stabilizes the system if A is not asymptotically (Hurwitz) stable.

2) Dynamic output-feedback control

In the dynamic case, each local controller, dedicated to the channel (u_i, y_i) has the following structure

$$\mathcal{C}_i : \begin{cases} \dot{\bar{x}}_i(t) = & A_c^i \bar{x}_i(t) + B_c^i y_i(t) \\ u_i(t) = & K_x^i \bar{x}_i(t) + K_y^i y_i(t) \end{cases}$$

Note that the state of each controller $\mathcal{C}_i$ (i.e., $\bar{x}_i$) may not be strictly related to the state x of the system. For example, it may be merely the state of an ad-hoc PID-type controller.

The overall controller displays a *decentralized – block diagonal - structure* (i.e., it is *structured*). In fact, if we define

$$\bar{x}(t) = \begin{bmatrix} \bar{x}_1(t) \\ \vdots \\ \bar{x}_N(t) \end{bmatrix}$$

we obtain that, collectively

$$\mathcal{C}: \begin{cases} \dot{\bar{x}}(t) = & A_c^{\mathrm{d}} \bar{x}(t) + B_c^{\mathrm{d}} y(t) \\ u(t) = & K_x^{\mathrm{d}} \bar{x}(t) + K_y^{\mathrm{d}} y(t) \end{cases}$$

where the main difference with respect to (ii) lies in the fact that $A_c^{\mathrm{d}} = \mathrm{diag}(A_c^1, \dots, A_c^N), B_c^{\mathrm{d}} = \mathrm{diag}(B_c^1, \dots, B_c^N), K_x^{\mathrm{d}} = \mathrm{diag}(K_x^1, \dots, K_x^N)$, $K_y^{\mathrm{d}} = \mathrm{diag}(K_y^1, \dots, K_y^N)$, i.e., all the involved matrices are constrained to be block-diagonal.

3) *Static state-feedback control*

In decentralized control, the assumption that the overall state x of the system $\mathcal{S}$ under control is available to (i.e., measured by) each single local controller $\mathcal{C}_i$ is commonly disregarded.

Normally, the idea of state-feedback decentralized control implies the existence of a partition of the state vector x into subvectors x_i, $i = 1, \dots, N$.

- These subvectors are non-overlapping in decentralized control, i.e., they do not have states in common.
- Up to a suitable permutation of the elements of x, we can write
$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$
where $x_i \in \mathbb{R}^{n_i}$ and $\sum_{i=1}^{N} n_i = n$
- Especially where the system model is developed based on physical considerations, the decomposition of vector x into subvectors x_i is simple. Consider the examples introduced in the previous Chapter II.1:
    - In examples 1,2,3 (carts on a railway), $x_i = [p_i, v_i]$ for all $i = 1,2,3$.
    - In example 4 (water tanks), $x_i = h_i, i = 1,2$.

State-feedback decentralized control structures are, as a matter of fact, special cases of output-feedback decentralized control structures. In fact it is set, for each controller,

$$u_i(t) = K_x^i x_i(t)$$

Since x_i is a subvector of x, we can write

$$x_i = C_i x = \underbrace{[0 \quad \cdots \quad 0 \quad I \quad 0 \quad \cdots \quad 0]}_{\text{only the } i-th \text{ block is} \neq 0} x$$

Collectively, we write

$$C = \begin{bmatrix} C_1 \\ \vdots \\ C_N \end{bmatrix} = I, K_x^d = \mathrm{diag}(K_x^1, \dots, K_x^N)$$

and the closed-loop system dynamics is

$$\dot{x}(t) = Ax(t) + BK_x^d Cx(t) = (A + BK_x^d)x(t)$$

4) *Decentralized fixed modes*

The question, addressed in this section, is: given a system $\mathcal{S}$, when is it possible to place the eigenvalues of the closed-loop system at an arbitrary desired position with a decentralized feedback controller, possibly a dynamic output-feedback one? The notion of decentralized fixed modes allows to answer this question.

4.1) Definition of decentralized fixed modes

To define the decentralized fixed modes (DFM), we start considering a decentralized output-feedback static control law of the type $u_i(t) = K_y^i y_i(t)$, for all $i = 1, \dots, N$. We define the set $\mathcal{K}^d$ as the set of all block-diagonal matrices of the type $K_y^{\mathrm{d}} = \mathrm{diag}\left(K_y^1, \dots, K_y^{\mathrm{N}}\right)$, $K_y^{\mathrm{i}} \in \mathbb{R}^{m_i \times p_i}$.

Definition 2 (decentralized fixed modes)

The elements of the set

$$\Lambda_{DFM} = \bigcap_{K_y^d \in \mathcal{K}^d} \mathrm{spec}(A + BK_y^d C)$$

are the decentralized fixed modes (DFM).

Remark

Recalling Definition 1, note that the only difference between CFMs and DFMs is the fact that the gain matrix K_y , for CFMs, is free to take any structure while, for DFMs, it is constrained to be block diagonal (i.e., it lies in the set $\mathcal{K}^d$). Therefore, if a mode is a CFM, then there does not exist a matrix $K_y \in \mathbb{R}^{m \times p}$ that "moves" its position, including block-diagonal gain matrices $K_y \in \mathcal{K}^d$. Therefore, a CFM must also be a DFM. This proves that

$$\Lambda_{CFM} \subseteq \Lambda_{DFM}$$

4.2) Decentralized fixed modes and eigenvalue assignment using a decentralized dynamic, output feedback controller (linear and time invariant case)

As in the centralized case, a decentralized static output feedback control law of the type

$$u_i(t) = K_y^i y_i(t)$$

for all $i = 1, \dots, N$ is able to "move" the (non-fixed) eigenvalues of the system $\mathcal{S}$, but does not allow to place them in arbitrary prescribed positions. However, it can be proved that an output-feedback dynamic controller can do it. This is stated in the following theorem.

Theorem 2

The following statements hold.

- There exists a decentralized (linear, time-invariant, and finite-dimensional) output feedback dynamic controller such that the closed loop system is asymptotically stable if and only if all the DFM have strictly negative real part.
- There exists a decentralized (linear, time-invariant, and finite-dimensional) output feedback dynamic controller such that the closed loop system has an arbitrary prescribed set of eigenvalues if and only if the system has no DFMs.

Remark

As discussed above, it is worth remarking that the decentralized "state"-feedback control law consisting of setting

$$u_i(t) = K_x^i x_i(t)$$

for all $i = 1, \dots, N$, is not really a state-feedback control law, but rather an output-feedback one, where

$$y_i = x_i = C_i x = \underbrace{\begin{bmatrix} 0 & \cdots & 0 & I & 0 & \cdots & 0 \end{bmatrix}}_{\text{only the } i-th\ block\ is \neq 0} x$$

for all $i = 1, \dots, N$. Therefore, according to Theorem 2, it is not sufficient for placing the eigenvalues of the closed loop system at will. Rather, a dynamic controller must be used also in this case.

4.3) Necessary conditions for decentralized fixed modes

Remark the following simple fact: if a mode is both observable and controllable by a single channel (u_i, y_i), it cannot be a DFM. In fact, by closing just the i-th control loop and setting $K_y^j = 0$ for all $j \neq i$, its position changes.

This remark can be rephrased as follows: mode λ can be fixed only if it is either uncontrollable or unobservable (or both) from each channel. Therefore, mathematically, a necessary condition for λ (eigenvalue of A) to be a DFM is that, for all $i = 1, \dots, N$

- $\operatorname{rank}(\begin{bmatrix} \lambda I - A & B_i \end{bmatrix}) < n$ (uncontrollable from input u_i), or/and
- $\operatorname{rank}\left(\begin{bmatrix} \lambda I - A \\ C_i \end{bmatrix}\right) < n$ (unobservable from output y_i).

The following lemma is a reformulation of this condition.

Lemma 1

A necessary condition for λ (eigenvalue of A) to be a DFM is that there exists a disjoint partition of the index set $\mathcal{I} = \{1, \dots, N\}$, i.e., $\mathcal{D} = \{i_1, \dots, i_k\}$, $\mathcal{H} = \{i_{k+1}, \dots, i_N\}$ (where $\mathcal{D} \cap \mathcal{H} = \emptyset$ and $\mathcal{D} \cup \mathcal{H} = \mathcal{I}$) such that, at the same time:

- $\mathrm{rank}([\lambda I - A \quad B_i]) < n$ for all $i \in \mathcal{D}$ and
- $\mathrm{rank}\left(\begin{bmatrix} \lambda I - A \\ C_i \end{bmatrix}\right) < n$ for all $i \in \mathcal{H}$.

A slightly stronger necessary condition for λ (eigenvalue of A) to be a DFM is provided by the following lemma, under a similar line of reasoning. To state it, we need to define the following matrices.

- For $\mathcal{D} = \{i_1, \dots, i_k\}$,

$$B_{\mathcal{D}} = [B_{i_1} \quad \cdots \quad B_{i_k}],\; C_{\mathcal{D}} = \begin{bmatrix} C_{i_1} \\ \vdots \\ C_{i_k} \end{bmatrix}$$

- For $\mathcal{H} = \{i_{k+1}, \dots, i_N\}$,

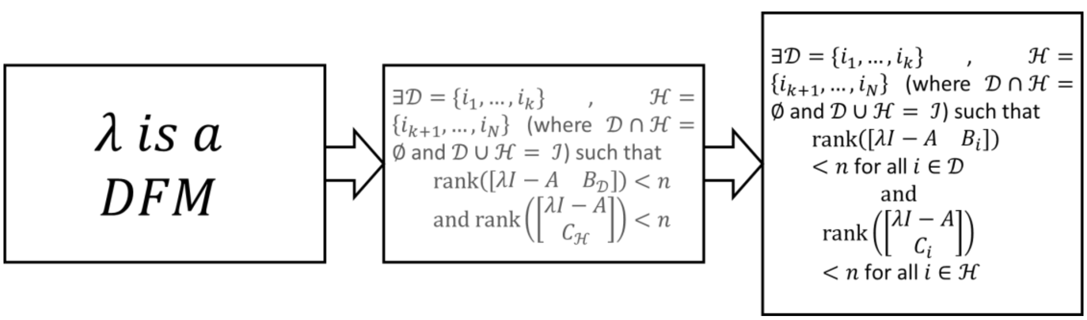
$$B_{\mathcal{H}} = [B_{i_{k+1}} \quad \cdots \quad B_{i_N}],\; C_{\mathcal{H}} = \begin{bmatrix} C_{i_{k+1}} \\ \vdots \\ C_{i_N} \end{bmatrix}$$

Lemma 2

A necessary condition for λ (eigenvalue of A) to be a DFM is that there exists a disjoint partition of the index set $\mathcal{I} = \{1, \dots, N\}$, i.e., $\mathcal{D} = \{i_1, \dots, i_k\}$, $\mathcal{H} = \{i_{k+1}, \dots, i_N\}$ (where $\mathcal{D} \cap \mathcal{H} = \emptyset$ and $\mathcal{D} \cup \mathcal{H} = \mathcal{I}$) such that, at the same time

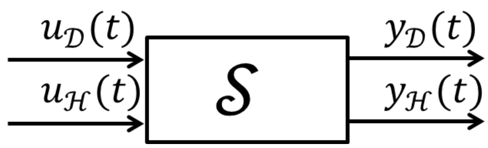
$$\mathrm{rank}([\lambda I - A \quad B_{\mathcal{D}}]) < n \text{ and } \mathrm{rank}\left(\begin{bmatrix} \lambda I - A \\ C_{\mathcal{H}} \end{bmatrix}\right) < n$$

To clarify the conservativeness of the stated conditions, see the following figure.



4.4) A Necessary and sufficient condition for decentralized fixed modes

The necessary condition stated in Section 4.3 implies that we can regard system $\mathcal{S}$ as in the figure below, having two collective inputs ($u_{\mathcal{D}}$ and $u_{\mathcal{H}}$) and two collective outputs ($y_{\mathcal{D}}$ and $y_{\mathcal{H}}$)

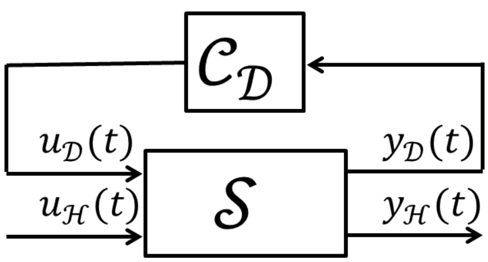


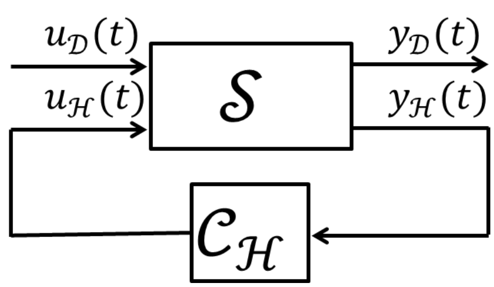
defined as

$$u_{\mathcal{D}} = \begin{bmatrix} u_{i_1} \\ \vdots \\ u_{i_k} \end{bmatrix}, y_{\mathcal{D}} = \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_k} \end{bmatrix}, u_{\mathcal{H}} = \begin{bmatrix} u_{i_{k+1}} \\ \vdots \\ u_{i_N} \end{bmatrix}, y_{\mathcal{H}} = \begin{bmatrix} y_{i_{k+1}} \\ \vdots \\ y_{i_N} \end{bmatrix}$$

The condition of Lemma 2 states in practice that

- λ cannot be "placed" by merely closing one loop on "channel" $\mathcal{D}$ because it is uncontrollable from $u_{\mathcal{D}}$, i.e., $\text{rank}([\lambda I - A \quad B_{\mathcal{D}}]) < n$;
- λ cannot be "placed" by merely closing one loop on "channel" $\mathcal{H}$ because it is unobservable from $y_{\mathcal{H}}$, i.e., $\text{rank}\left(\begin{bmatrix} \lambda I - A \\ C_{\mathcal{H}} \end{bmatrix}\right) < n$.





This condition, however, is not sufficient for λ to be a DFM. In fact

- there are cases in which, by closing a loop on channel $\mathcal{D}$ (left figure above), i.e., by setting
$$u_{\mathcal{D}}(t) = \text{diag}\left(K_y^{i_1}, \dots, K_y^{i_k}\right) y_{\mathcal{D}}(t)$$
the "mode" λ becomes observable by $y_{\mathcal{H}}$;
- there are cases in which, by closing a loop on channel $\mathcal{H}$ (right figure above), i.e., by setting
$$u_{\mathcal{H}}(t) = \text{diag}\left(K_y^{i_{k+1}}, \dots, K_y^{i_N}\right) y_{\mathcal{H}}(t)$$
the mode λ becomes controllable by $u_{\mathcal{D}}$.

Example

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1 + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_2$$

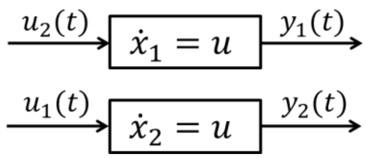
$$y_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Therefore

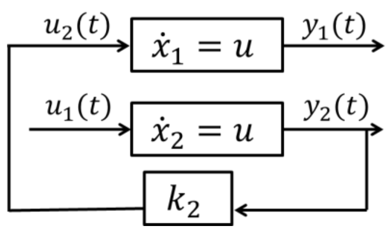
$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, B_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, B_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, C_1 = \begin{bmatrix} 1 & 0 \end{bmatrix}, C_2 = \begin{bmatrix} 0 & 1 \end{bmatrix}, C = I$$

The system has no centralized fixed modes because the pair (A, B) is controllable and the pair (A, C) is observable. Let us analyze this system from a graph perspective, as in figure.



Clearly, x_1 cannot be placed, neither from the pair (u_1, y_1) (since it is uncontrollable from u_1), nor from the pair (u_2, y_2) (since it is unobservable from y_2). Similarly, x_2 cannot be placed, neither from the pair (u_1, y_1) (since it is unobservable from y_1), nor from the pair (u_2, y_2) (since it is uncontrollable from u_2). These facts are necessary for x_1 and x_2 to be DFMs, but not sufficient. In fact

- If we close the loop on channel 2, x_1 becomes both controllable from u_1 and observable from y_1, and so it can be placed through the pair (u_1, y_1), as it is possible to see in the figure below.



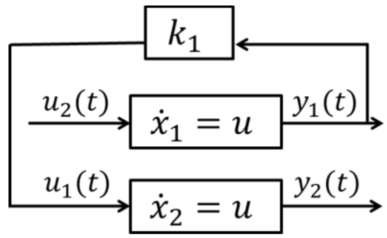
  This can be proved analytically by writing the dynamics of the system when loop 2 is closed, i.e.,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & k_2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_1$$

$$y_1 = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

  The pair $\left(\begin{bmatrix} 0 & k_2 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$ is controllable and the pair $\left(\begin{bmatrix} 0 & k_2 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 \end{bmatrix}\right)$ is observable $\forall k_2 \neq 0$.

- If we close the loop on channel 1, x_2 becomes both controllable from u_2 and observable from y_2, and so it can be placed through the pair (u_2, y_2), as it is possible to see in the figure below.



This can be proved analytically by writing the dynamics of the system when loop 2 is closed, i.e.,

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ k_1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_2$$

$$y_2 = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

The pair $\left(\begin{bmatrix} 0 & 0 \\ k_1 & 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ is controllable and the pair $\left(\begin{bmatrix} 0 & 0 \\ k_1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \end{bmatrix}\right)$ is observable $\forall k_1 \neq 0$.

We conclude that the system has no DFMs.

Summing up, so far we have shown that a mode λ is decentralized fixed only if (**necessary condition**) there exists a disjoint partition of the index set $\mathcal{I} = \{1, \dots, N\}$, i.e., $\mathcal{D} = \{i_1, \dots, i_k\}$, $\mathcal{H} = \{i_{k+1}, \dots, i_N\}$ (where $\mathcal{D} \cap \mathcal{H} = \emptyset$ and $\mathcal{D} \cup \mathcal{H} = \mathcal{I}$) such that, at the same time

- λ is uncontrollable from channel $\mathcal{D}$ in case channel $\mathcal{H}$ is both **open** and **closed** (with a decentralized controller);
- λ is unobservable from channel $\mathcal{H}$ in case channel $\mathcal{D}$ is both **open** and **closed** (with a decentralized controller).

It can be proved that this is both necessary and sufficient for λ to be a DFM, and that this condition can also be formulated in a mathematically simple way, as stated by the following Theorem 1.

Theorem 1

A necessary and sufficient condition for λ (eigenvalue of A) to be a DFM is that there exists a disjoint partition of the index set $\mathcal{I} = \{1, \dots, N\}$, i.e., $\mathcal{D} = \{i_1, \dots, i_k\}$, $\mathcal{H} = \{i_{k+1}, \dots, i_N\}$ (where $\mathcal{D} \cap \mathcal{H} = \emptyset$ and $\mathcal{D} \cup \mathcal{H} = \mathcal{I}$) such that

$$\operatorname{rank}\left(\begin{bmatrix} \lambda I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) < n$$

4.4) Computation of decentralized fixed modes

The condition given by Theorem 1, despite it definitely clarifies the physical meaning of DFMs, is rather impractical for the numerical point of view. In fact, for systems with a large number of input/output channels, this makes the computation of DFMs intensive: we should generate, for all eigenvalues λ of A, all possible partitions $\mathcal{D}$ and $\mathcal{H}$, for which we must evaluate the rank of the corresponding matrix

$$\begin{bmatrix} \lambda I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}$$

A more computationally simple procedure can be derived from Definition 2, and is the following.

1. Compute the spectrum of matrix A, i.e., $\mathrm{spec}(A)$.
2. Select the block-diagonal matrix K_y^d randomly.
3. Compute $\mathrm{spec}(A + BK_y^d C)$.
4. The set of DFMs is given by $\mathrm{spec}\left(A + BK_y^d C\right) \cap \mathrm{spec}(A)$ with probability 1.

5) *Structural decentralized fixed modes and discrete-time decentralized controllers*

5.1) Structural decentralized fixed modes

In Section 4, a major result has been established: **even if the system $\mathcal{S}$ is (centrally) both observable and controllable, it may not be possible to stabilize it using a decentralized** (linear time-invariant and finite-dimensional) **control law due to the presence of the *decentralized fixed modes***.

However, it has been more recently discovered that decentralized fixed modes are of two types

- *Structural decentralized fixed modes*;
- *Non-structural decentralized fixed modes*.

This distinction turns out to be extremely important. In fact, it can be shown by simple examples that, if the system does not have CFMs and has only non-structural DFM, a suitable decentralized control law can indeed be devised, that makes the closed loop system asymptotically stable (in some sense), provided that it is either non-linear, time-varying, or non finite-dimensional. On the other hand, it can be proved that, if the system has unstable structural DFMs, it cannot be made asymptotically stable through any decentralized output feedback controller.

Recall that the decentralized static output-feedback control law consists of setting

$$u_i(t) = K_y^i y_i(t)$$

for all $i = 1, \dots, N$. We collectively have that

$$u(t) = K_y^{\mathrm{d}} y(t)$$

where $K_y^{\mathrm{d}} \in \mathcal{K}^d$, i.e., is block-diagonal. The closed-loop system is described by the dynamical model

$$\dot{x}(t) = Ax(t) + BK_y^{\mathrm{d}} y(t) = (A + BK_y^{\mathrm{d}} C)x(t)$$

We define $A_{cl}\left(K_y^d\right) = A + BK_y^d C$. It is possible to prove that if, for a given $K_y^{\mathrm{d}} \in \mathcal{K}^d$, the transfer function

$$C_j \left(sI - A_{cl}(K_y^d)\right)^{-1} B_i \neq 0$$

channel i can transmit any information to channel j in finite time.

To define the structural decentralized fixed modes, we define *a graph* $\mathcal{G}$, which represents all the possible connections that can occur between the system channels by closing the control loops in a decentralized way. We proceed as follows

- The nodes of graph $\mathcal{G}$ are N (i.e., defined by indices $1, \dots, N$), each representing a channel.
- For all j, i with $j \neq i$, an arc $i \to j$ exists if and only if
$$C_j \left(sI - A_{cl}(K_y^d)\right)^{-1} B_i \neq 0$$
for some $K_y^{\mathrm{d}} \in \mathcal{K}^d$.

Now that the *graph* $\mathcal{G}$ is defined, we define the $\tilde{N}$ (with $\tilde{N} \leq N$) strongly connected subgraphs $\mathcal{G}_k$, ($k = 1, \dots, \tilde{N}$) of $\mathcal{G}$ according to the following rule: nodes i and j belong to the same subgraph $\mathcal{G}_k$ if and only if $j \to i$ and $i \to j$. We then denote as $\mathcal{I}_k$ ($k = 1, \dots, \tilde{N}$) the set of nodes belonging to the subgraph $\mathcal{G}_k$.

The main idea is that, if two nodes are connected (i.e., if $j \to i$ and $i \to j$), they must be considered as belonging to a unique channel ($\begin{bmatrix} u_i \\ u_j \end{bmatrix}, \begin{bmatrix} y_i \\ y_j \end{bmatrix}$), not two separated ones, since enough information exchange is possible between them. This, in general, allows to define the so-called *quotient system* as

$$\tilde{\mathcal{S}}: \begin{cases} \dot{x} = & Ax + \tilde{B}_1 \tilde{u}_1 + \cdots + \tilde{B}_{\tilde{N}} \tilde{u}_{\tilde{N}} \\ \tilde{y}_k = & \tilde{C}_k x \qquad\qquad k = 1, \dots, \tilde{N} \end{cases}$$

where

$$\tilde{u}_k = \begin{bmatrix} u_{i_1} \\ \vdots \\ u_{i_{r_k}} \end{bmatrix}, \tilde{y}_k = \begin{bmatrix} y_{i_1} \\ \vdots \\ y_{i_{r_k}} \end{bmatrix}$$

where $\mathcal{I}_k := \{i_1, \dots, i_{r_k}\}$ and, consistently,

$$\tilde{B}_k = \begin{bmatrix} B_{i_1} \cdots B_{i_{r_k}} \end{bmatrix}, \tilde{C}_k = \begin{bmatrix} C_{i_1} \\ \vdots \\ C_{i_{r_k}} \end{bmatrix}$$

Now we are in the position of defining the structural decentralized fixed modes of system $\mathcal{S}$.

Definition 3

The *structural decentralized fixed modes* (SDFMs) of system $\mathcal{S}$ are the DFMs of the quotient system $\tilde{\mathcal{S}}$.

We will denote with Λ_{SDFM} the set of SDFMs. It is easy to prove that the following inclusions hold

$$\Lambda_{CFM} \subseteq \Lambda_{SDFM} \subseteq \Lambda_{DFM}$$

The following theorem can be proved.

Theorem 3

There exists a time interval T and a decentralized (possibly time-varying, non-linear, or infinite-dimensional) control law such that $x(T) = 0$ for any unknown initial condition $x(0)$ if and only if the system has no SDFMs.

5.2) Decentralized fixed modes and digital decentralized controllers

The fact that we can devise a decentralized controller also when DFMs are present (provided that there are no unstable SDFMs) has also a major impact on the possibility of controlling a system using digital controllers. Note, in fact, that a digital control law of the type

$$u_k = K_y^d y_k, \text{i.e.}, u(t) = K_y^d y(kh), \text{for } t \in [kh, (k+1)h)$$

can be regarded (at the "continuous-time system side") as a time-varying and infinite-dimensional (i.e., with delays) control law. As such, the modes that are decentralized fixed under a continuous-time (LTI) control law (but are not structurally fixed) may become non-fixed when a digital control law is used.

First we define the following.

Definition 4 (decentralized fixed modes under digital control)

The elements of the set

$$\Lambda_{DFM}^{\text{discrete-time}} = \bigcap_{K_y \in \mathcal{K}^d} \text{spec}(F + GK_y H)$$

are the *decentralized fixed modes under digital control* (dtDFM).

In the dedicated research literature it has been shown that (at least provided that matrix A is diagonalizable), the set $\Lambda_{DFM}^{\text{discrete-time}}$ has cardinality smaller or equal to the cardinality of Λ_{DFM}. More specifically, the elements of the set $\Lambda_{DFM}^{\text{discrete-time}}$ are equal to $e^{\lambda_i h}$, where $\lambda_i \in \Lambda_{DFM}$. With some abuse of notation, we write $\Lambda_{DFM}^{\text{discrete-time}} \subseteq e^{\Lambda_{DFM}h}$. Of course, by Theorem 3, it must also hold that $e^{\Lambda_{SDFM}h} \subseteq \Lambda_{DFM}^{\text{discrete-time}}$. Overall, we obtain that

$$e^{\Lambda_{SDFM}h} \subseteq \Lambda_{DFM}^{\text{discrete-time}} \subseteq e^{\Lambda_{DFM}h}$$

6) Examples

In this section we analyze in details some examples, drawn from the literature.

EXAMPLE 1

Consider the following system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & -1 & 0\\ 0 & -3 & 0\\ 0 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}1\\ 1\\ 0\end{bmatrix}u_1 + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u_2$$

$$y_1 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

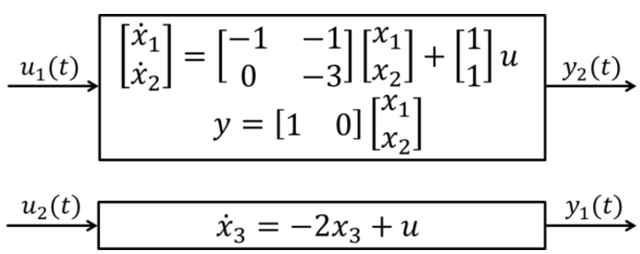
$$y_2 = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Therefore

$$A = \begin{bmatrix}-1 & -1 & 0\\ 0 & -3 & 0\\ 0 & 0 & -2\end{bmatrix}, B_1 = \begin{bmatrix}1\\ 1\\ 0\end{bmatrix}, B_2 = \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}, B = \begin{bmatrix}1 & 0\\ 1 & 0\\ 0 & 1\end{bmatrix}, C_1 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}, C_2 = \begin{bmatrix}1 & 0 & 0\end{bmatrix},$$

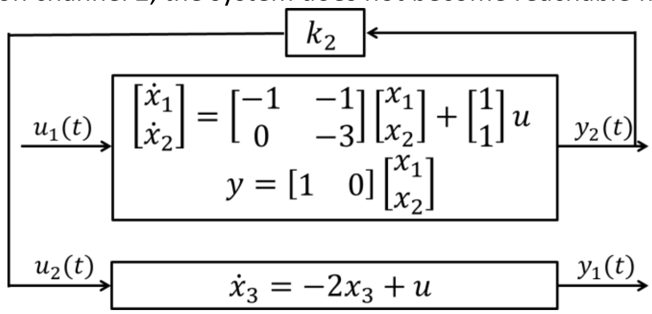
$$C = \begin{bmatrix}0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix}$$

The system has no centralized fixed modes because the pair (A, B) is controllable and the pair (A, C) is observable. Let us analyze this system from a graph perspective, as in figure.



Clearly, x_1 and x_2 cannot be placed, neither from the pair (u_1, y_1) (since they are unobservable from y_1), nor from the pair (u_2, y_2) (since they are uncontrollable from u_2). Similarly, x_3 cannot be placed, neither from the pair (u_1, y_1) (since it is uncontrollable from u_1), nor from the pair (u_2, y_2) (since it is unobservable from y_2). Also, note the following facts.

- If we close the loop on channel 2, the system does not become reachable from input u_1.



This can be proved analytically by writing the dynamics of the system when loop 2 is closed, i.e.,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & -1 & 0\\ 0 & -3 & 0\\ k_2 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}1\\ 1\\ 0\end{bmatrix}u_1$$

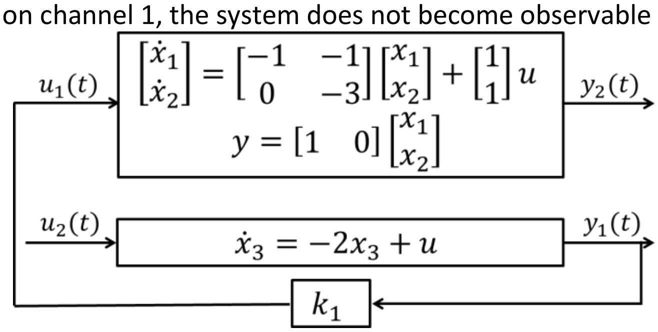
$$y_1 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The reachability matrix of the system above is

$$\mathcal{R} = \begin{bmatrix}1 & -2 & 5\\ 1 & -3 & 9\\ 0 & k_2 & -4k_2\end{bmatrix}$$

which has rank 2 for all k_2.

- If we close the loop on channel 1, the system does not become observable from y_2.



This can be proved analytically by writing the dynamics of the system when loop 1 is closed, i.e.,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & -1 & k_1\\ 0 & -3 & k_1\\ 0 & 0 & -2\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u_2$$

$$y_2 = \begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The observability matrix of the system above is

$$\mathcal{O} = \begin{bmatrix}1 & 0 & 0\\ -1 & -1 & k_1\\ 1 & 4 & -4k_1\end{bmatrix}$$

which has rank 2 for all k_1.

We conclude that the system has DFMs.

To clarify which are the system DFMs, we resort to Theorem 1. We consider the three eigenvalues:

- Consider $\lambda_1 = -1$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\operatorname{rank}\left(\begin{bmatrix} \lambda_1 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}\right) = 3 = n$$

  Therefore, $\lambda_1 = -1$ is not a DFM.
- Consider $\lambda_2 = -3$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

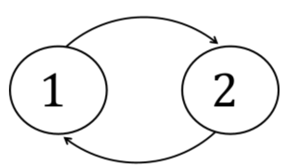
$$\operatorname{rank}\left(\begin{bmatrix} \lambda_2 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix} -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}\right) = 3 = n$$

  Therefore, $\lambda_2 = -3$ is not a DFM.
- Consider $\lambda_3 = -2$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\operatorname{rank}\left(\begin{bmatrix} \lambda_3 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix} -1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}\right) = 2 < n$$

  Therefore, $\lambda_3 = -2$ is a DFM.

It is now of interest to understand if $\lambda_3 = -2$ is also a structural DFM. To do it, we adopt the procedure sketched in Section 5.2. The corresponding graph $\mathcal{G}$ is the following



It is clear from the figure that the channels 1 and 2 are connected, and therefore they belong to the same subgraph. In view of this, the corresponding quotient system is the whole centralized system, which has no fixed modes. Therefore, the system has no structural fixed modes.

In view of the previous result, it is possible that the system has no *decentralized fixed modes under digital control.* We carry out the analysis conducted so far (consistent with Theorem 1), but considering the discrete-time system (obtained with h=0.1 s)

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} 0.9048 & -0.0820 & 0 \\ 0 & 0.7408 & 0 \\ 0 & 0 & 0.8187 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0.0908 \\ 0.0864 \\ 0 \end{bmatrix} u_1(k) + \begin{bmatrix} 0 \\ 0 \\ 0.0906 \end{bmatrix} u_2(k)$$

$$y_1(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

$$y_2(k) = [1 \quad 0 \quad 0] \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

We consider the three eigenvalues:

- Consider $\lambda_1 = 0.9048$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute
$$\mathrm{rank}\left(\begin{bmatrix} \lambda_1 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix}\right) = 3 = n$$
Therefore, λ_1 is not a dtDFM.
- Consider $\lambda_2 = 0.7408$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute
$$\mathrm{rank}\left(\begin{bmatrix} \lambda_2 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix}\right) = 3 = n$$
Therefore, λ_2 is not a dtDFM.
- Consider $\lambda_3 = 0.8187$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute
$$\mathrm{rank}\left(\begin{bmatrix} \lambda_3 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix}\right) = 3 = n$$
Therefore, λ_3 is NOT a dtDFM.

This shows that the absence of structural fixed modes allows to make the system decentrally controllable with a digital (linear time-invariant) control law.

EXAMPLE 2

Consider the following system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

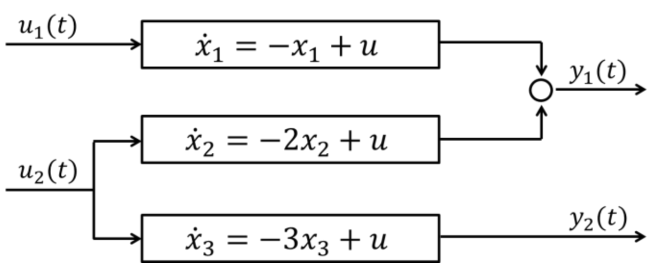
$$y_2 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Therefore

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}, B_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, B_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix}, C_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix}, C_2 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix},$$

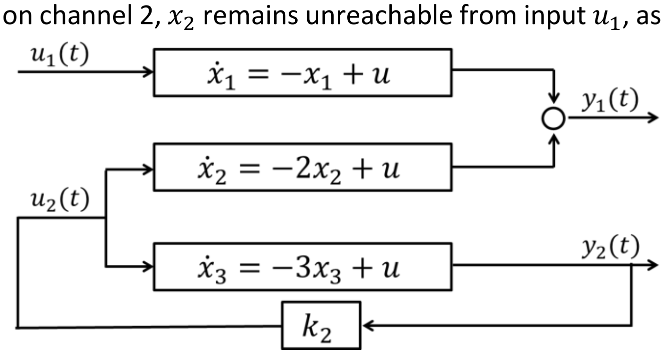
$$C = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The system has no centralized fixed modes because the pair (A, B) is controllable and the pair (A, C) is observable. Let us analyze this system from a graph perspective, as in figure.

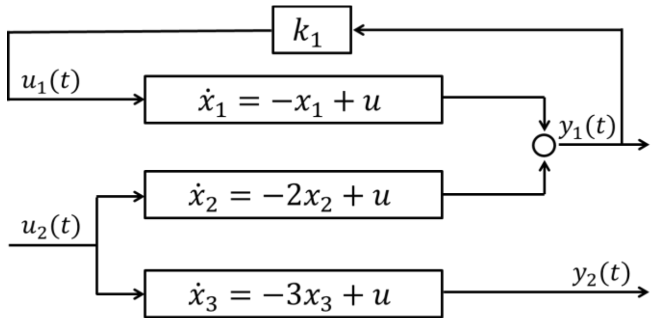


Clearly, x_2 cannot be placed, neither from the pair (u_1, y_1) (since it is uncontrollable from u_1), nor from the pair (u_2, y_2) (since it is unobservable from y_2). Also, consider the following.

- If we close the loop on channel 2, x_2 remains unreachable from input u_1, as clear from the figure.



- If we close the loop on channel 1, x_2 remains unobservable from output y_2, as clear from the figure.



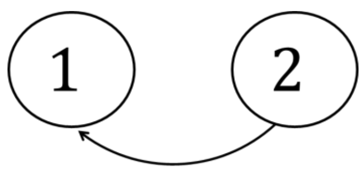
We conclude that x_2 is a DFM.

This can be validated by resorting to Theorem 1. Considering $\lambda_2 = -2$, the only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\operatorname{rank}\left(\begin{bmatrix} \lambda_2 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\right) = 2 < n$$

Therefore, $\lambda_2 = -2$ is a DFM.

It is now of interest to understand if $\lambda_2 = -2$ is also a structural DFM. To do it, we adopt the procedure sketched in Section 5.2. The corresponding graph $\mathcal{G}$ is the following



It is clear from the figure that the channels 1 and 2 do not belong to a unique connected graph. In view of this, the corresponding quotient system corresponds with the original decomposed model. Therefore the structural decentralized fixed modes are the decentralized fixed modes, i.e., $\lambda_2 = -2$. In view of the previous result, the system has also a *decentralized fixed mode under digital control*, i.e., e^{-2h}.

EXAMPLE 3

Consider the following system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u_1 + \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}u_2$$

$$y_1 = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

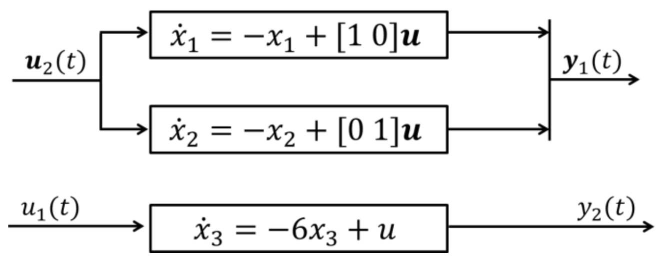
$$y_2 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Therefore

$$A = \begin{bmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -6\end{bmatrix}, B_1 = \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}, B_2 = \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}, B = \begin{bmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 1 & 0 & 0\end{bmatrix},$$

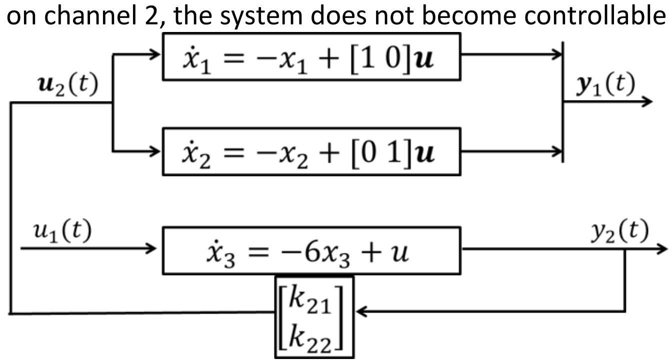
$$C_1 = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}, C_2 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}, C = I$$

The system has no centralized fixed modes because the pair (A, B) is controllable and the pair (A, C) is observable. Let us analyze this system from a graph perspective, as in figure.



Clearly, x_1 and x_2 cannot be placed, neither from the pair (u_1, y_1) (since they are uncontrollable from u_1), nor from the pair (u_2, y_2) (since they are unobservable from y_2). Similarly, x_3 cannot be placed, neither from the pair (u_1, y_1) (since it is unobservable from from y_1), nor from the pair (u_2, y_2) (since it is uncontrollable from u_2). Also, note the following facts.

- If we close the loop on channel 2, the system does not become controllable from input u_1.



This can be proved analytically by writing the dynamics of the system when loop 2 is closed, i.e.,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & 0 & k_{21}\\ 0 & -1 & k_{22}\\ 0 & 0 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u_1$$

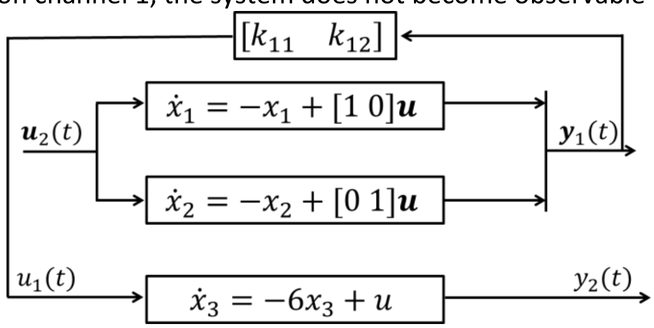
$$y_1 = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The reachability matrix of the system above is

$$\mathcal{R} = \begin{bmatrix}0 & k_{21} & -7k_{21}\\ 0 & k_{22} & -7k_{22}\\ 1 & -6 & -36\end{bmatrix}$$

which has rank 2 for all k_{21}, k_{22}.

- If we close the loop on channel 1, the system does not become observable from y_2.



This can be proved analytically by writing the dynamics of the system when loop 1 is closed, i.e.,

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ k_{11} & k_{12} & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}u_2$$

$$y_2 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

The observability matrix of the system above is

$$\mathcal{O} = \begin{bmatrix}0 & 0 & 1\\ k_{11} & k_{12} & -6\\ -7k_{11} & -7k_{12} & -36\end{bmatrix}$$

which has rank 2 for all k_{11}, k_{12}.

We conclude that the system has DFMs.

To clarify which are the system DFMs, we resort to Theorem 1. We consider the two different eigenvalues:

- Consider $\lambda_1 = -1$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\operatorname{rank}\left(\begin{bmatrix}\lambda_1 I - A & B_{\mathcal{D}}\\ C_{\mathcal{H}} & 0\end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 0 & 0 & 5 & 1\\ 0 & 0 & 1 & 0\end{bmatrix}\right) = 2 < n$$

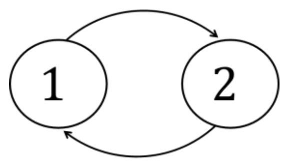
Therefore, $\lambda_1 = -1$ is a DFM.

- Consider $\lambda_2 = -6$. The only candidate partition of the index set is $\mathcal{D} = \{2\}$, $\mathcal{H} = \{1\}$. We compute

$$\mathrm{rank}\left(\begin{bmatrix} \lambda_2 I - A & B_{\mathcal{D}} \\ C_{\mathcal{H}} & 0 \end{bmatrix}\right) = 3 = n$$

Therefore, $\lambda_2 = -3$ is not a DFM.

It is now of interest to understand if $\lambda_1 = -1$ is also a structural DFM. To do it, we adopt the procedure sketched in Section 5.2. The corresponding graph $\mathcal{G}$ is the following



It is clear from the figure that the channels 1 and 2 are connected, and therefore belong to the same subgraph. In view of this, the corresponding quotient system is the centralized one, which has no fixed modes. Therefore, the system has no structural fixed modes.

In view of the previous result, it is possible that the system has no *decentralized fixed modes under digital control.* We carry out the analysis conducted so far (consistent with Theorem 1), but considering the discrete-time system (obtained with h=0.1 s)

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \end{bmatrix} = \begin{bmatrix} -0.8465 & 0 & 0 \\ 0 & -0.8465 & 0 \\ 0 & 0 & -0.3680 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0.1053 \end{bmatrix} u_1(k) + \begin{bmatrix} 0.1535 & 0 \\ 0 & 0.1535 \\ 0 & 0 \end{bmatrix} u_2(k)$$

$$y_1(k) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

$$y_2(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \end{bmatrix}$$

We consider just the critical eigenvalue, i.e., $\lambda_1 = 0.8465$. The only candidate partition of the index set is $\mathcal{D} = \{1\}$, $\mathcal{H} = \{2\}$. We compute

$$\mathrm{rank}\left(\begin{bmatrix} \lambda_1 I - F & G_{\mathcal{D}} \\ H_{\mathcal{H}} & 0 \end{bmatrix}\right) = \mathrm{rank}\left(\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0.4785 & 0.1053 \\ 0 & 0 & 1 & 0 \end{bmatrix}\right) = 2 < n$$

Therefore, λ_1 is a DFM under digital (time-invariant) control. It is finally worth pointing out that a time-varying control law could however make the system controllable, in view of the fact that the system has no SDFMs.

C. DISTRIBUTED CONTROL

A distributed controller is characterized by the fact that each input u_i is a function, not only of the corresponding output y_i, but also of pieces of information from other channels (typically y_j, with $j \neq i$), compatibly with the *information structure constraints* (i.e., provided that communication of y_j to the controller $\mathcal{C}_i$ is practically possible). All the channels from which controller $\mathcal{C}_i$ can receive information are collected in the set $\mathcal{N}_i := \{j \in \mathcal{I}: y_j \text{ is available to controller } \mathcal{C}_i\}$. Note that it is assumed that $i \in \mathcal{N}_i$ for all $i = 1, \dots, N$.

For later use, we define $\boldsymbol{\mathcal{N}} = \{\mathcal{N}_1, \dots, \mathcal{N}_N\}$, which completely defines the information structure constraints of the whole system.

1) Static output-feedback control

A distributed static output feedback control law is of the type, for all $i = 1, \dots, N$

$$u_i(t) = \sum_{j \in \mathcal{N}_i} K_y^{ij} y_j(t)$$

where $K_y^{ij} \in \mathbb{R}^{m_i \times p_j}$. In matrix form

$$u_i(t) = \begin{bmatrix} K_y^{i1} & \cdots & K_y^{iN} \end{bmatrix} y(t)$$

where $K_y^{ij} = 0$ if $j \notin \mathcal{N}_i$. Collectively, the whole vector $u(t)$ is defined as follows

$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_N(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & \cdots & K_y^{1N} \\ \vdots & \ddots & \vdots \\ K_y^{N1} & \cdots & K_y^{NN} \end{bmatrix} \begin{bmatrix} y_1(t) \\ \vdots \\ y_N(t) \end{bmatrix} = K_y \; y(t)$$

where the main peculiarity of the control law above lies in the fact that the overall gain matrix K_y is sparse, and its structure (i.e., the blocks that are non-zero) depends upon the available communication links. The closed-loop system is described by the dynamical model

$$\dot{x}(t) = Ax(t) + BK_y \; y(t) = (A + BK_y \; C)x(t)$$

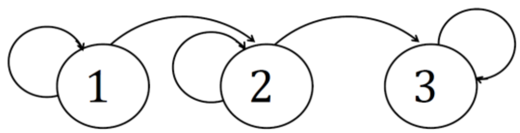
Similarly to the centralized and decentralized cases, there is never a guarantee that a gain K_y exists, that stabilizes the system if A is not asymptotically (Hurwitz) stable.

Examples

It is worth, at this point, showing how common information exchange patterns affect the sparsity structure of the gain matrix K_y .

1) String with unidirectional communication

Consider the “linear” transmission graph sketched in the following figure.



Specifically, an arrow exists between two nodes $j \to i$ if $j \in \mathcal{N}_i$, i.e., output y_j can be transmitted to controller $\mathcal{C}_i$. In this case

$$u_1(t) = K_y^{11} y_1(t)$$
$$u_2(t) = K_y^{21} y_1(t) + K_y^{22} y_2(t)$$
$$u_3(t) = K_y^{32} y_2(t) + K_y^{33} y_3(t)$$

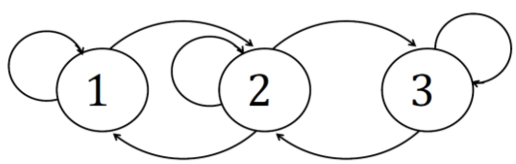
Collectively, the whole vector $u(t)$ is defined as

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & 0 & 0 \\ K_y^{21} & K_y^{22} & 0 \\ 0 & K_y^{32} & K_y^{33} \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

resulting in a triangular banded gain matrix.

2) String with bidirectional communication

Consider the “linear” transmission graph sketched in the following figure.



We can write that

$$u_1(t) = K_y^{11} y_1(t) + K_y^{12} y_2(t)$$
$$u_2(t) = K_y^{21} y_1(t) + K_y^{22} y_2(t) + K_y^{23} y_3(t)$$
$$u_3(t) = K_y^{32} y_2(t) + K_y^{33} y_3(t)$$

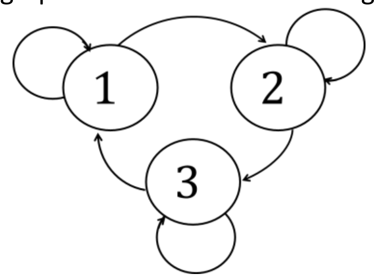
Collectively, the whole vector $u(t)$ is defined as

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & K_y^{12} & 0 \\ K_y^{21} & K_y^{22} & K_y^{23} \\ 0 & K_y^{32} & K_y^{33} \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

resulting in a banded (but not triangular) gain matrix.

3) Cycle

Consider the cyclic transmission graph sketched in the following figure.



We can write that

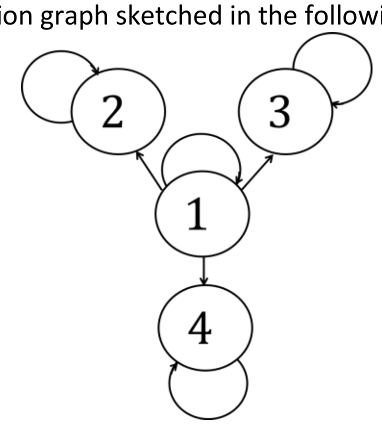
$$u_1(t) = K_y^{11} y_1(t) + K_y^{13} y_3(t)$$
$$u_2(t) = K_y^{21} y_1(t) + K_y^{22} y_2(t)$$
$$u_3(t) = K_y^{32} y_2(t) + K_y^{33} y_3(t)$$

Collectively, the whole vector $u(t)$ is defined as

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & 0 & K_y^{13} \\ K_y^{21} & K_y^{22} & 0 \\ 0 & K_y^{32} & K_y^{33} \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

4) Star with unidirectional communication (1)

Consider the star-like transmission graph sketched in the following figure.



In this case

$$u_1(t) = K_y^{11} y_1(t)$$
$$u_2(t) = K_y^{21} y_1(t) + K_y^{22} y_2(t)$$
$$u_3(t) = K_y^{31} y_1(t) + K_y^{33} y_3(t)$$
$$u_4(t) = K_y^{41} y_1(t) + K_y^{44} y_4(t)$$

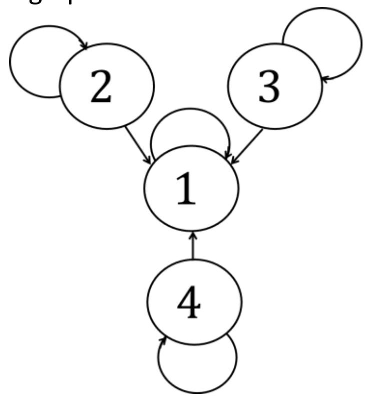
Collectively, the whole vector $u(t)$ is defined as

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \\ u_4(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & 0 & 0 & 0 \\ K_y^{21} & K_y^{22} & 0 & 0 \\ K_y^{31} & 0 & K_y^{33} & 0 \\ K_y^{41} & 0 & 0 & K_y^{44} \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \\ y_4(t) \end{bmatrix}$$

The structure of matrix K_y is denoted bordered block diagonal (BBD).

5) Star with unidirectional communication (2)

Consider the star-like transmission graph sketched in the following figure.



In this case

$$u_1(t) = K_y^{11}y_1(t) + K_y^{12}y_2(t) + K_y^{13}y_3(t) + K_y^{14}y_4(t)$$
$$u_2(t) = K_y^{22}y_2(t)$$
$$u_3(t) = K_y^{33}y_3(t)$$
$$u_4(t) = K_y^{44}y_4(t)$$

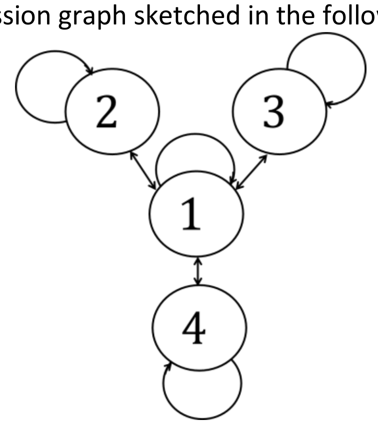
Collectively, the whole vector $u(t)$ is defined as

$$u(t) = \begin{bmatrix} u_1(t) \\ u_2(t) \\ u_3(t) \\ u_4(t) \end{bmatrix} = \begin{bmatrix} K_y^{11} & K_y^{12} & K_y^{13} & K_y^{14} \\ 0 & K_y^{22} & 0 & 0 \\ 0 & 0 & K_y^{33} & 0 \\ 0 & 0 & 0 & K_y^{44} \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \\ y_4(t) \end{bmatrix}$$

Also in this case, the structure of matrix K_y is denoted bordered block diagonal (BBD).

6) Star with bidirectional communication

Consider the star-like transmission graph sketched in the following figure.



In this case

$$u_1(t) = K_y^{11}y_1(t) + K_y^{12}y_2(t) + K_y^{13}y_3(t) + K_y^{14}y_4(t)$$

$$u_2(t) = K_y^{21}y_1(t) + K_y^{22}y_2(t)$$

$$u_3(t) = K_y^{31}y_1(t) + K_y^{33}y_3(t)$$

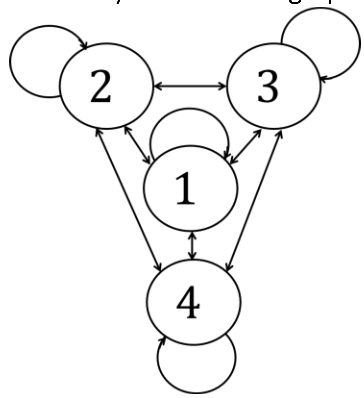
$$u_4(t) = K_y^{41}y_1(t) + K_y^{44}y_4(t)$$

Collectively, the whole vector $u(t)$ is defined as

$$u(t)=\begin{bmatrix}u_1(t)\\u_2(t)\\u_3(t)\\u_4(t)\end{bmatrix}=\begin{bmatrix}K_y^{11} & K_y^{12} & K_y^{13} & K_y^{14}\\K_y^{21} & K_y^{22} & 0 & 0\\K_y^{31} & 0 & K_y^{33} & 0\\K_y^{41} & 0 & 0 & K_y^{44}\end{bmatrix}\begin{bmatrix}y_1(t)\\y_2(t)\\y_3(t)\\y_4(t)\end{bmatrix}$$

Again, the structure of matrix K_y is (doubly) bordered block diagonal (BBD).

7) All to all transmission graph (no information structure constraints)
Consider the all-to-all (i.e., fully connected) transmission graph sketched in the following figure.



In this case

$$\begin{aligned}u_1(t)&=K_y^{11}y_1(t)+K_y^{12}y_2(t)+K_y^{13}y_3(t)+K_y^{14}y_4(t)\\u_2(t)&=K_y^{21}y_1(t)+K_y^{22}y_2(t)+K_y^{23}y_3(t)+K_y^{24}y_4(t)\\u_3(t)&=K_y^{31}y_1(t)+K_y^{32}y_2(t)+K_y^{33}y_3(t)+K_y^{34}y_4(t)\\u_4(t)&=K_y^{41}y_1(t)+K_y^{42}y_2(t)+K_y^{43}y_3(t)+K_y^{44}y_4(t)\end{aligned}$$

Collectively, the whole vector $u(t)$ is defined as

$$u(t)=\begin{bmatrix}u_1(t)\\u_2(t)\\u_3(t)\\u_4(t)\end{bmatrix}=\begin{bmatrix}K_y^{11} & K_y^{12} & K_y^{13} & K_y^{14}\\K_y^{21} & K_y^{22} & K_y^{23} & K_y^{24}\\K_y^{31} & K_y^{32} & K_y^{33} & K_y^{34}\\K_y^{41} & K_y^{42} & K_y^{43} & K_y^{44}\end{bmatrix}\begin{bmatrix}y_1(t)\\y_2(t)\\y_3(t)\\y_4(t)\end{bmatrix}$$

which is full, i.e., equivalent to the gain matrix in the centralized case.
Note, however, that there are fundamental differences with respect to the centralized control case:

- *N* controllers are present here, while in centralized control the control we have 1 control station;
- the implementation is distributed, i.e., each local regulator is in charge of part of the control law computation;
- we need an all-to-all transmission graph, while in centralized control a (birectional) star-like transmission graph is sufficient.

2) *Dynamic output-feedback control*

In the dynamic case, each local controller, dedicated to the channel (u_i, y_i) has the following structure

$$\mathcal{C}_i\colon \begin{cases} \dot{\bar{x}}_i(t) = & A_c^i \bar{x}_i(t) + \sum_{j\in\mathcal{N}_i} B_c^{ij} y_j(t) \\ u_i(t) = & K_x^i \bar{x}_i(t) + \sum_{j\in\mathcal{N}_i} K_y^{ij} y_j(t) \end{cases}$$

If we define

$$\bar{x}(t) = \begin{bmatrix} \bar{x}_1(t) \\ \vdots \\ \bar{x}_N(t) \end{bmatrix}$$

we obtain that, collectively

$$\mathcal{C}_i\colon \begin{cases} \dot{\bar{x}}(t) = & A_c^{\mathrm{d}} \bar{x}(t) + B_c\ y(t) \\ u(t) = & K_x^{\mathrm{d}} \bar{x}(t) + K_y\ y(t) \end{cases}$$

where $A_c^{\mathrm{d}} = \mathrm{diag}(A_c^1, \dots, A_c^N), K_x^{\mathrm{d}} = \mathrm{diag}(K_x^1, \dots, K_x^N)$ are block-diagonal, while B_c and K_y have a structure derived from the information structure constraints, as in the case of static output-feedback control.

3) *Static state-feedback control*

As for *decentralized control,* we cannot assume in general that the state x of the system $\mathcal{S}$ may be available to each local controller. Therefore, we again make the assumption that a suitable partition of the state vector x into subvectors x_i, $i = 1, \dots, N$, is possible. That is, we assume that, possibly up to a permutation of the elements of x, we can write

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_N \end{bmatrix}$$

where $x_i \in \mathbb{R}^{n_i}$ and $\sum_{i=1}^{N} n_i = n$.

For each controller the control law is

$$u_i(t) = \sum_{j\in\mathcal{N}_i} K_x^{ij} x_j(t)$$

where $K_x^{ij} \in \mathbb{R}^{m_i \times p_j}$. In matrix form

$$u_i(t) = [K_x^{i1} \quad \cdots \quad K_x^{iN}] x(t)$$

where $K_x^{ij} = 0$ if $j \notin \mathcal{N}_i$. Collectively, the whole vector $u(t)$ is defined as follows

$$u(t) = \begin{bmatrix} u_1(t) \\ \vdots \\ u_N(t) \end{bmatrix} = \begin{bmatrix} K_x^{11} & \cdots & K_x^{1N} \\ \vdots & \ddots & \vdots \\ K_x^{N1} & \cdots & K_x^{NN} \end{bmatrix} x(t) = K_x\ x(t)$$

4) *Distributed fixed modes*

The question, addressed in this section, is similar to the one that has previously motivated the discussion about *decentralized fixed modes*. Given a system $\mathcal{S}$, is it possible to place the eigenvalues of the closed-loop system at arbitrary desired positions with a distributed feedback controller, e.g., a dynamic output-feedback one? In the following we introduce the notion of *distributed fixed modes*.

To define the distributed fixed modes (DiFM), we start considering a distributed output-feedback static control law of the type $u_i(t) = \sum_{j\in\mathcal{N}_i} K_y^{ij} y_j(t)$ for all $i = 1, \dots, N$. Another important ingredient, necessary to properly define DiFMs, is the class of all possible control gains compatible with the given information structure constrains, defined by the set $\boldsymbol{\mathcal{N}}$: it will be denoted by $\mathcal{K}^{\boldsymbol{\mathcal{N}}}$.

More specifically, $\mathcal{K}^{\boldsymbol{\mathcal{N}}}$ is the set of all matrices

$$K_y = \begin{bmatrix} K_y^{11} & \cdots & K_y^{1N} \\ \vdots & \ddots & \vdots \\ K_y^{N1} & \cdots & K_y^{NN} \end{bmatrix}$$

such that, for all $i = 1, \dots, N$, $K_y^{ij} = 0$ if $j \notin \mathcal{N}_i$.

Definition 5 (distributed fixed modes)

The elements of the set

$$\Lambda_{DiFM} = \bigcap_{K_y \in \mathcal{K}^{\boldsymbol{\mathcal{N}}}} \mathrm{spec}(A + BK_yC)$$

are the distributed fixed modes (DiFM).

Remark that, If a mode is a CFM, then there does not exist a matrix $K_y \in \mathbb{R}^{m\times p}$ that "moves" its position, including "structured" gain matrices $K_y \in \mathcal{K}^{\boldsymbol{\mathcal{N}}}$. Therefore, a CFM must also be a DiFM, i.e., $\Lambda_{CFM} \subseteq \Lambda_{DiFM}$. On the other hand, it is clear that the class $\mathcal{K}^d$ is a subset of the class $\mathcal{K}^{\boldsymbol{\mathcal{N}}}$, from which it follows that $\Lambda_{DiFM} \subseteq \Lambda_{DFM}$. Overall

$$\Lambda_{CFM} \subseteq \Lambda_{DiFM} \subseteq \Lambda_{DFM}$$

A computationally simple procedure can be derived from Definition 5:

1. Compute the spectrum of matrix A, i.e., $\mathrm{spec}(A)$.
2. Select the matrix $K_y \in \mathcal{K}^{\boldsymbol{\mathcal{N}}}$ randomly.
3. Compute $\mathrm{spec}(A + BK_yC)$.
4. The set of DiFMs is given by $\mathrm{spec}\big(A + BK_yC\big) \cap \mathrm{spec}(A)$ with probability 1.

Similarly to the decentralized control case, in the case DiFM are absent, from Definition 5 we infer that a distributed static output feedback control law is able to “move” the eigenvalues of the system $\mathcal{S}$, but does not allow to place them in arbitrary prescribed positions. However, it can be proved that an output-feedback dynamic controller can do it. This is stated in the following theorem.

Theorem 4

The following statements hold.

- There exists a distributed (linear, time-invariant, and finite-dimensional) output feedback dynamic controller such that the closed loop system is asymptotically stable if and only if all the DiFM have strictly negative real part.
- There exists a distributed (linear, time-invariant, and finite-dimensional) output feedback dynamic controller such that the closed loop system has an arbitrary prescribed set of eigenvalues if and only if the system has no DiFMs.

Examples

In this section we consider the examples introduced in the in Section C.6 showing that, by allowing communication, we may “remove” the fixed modes. Specifically, we show that we are able to define a set $\boldsymbol{\mathcal{N}}$ such that the corresponding set of DiFM is empty.

EXAMPLE 1

Consider the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & -1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Recall that $\lambda_3 = -2$ is a DFM (although not structural). To remove the fixed mode, we try to remove some information structure constraints, by allowing some communication to take place. We use the numerical test sketched above in two different cases.

1. We define $\mathcal{N}_1 = \{1,2\}$ and $\mathcal{N}_2 = \{2\}$. Indeed, we allow matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & K_y^{12} \\ 0 & K_y^{22} \end{bmatrix}$$

In this case $\lambda_3 = -2$ is a DiFM.

2. We define $\mathcal{N}_2 = \{1\}$ and $\mathcal{N}_2 = \{1,2\}$, allowing matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & 0 \\ K_y^{21} & K_y^{22} \end{bmatrix}$$

In this case the system has no DiFMs.

EXAMPLE 2

Consider the system

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u_1 + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u_2$$

$$y_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$y_2 = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Recall that $\lambda_2 = -2$ is a structural DFM. To remove the fixed mode, we try to remove some information structure constraints, by allowing some communication to take place. We use the numerical test sketched above in two different cases.

1. We define $\mathcal{N}_1 = \{1,2\}$ and $\mathcal{N}_2 = \{2\}$. Indeed, we allow matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & K_y^{12} \\ 0 & K_y^{22} \end{bmatrix}$$

In this case $\lambda_2 = -2$ is a DiFM.

2. We define $\mathcal{N}_2 = \{1\}$ and $\mathcal{N}_2 = \{1,2\}$, allowing matrix K_y to be of the form

$$K_y = \begin{bmatrix} K_y^{11} & 0 \\ K_y^{21} & K_y^{22} \end{bmatrix}$$

In this case the system has no DiFMs.

EXAMPLE 3

Consider the system

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\\ \dot{x}_3\end{bmatrix} = \begin{bmatrix}-1 & 0 & 0\\ 0 & -1 & 0\\ 0 & 0 & -6\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix} + \begin{bmatrix}0\\ 0\\ 1\end{bmatrix}u_1 + \begin{bmatrix}1 & 0\\ 0 & 1\\ 0 & 0\end{bmatrix}u_2$$

$$y_1 = \begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

$$y_2 = \begin{bmatrix}0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\\ x_3\end{bmatrix}$$

Recall that $\lambda_1 = -1$ is a DFM, although not structural. To remove the fixed mode, we try to remove some information structure constraints, by allowing some communication to take place. We use the numerical test sketched above in two different cases.

1. We define $\mathcal{N}_1 = \{1,2\}$ and $\mathcal{N}_2 = \{2\}$. Indeed, we allow matrix K_y to be of the form

$$K_y = \begin{bmatrix}K_y^{11} & K_y^{12}\\ 0 & K_y^{22}\end{bmatrix}$$

   In this case $\lambda_1 = -1$ is a DiFM.

2. We define $\mathcal{N}_2 = \{1\}$ and $\mathcal{N}_2 = \{1,2\}$, allowing matrix K_y to be of the form

$$K_y = \begin{bmatrix}K_y^{11} & 0\\ K_y^{21} & K_y^{22}\end{bmatrix}$$

   In this case the system has no DiFMs.

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II.3: DESIGN OF STATIC CONTROL LAWS USING LMIS

A. STATIC CENTRALIZED STATE-FEEDBACK CONTROL FOR ASYMPTOTIC STABILITY

1) Continuous-time systems

In this section we consider the control of continuous-time systems which can be described using the following dynamic model

$$\dot{x}(t) = Ax(t) + Bu(t)$$

The static state-feedback control law is

$$u(t) = K_x x(t)$$

where $K_x \in \mathbb{R}^{m\times n}$ is the *control gain.* In this way that the closed-loop system dynamics is

$$\dot{x}(t) = (A + BK_x)x(t)$$

As discussed, the matrix $A + BK_x$ is asymptotically (i.e., Hurwitz) stable if and only if there exists a symmetric matrix $P > 0$ such that

$$(A + BK_x)^T P + P(A + BK_x) < 0$$

In control design, the unknowns are P and the gain K_x. The inequality above can be rewritten as

$$\underbrace{A^T P + PA}_{\text{linear in } P} + \underbrace{K_x^T B^T P + PBK_x}_{\text{non linear in } P \text{ and } K_x} < 0$$

The above matrix inequality is not linear. However, it is possible to cast it, under a suitable reformulation, as a linear one. To do so, we define

$$Y = P^{-1}$$

$$L = K_x P^{-1}$$

and we rewrite the inequality as

$$P(YA^T + AY + L^T B^T + BL)P < 0$$

which holds true if and only if

$$YA^T + AY + L^T B^T + BL < 0$$

Note that, now, the inequality (having the independent variables Y and L as free/design variables) is linear.

LMI FEASIBILITY PROBLEM CT1

To summarize, the LMI feasibility problem to be solved to design a control gain K_x that guarantees Hurwitz stability of the closed-loop system matrix $A + BK_x$ simply consists of checking if $\exists\, Y = Y^T > 0, L$ such that

$$\begin{gathered} YA^T + AY + L^T B^T + BL < 0 \\ Y > 0 \end{gathered}$$

If such solution exists, the gain matrix $K_x = LY^{-1}$ guarantees such stability properties. Note that, being the latter a mere feasibility problem without any account on the system performances, we cannot expect any optimality features by the closed-loop system.

2) *Discrete-time systems*

In this section we consider the control of discrete-time systems (e.g., obtained from the continuous-time system with a sample and hold discretization) which can be described using the following dynamic model

$$x_{k+1} = Fx_k + Gu_k$$

The static state-feedback control law is

$$u_k = K_x x_k$$

where $K_x \in \mathbb{R}^{m\times n}$ is the *control gain.* In this way that the closed-loop system dynamics is

$$x_{k+1} = (F + GK_x)x_k$$

As discussed, the matrix $F + GK_x$ is asymptotically (Schur) stable if and only if there exists a symmetric matrix $P > 0$ such that

$$(F + GK_x)^T P(F + GK_x) - P < 0$$

This is equivalent to require that there exists a symmetric matrix $P > 0$ such that

$$(F + GK_x)P(F + GK_x)^T - P < 0$$

In control design, the unknowns are P and the gain K_x. The inequality above can be rewritten as

$$\underbrace{FPF^T - P}_{\text{linear in } P} + \underbrace{FPK_x^T G^T + GK_x PF^T + GK_x PK_x^T G^T}_{\text{non linear in } P \text{ and } K_x} < 0$$

Note that the one above is not a linear matrix inequality. However, it is possible to cast it, under a suitable reformulation, as a linear one. To do so, we first define

$$L = K_x P$$

By doing this, we rewrite the inequality above as

$$\underbrace{P - FPF^T - FL^T G^T - GLF^T}_{\text{linear in } P \text{ and } L} - \underbrace{GLP^{-1}L^T G^T}_{\text{non linear in } P \text{ and } L} > 0$$

This inequality is still nonlinear. However, it can be shown that, in view of the Schur complement, it is equivalent to

$$\begin{bmatrix} P - FPF^T - FL^T G^T - GLF^T & GL \\ L^T G^T & P \end{bmatrix} > 0$$

which is linear in both P and L.

Remark 1

The Schur complement is formulated as follows: the LMI

$$\begin{bmatrix} Q & S \\ S^T & R \end{bmatrix} > 0$$

where Q and R are symmetric, is equivalent to

$$\begin{cases} R > 0 \\ Q - SR^{-1}S^T > 0 \end{cases}$$

Remark 2

An alternative but equivalent rule with respect to the one introduced in Remark 1 will be used in the following sections. More specifically, assume that (being $R > 0$) we need to impose

$$Q + SR^{-1}S^T < 0$$

We rewrite the latter inequality as $-Q - (-S)R^{-1}(-S^T) > 0$. We now resort to the Schur complement formulation introduced in Remark 1 and we rewrite it as

$$\begin{bmatrix} -Q & -S \\ -S^T & R \end{bmatrix} > 0$$

which also reads as

$$\begin{bmatrix} Q & S \\ S^T & -R \end{bmatrix} < 0$$

LMI FEASIBILITY PROBLEM DT1

To summarize, the LMI feasibility problem to be solved to design a control gain K_x that guarantees Schur stability of the closed-loop system matrix $F + GK_x$ simply consists of checking if $\exists\ P = P^T > 0, L$ such that

$$\begin{bmatrix} P - FPF^T - FL^T G^T - GLF^T & GL \\ L^T G^T & P \end{bmatrix} > 0$$

If such solution exists, the gain matrix $K_x = LP^{-1}$ guarantees such stability properties. As in the continuous-time case, being the latter a mere feasibility problem without any account on the system performances, we cannot expect any optimality features by the closed-loop system.

LMI PROBLEM COMPUTATIONAL EFFORT

A meaningful, although quite rough, measure of the computational effort associated with the centralized LMI feasibility problems CT1 and DT1 above is the total number of LMI variables (i.e., the $n(n+1)/2$ free variables in the symmetric matrix Y – or P – and the nm free variables in matrix L), i.e.,

$$\eta_{CENT} = \frac{n(n+1)}{2} + nm$$

In applications to large-scale systems, the computational effort η may become critical, e.g., in view of the fact that η is proportional to n^2. This constraints the application of the approach to systems of relatively small/medium scale. This limitation will be overcome in designing distributed and decentralized controllers, where the number of degrees of freedom of the optimization problem will be restricted by restricting the structure of the matrices Y – or P – and L.

B. ADDITIONAL INGREDIENTS TO THE LMI PROBLEM

As remarked in the previous section, the feasibility problems defined so far, although they guarantee asymptotic stability, are not tailored to optimize any performance index or to place the eigenvalues of the closed-loop system in desired positions or regions. To overcome this limitation, in this section we introduce and discuss additional ingredients, that can be added to any LMI problem (e.g., the ones discussed in Section A), to confer some additional properties to the closed loop system, both in the continuous-time and in the discrete-time domain.

1) Reduction of the control effort

The control gains defined in the previous section are derived in a similar way, i.e.,

- $K_x = LY^{-1}$ in the continuous-time framework, where both Y and L are optimization variables.
- $K_x = LP^{-1}$ in the discrete-time framework, where both P and L are optimization variables.

It would be desirable to limit the control effort, which is obtained by limiting/minimizing the size (i.e., the 2-norm) of the control gain K_x. This is possible by imposing $\|K_x\| \leq \kappa_x$. To do this, we recall that $\|K_x\| \leq \|L\|\|Y^{-1}\|$ (respectively, $\|K_x\| \leq \|L\|\|P^{-1}\|$) and we enforce, at the same time, a limitation on $\|L\|$ and $\|Y^{-1}\|$ (respectively, $\|P^{-1}\|$).

- To obtain $\|L\| \leq \sqrt{\kappa_L}$ (for some positive scalar variable k_L) we impose $L^T L \leq \kappa_L I$. If we rewrite it as $\kappa_L I - L^T (I)^{-1} L \geq 0$, we note that it can be rewritten as a linear inequality (by resorting to the Schur complement) as follows

$$\begin{bmatrix} \kappa_L I & L^T \\ L & I \end{bmatrix} \geq 0$$

- To obtain $\|Y^{-1}\| \leq \kappa_Y$ (for some positive scalar variable κ_Y) we impose (being Y symmetric and positive definite) $Y^{-1} \leq \kappa_Y I$. The latter can be rewritten as $\kappa_Y I - IY^{-1}I \geq 0$ and, by resorting to the Schur complement, as

$$\begin{bmatrix} \kappa_Y I & I \\ I & Y \end{bmatrix} \geq 0$$

  Of course, the latter idea can be applied, in the discrete-time case, to matrix P.

To summarize, the ingredients to be added to the LMI problem are

- Cost $J = a_L\kappa_L + a_Y\kappa_Y$, where a_L, a_Y are positive parameters to be properly selected. In the literature, a suggested choice is $a_L = 0.01, a_Y = 10$.
- Additional LMIs defined above.

Clearly, the latter procedure allows to minimize κ_L and κ_Y such that $\|K_x\| \leq \sqrt{\kappa_L}\kappa_Y$.

2) Limitation on the spectral abscissa of $A + BK_x$ (α-stability)

Assume that we aim to make all the eigenvalues of $A + BK_x$ satisfy $Re(\lambda) < -\alpha$. This is possible by simply guaranteeing that matrix $A + BK_x + \alpha I$ is Hurwitz stable. In fact, the eigenvalues of $A + BK_x + \alpha I$ are $\lambda_i + \alpha$, being $\lambda_i, i = 1, \dots, N$ the eigenvalues of $A + BK_x$. Therefore, the Hurwitz stability of $A + BK_x + \alpha I$ guarantees that $Re(\lambda_i + \alpha) < 0$ for all $i = 1, \dots, N$, i.e., that $Re(\lambda_i) < -\alpha$ for all $i = 1, \dots, N$. To guarantee that matrix $A + BK_x + \alpha I$ is Hurwitz stable, by resorting to the results reported in Section A.1, we need to impose that $Y(A + BK_x + \alpha I)^T + (A + BK_x + \alpha I)Y = Y(A + \alpha I)^T + (A + \alpha I)Y + L^T B^T + BL < 0$, which reads as

$$YA^T + AY + L^T B^T + BL + 2\alpha Y < 0$$

3) Limitation on the spectral radius of $A + BK_x$.

Assume that we aim to make all the eigenvalues of $A + BK_x$ satisfy $|\lambda| < \rho$. This is possible by simply requiring that matrix $\frac{1}{\rho}(A + BK_x)$ is Schur stable. In fact, the eigenvalues of $\frac{1}{\rho}(A + BK_x)$ are $\frac{\lambda_i}{\rho}$, being $\lambda_i, i = 1, \dots, N$ the eigenvalues of $A + BK_x$. Therefore, the Schur stability of $\frac{1}{\rho}(A + BK_x)$guarantees that $|\frac{\lambda_i}{\rho}| < 1$ for all $i = 1, \dots, N$, i.e., that $|\lambda_i| < \rho$ for all $i = 1, \dots, N$. To guarantee that matrix $\frac{1}{\rho}(A + BK_x)$ is Schur stable, by resorting to the results reported in Section A.2, we need to impose that $\frac{1}{\rho^2}(A + BK_x)P(A + BK_x)^T - P < 0$, i.e., that $(A + BK_x)P(A + BK_x)^T - \rho^2 P < 0$, which can be rephrased as

$$\begin{bmatrix} \rho^2 P - APA^T - AL^T B^T - BLA^T & BL \\ L^T B^T & P \end{bmatrix} > 0$$

Note, in passing, that the above requirement is generally desirable in the discrete-time framework. In this case, we apply this result by trivially recalling that A has to be replaced with F and B has to be replaced with G.

4) Eigenvalues of $A + BK_x$ in a disk

As a generalization of the results presented in the previous subsections, we may want to place all the eigenvalues of the closed-loop system matrix in a disk of center α and radius ρ. This is simply obtained by requiring that the matrix $\frac{1}{\rho}(A + BK_x + \alpha I)$ is Schur stable. This is guaranteed if and only if $\frac{1}{\rho^2}(A + BK_x + \alpha I)P(A + BK_x + \alpha I)^T - P < 0$, which reads

$$(A + BK_x)P(A + BK_x)^T + \alpha P(A + BK_x)^T + \alpha(A + BK_x)P - (\rho^2 - \alpha^2)P < 0$$

By defining $L = K_x P$ and applying the Schur complement, the latter is rewritten as

$$\begin{bmatrix} (\rho^2-\alpha^2)P - APA^T - \ AL^TB^T - BLA^T - \alpha(PA^T + AP + L^TB^T + BL) & BL \\ L^TB^T & P \end{bmatrix} > 0$$

5) Eigenvalues of $A + BK_x$ in a region $\mathcal{D}$

In this section, we consider the case in which we require the eigenvalues of a matrix to lie in the region $\mathcal{D}$ of the complex plane, i.e.

$$\mathcal{D} = \{z \in \mathbb{C}: f_D(z) < 0\}$$

where $f_D(z) = \Lambda + z\Theta + z^*\Theta^T$ and where Λ is a symmetric real matrix, while Θ is a real matrix of the same dimension. In the definition of $\mathcal{D}$, the symbol '<' requires $f_D(z)$ to be a negative definite matrix.

The previous definition implies that $\mathcal{D}$ is convex and symmetric with respect to the real axis. It allows to describe, e.g., conic sectors (i.e., such that $aRe(z) + b < -|Im(z)|$), vertical half planes (i.e., such that $Re(z) + \alpha < 0$ or $Re(z) + \alpha > 0$), vertical strips (i.e., such that $h_1 < Re(z) < h_2$), horizontal strips (i.e., such that $|Im(z)| < b$), ellipses, parabolas, and hyperbolic sectors, etc.

EXAMPLE 1. The disk of radius ρ and center α can be described as a region $\mathcal{D}$ where

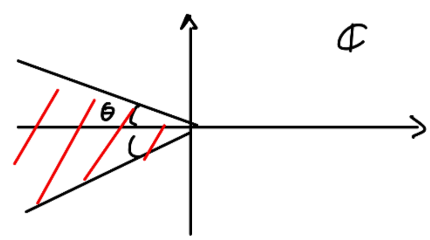
$$f_D(z) = \begin{bmatrix} -\rho & \alpha + z \\ \alpha + z^* & -\rho \end{bmatrix}$$

i.e., with

$$\Lambda = \begin{bmatrix} -\rho & \alpha \\ \alpha & -\rho \end{bmatrix}, \Theta = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

In fact, $f_D(z) < 0$ if and only if its eigenvalues are negative, i.e., iff $|z + \alpha| < r$. This comes from the requirements that $\det(f_D(z)) > 0$ and $\operatorname{tr}(f_D(z)) < 0$.

EXAMPLE 2. The sector region depicted in the figure can be described by



$$f_D(z) = \begin{bmatrix} \sin(\theta)(z + z^*) & \cos(\theta)(z - z^*) \\ \cos(\theta)(z^* - z) & \sin(\theta)(z + z^*) \end{bmatrix}$$

i.e., with

$$\Lambda = 0, \Theta = \begin{bmatrix} \sin(\theta) & \cos(\theta) \\ -\cos(\theta) & \sin(\theta) \end{bmatrix}.$$

In fact, $f_D(z) < 0$ if and only if $\det(f_D(z)) > 0$ and $\operatorname{tr}(f_D(z)) < 0$, i.e., $4\sin(\theta)^2 \left(Re(z)\right)^2 - 4\cos(\theta)^2 \left(Im(z)\right)^2 > 0$ and $4\sin(\theta)\left(Re(z)\right) < 0$. Overall, we obtain that

$$\begin{cases} Re(z) < 0 \\ |Im(z)| < \tan(\theta)|Re(z)| \end{cases}$$

Note that, having the eigenvalues in the region defined here is very important, especially in the continuous-time framework. In fact complex conjugate eigenvalues $-\xi\varpi \pm j\sqrt{1-\xi^2}\varpi$ satisfying the above condition have a damping factor $\xi \in (\cos(\theta), 1)$, significantly reducing overshoots in the system response.

Provided that the region $\mathcal{D}$ can be defined as above stated, the following Theorem can be applied to obtain an LMI guaranteeing that the eigenvalues of $A + BK_x$ lie in $\mathcal{D}$.

Theorem ($\mathcal{D}$-stability)

The eigenvalues of $A + BK_x$ are in $\mathcal{D}$ (in this case we say that $A + BK_x$ is $\mathcal{D}$-stable) if and only if there exists a symmetric matrix $Y > 0$ such that $\mathcal{M}_D(A + BK_x, Y) < 0$, where $\mathcal{M}_D(A + BK_x, Y)$ is a matrix whose (i, j)-th block is

$$\mathcal{M}_D(A + BK_x, Y)|_{i,j} = \lambda_{ij}Y + \theta_{ij}(A + BK_x)Y + \theta_{ji}Y(A + BK_x)^T$$

where λ_{ij} and θ_{ij} are the (i, j)-th elements of matrices Λ and Θ, respectively.

Consider the examples shown previously. Regarding Example 1, the theorem provides an alternative way, with respect to the one derived in the previous Section 4 to constrain the eigenvalues of $A + BK_x$ to lie in a specified disk. On the other hand, concerning Example 2, the theorem provides the following matrix inequality that guarantees the eigenvalues of the closed-loop system matrix to lie in the sector shown in the figure, i.e.,

$$\begin{bmatrix} \sin(\theta)\{(A + BK_x)Y + Y(A + BK_x)^T\} & \cos(\theta)\{(A + BK_x)Y - Y(A + BK_x)^T\} \\ \cos(\theta)\{-(A + BK_x)Y + Y(A + BK_x)^T\} & \sin(\theta)\{(A + BK_x)Y + Y(A + BK_x)^T\} \end{bmatrix} < 0$$

As done before, the inequality can be cast as a linear one provided that we define $L = K_xY$ and we rephrase it as follows

$$\begin{bmatrix} \sin(\theta)\{AY + YA^T + BL + L^TB^T\} & \cos(\theta)\{AY - YA^T + BL - L^TB^T\} \\ \cos(\theta)\{-AY + YA^T - BL + L^TB^T\} & \sin(\theta)\{AY + YA^T + BL + L^TB^T\} \end{bmatrix} < 0$$

C. STATIC CENTRALIZED STATE-FEEDBACK CONTROL FOR OPTIMAL PERFORMANCE GUARANTEES

Besides guaranteeing stability and eigenvalue placement in prescribed positions, it is possible to formulate as LMIs also optimality requirements, e.g., in terms of minimization of $\mathcal{H}_2$ and/or $\mathcal{H}_\infty$ norms of closed-loop transfer functions between real/fictitious noises/errors and performance outputs. As well known, we can cast LQ control design as $\mathcal{H}_2$ minimization problems, while robust control problems can be formulated as $\mathcal{H}_\infty$ minimization ones.

1) Continuous-time systems, $\mathcal{H}_2$ control design

In this section we show how to design a control gain K_x such that the control law $u(t) = K_xx(t)$ allows to minimize the $\mathcal{H}_2$ norm (also said $\mathcal{L}_2$-to-$\mathcal{L}_\infty$ gain) of the transfer function $G_{zw}(s)$ of the system

$$\begin{array}{rcl}\dot{x}(t) & = & Ax(t) + Bu(t) + B_w w(t) \\ z(t) & = & Cx(t) + D_u u(t)\end{array}$$

Notably, by imposing

$$C = \begin{bmatrix}\sqrt{Q} \\ 0\end{bmatrix}, D_u = \begin{bmatrix}0 \\ \sqrt{R}\end{bmatrix}$$

it is possible to enforce LQ-like optimal performances. The $\mathcal{H}_2$ norm of the transfer function $G_{zw}(s)$ between $w(t)$ and $z(t)$, under $u(t) = K_x x(t)$, can be computed as

$$\|G_{zw}(s)\|_2^2 = \inf\{\mathrm{trace}\big((C + D_u K_x)Y(C + D_u K_x)^T\big): (A + BK_x)Y + Y(A + BK_x)^T + B_w B_w^T \leq 0\,\}$$

Let us define matrix S such that $S \geq (C + D_u K_x)Y(C + D_u K_x)^T$. If we define $L = K_x Y$, as usual, we rewrite is as $S - (CY + D_u L)Y^{-1}(CY + D_u L)^T \geq 0$. This eventually allows to cast the $\mathcal{H}_2$ norm minimization problem as follows.

$\min\limits_{Y,S,L} trace(S)$ subject to

$$YA^T + AY + BL + L^T B^T + B_w B_w^T \leq 0$$

$$\begin{bmatrix} S & CY + D_u L \\ L^T D_u^T + YC^T & Y \end{bmatrix} \geq 0$$

2) Continuous-time systems, $\mathcal{H}_\infty$ control design

In this section we show how to design a control gain K_x such that the control law $u(t) = K_x x(t)$ allows to minimize the $\mathcal{H}_\infty$ norm (also said $\mathcal{L}_2$-to-$\mathcal{L}_2$ gain) of the transfer function $G_{zw}(s)$ of the system

$$\begin{array}{rcl}\dot{x}(t) & = & Ax(t) + Bu(t) + B_w w(t) \\ z(t) & = & Cx(t) + D_u u(t) + D_w w(t)\end{array}$$

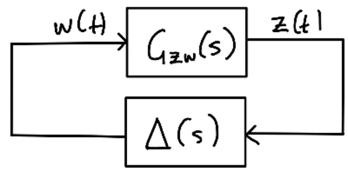
under $u(t) = K_x x(t)$. Note that, if we set $(t) = K_x x(t)$, the latter system can be reformulated as

$$\begin{array}{rcl}\dot{x}(t) & = & A_{cl} x(t) + B_w w(t) \\ z(t) & = & C_{cl} x(t) + D_w w(t)\end{array}$$

where $A_{cl} = A + BK_x$ and $C_{cl} = C + D_u K_x$. The limitation of the $\mathcal{H}_\infty$ of the latter transfer function is of particular importance in the framework of robust control, e.g., for guaranteeing asymptotic stability of the control system under possible structured and unstructured uncertainties. The robust control requirement reads as

$$\|G_{zw}(s)\|_\infty < \gamma$$

For instance this, in view of the small gain theorem, guarantees asymptotic stability of the uncertain system in the figure below, where $\Delta(s)$ is an uncertain transfer function where $||\Delta(s)||_\infty \leq \frac{1}{\gamma}$.



It is possible to show that the above scheme is asymptotically stable if there exists a symmetric and positive definite matrix Y such that

$$\begin{bmatrix} YA_{cl}^T + A_{cl}Y & B_w & YC_{cl}^T \\ B_w^T & -\gamma I & D_w^T \\ C_{cl}Y & D_w & -\gamma I \end{bmatrix} < 0$$

Remark. The statement above can be proved resorting to the Lyapunov Theorem. here we show the proof in case $\Delta(s)$ is a static function, but similar arguments can be applied also in case $\Delta(s)$ is the transfer function of a dynamical system. More specifically, we define the candidate Lyapunov function $V(x) = x^T Px$. Stability is guaranteed if $P > 0$ and

$$\dot{V}(x) = (A_{cl}x + B_w w)^T Px + x^T P(A_{cl}x + B_w w) = [x^T \quad w^T] \underbrace{\begin{bmatrix} A_{cl}^T P + PA_{cl} & PB_w \\ B_w^T P & 0 \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} x \\ w \end{bmatrix} < 0$$

Since $||\Delta(s)||_\infty \leq \frac{1}{\gamma}$ and noting that $\Delta(s)$ is the (static, in this case) function between variable $z(t)$ and variable $w(t)$, we have that $||w(t)||_2 \leq \frac{1}{\gamma}||z(t)||_2$, i.e.,

$$w^T w = [x^T \quad w^T] \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} \begin{bmatrix} x \\ w \end{bmatrix} \leq \frac{1}{\gamma^2} z^T z = \frac{1}{\gamma^2} [x^T \quad w^T] \begin{bmatrix} C_{cl}^T \\ D_w^T \end{bmatrix} [C_{cl} \quad D_{cl}] \begin{bmatrix} x \\ w \end{bmatrix}$$

which can be rewritten as

$$[x^T \quad w^T] \underbrace{\left\{ \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} - \frac{1}{\gamma^2} \begin{bmatrix} C_{cl}^T \\ D_w^T \end{bmatrix} [C_{cl} \quad D_{cl}] \right\}}_{\mathcal{B}} \begin{bmatrix} x \\ w \end{bmatrix} \leq 0$$

To proceed we resort to the following theorem, providing a sufficient condition for stability of the overall system.

Theorem. Let $\mathcal{A}$ and $\mathcal{B}$ symmetric matrices of the same dimensions. Then $y^T\mathcal{A}y < 0$ holds under the condition that $y^T\mathcal{B}y \leq 0$ if there exists a number $\tau > 0$ such that $\mathcal{A} - \tau\mathcal{B} < 0$.

Thanks to the above theorem, stability of the closed-loop system is verified if there exist $P > 0$ and $\tau > 0$ such that

$$\begin{bmatrix} A_{cl}^T P + P A_{cl} & P B_w \\ B_w^T P & 0 \end{bmatrix} - \tau \left\{ \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} - \frac{1}{\gamma^2} \begin{bmatrix} C_{cl}^T \\ D_w^T \end{bmatrix} [C_{cl} \quad D_{cl}] \right\}$$
$$= \begin{bmatrix} A_{cl}^T P + P A_{cl} & P B_w \\ B_w^T P & -\tau I \end{bmatrix} + \frac{\tau}{\gamma^2} \begin{bmatrix} C_{cl}^T \\ D_w^T \end{bmatrix} [C_{cl} \quad D_{cl}] < 0$$

If we define $P = \tau \tilde{P}$ and we multiply both sides of the inequality by γ/τ we obtain

$$\left\{ \begin{bmatrix} A_{cl}^T \tilde{P} + \tilde{P} A_{cl} & \tilde{P} B_w \\ B_w^T \tilde{P} & -\gamma I \end{bmatrix} + \frac{1}{\gamma} \begin{bmatrix} C_{cl}^T \\ D_w^T \end{bmatrix} [C_{cl} \quad D_{cl}] \right\} \frac{\tau}{\gamma} < 0$$

Applying the Schur complement, this is verified if

$$\begin{bmatrix} A_{cl}^T \tilde{P} + \tilde{P} A_{cl} & \tilde{P} B_w & C_{cl}^T \\ B_w^T \tilde{P} & -\gamma I & D_w^T \\ C_{cl} & D_w & -\gamma I \end{bmatrix} < 0$$

Eventually, we define $Y = \tilde{P}^{-1}$ and we pre and post-multiply the inequality by $\mathrm{diag}(Y, I, I)$, obtaining that the asymptotic stability of the system is verified if there exists a symmetric and positive definite matrix Y such that

$$\begin{bmatrix} Y A_{cl}^T + A_{cl} Y & B_w & Y C_{cl}^T \\ B_w^T & -\gamma I & D_w^T \\ C_{cl} Y & D_w & -\gamma I \end{bmatrix} < 0$$

which concludes the proof.

At this point, to derive the LMI design condition recall that $A_{cl} = A + BK_x$ and $C_{cl} = C + D_u K_x$, where K_x is a further degree of freedom. If we define $L = K_x Y$, we can cast the $\mathcal{H}_\infty$ norm minimization problem as follows.

$\min_{Y,L} \gamma$ subject to

$$\begin{bmatrix} YA^T + AY + BL + L^T B^T & B_w & YC^T + L^T D_u^T \\ B_w^T & -\gamma I & D_w^T \\ CY + D_u L & D_w & -\gamma I \end{bmatrix} < 0$$

3) Discrete-time systems, $\mathcal{H}_2$ control design

We now derive the optimal control law $u_k = K_x x_k$ that minimizes the $\mathcal{H}_2$ norm of the transfer function $G_{zw}(z)$ of the discrete-time system

$$\begin{aligned} x_{k+1} &= F x_k + G u_k + G_w w_k \\ z_k &= H x_k + D_u u_k \end{aligned}$$

In discrete-time, the $\mathcal{H}_2$ norm of the transfer function $G_{zw}(z)$ between w_k and z_k, under $u_k = K_x x_k$, can be computed as

$$\|G_{zw}(z)\|_2^2 = \inf\{trace\big((H + D_uK_x)P(H + D_uK_x)^T\big): (F + GK_x)P(F + GK_x)^T - P + G_wG_w^T \leq 0\,\}$$

Let us define matrix S such that $S \geq (H + D_uK_x)P(H + D_uK_x)^T$. If we define $L = K_xP$, as usual, we rewrite is as $S - (HP + D_uL)P^{-1}(HP + D_uL)^T \geq 0$. This eventually allows to cast the $\mathcal{H}_2$ norm minimization problem as follows.

$\min\limits_{Y,S,L} trace(S)$ subject to

$$\begin{bmatrix} P - FPF^T - \; FL^TG^T - GLF^T - G_wG_w^T & GL \\ L^TG^T & P \end{bmatrix} \geq 0$$

$$\begin{bmatrix} S & HP + D_uL \\ L^TD_u^T + PH^T & P \end{bmatrix} \geq 0$$

4) Discrete-time systems, $\mathcal{H}_\infty$ control design

In this section we show how to design a control gain K_x such that the control law $u_k = K_xx_k$ allows to minimize the $\mathcal{H}_\infty$ norm of the transfer function $G_{zw}(z)$ of the system

$$\begin{aligned} x_{k+1} &= Fx_k + Gu_k + G_ww_k \\ z_k &= Hx_k + D_uu_k + D_ww_k \end{aligned}$$

under $u_k = K_xx_k$. Note that, if we set $u_k = K_xx_k$, the latter system can be reformulated as

$$\begin{aligned} x_{k+1} &= F_{cl}x_k + G_ww_k \\ z_k &= H_{cl}\,x_k + D_ww_k \end{aligned}$$

where $F_{cl} = F + GK_x$ and $H_{cl} = H + D_uK_x$. Similarly to the case of continuous-time systems, it is possible to prove that $G_{zw}(z)$ is asymptotically stable and $\|G_{zw}(z)\|_\infty < \gamma$ if and only if there exists a symmetric and positive definite matrix P such that

$$\begin{bmatrix} P & F_{cl}P & G_w & 0 \\ PF_{cl}^T & P & 0 & PH_{cl}^T \\ G_w^T & 0 & \gamma I & D_w^T \\ 0 & H_{cl}P & D_w & \gamma I \end{bmatrix} > 0$$

Recall that $F = F + GK_x$ and $H_{cl} = H + D_uK_x$, where K_x is a further degree of freedom. If we define $L = K_xP$, we can cast the $\mathcal{H}_\infty$ norm minimization problem as follows.

$\min\limits_{P,L} \gamma$ subject to

$$\begin{bmatrix} P & FP + GL & G_w & 0 \\ PF^T + L^TG^T & P & 0 & PH^T + L^TD_u^T \\ G_w^T & 0 & \gamma I & D_w^T \\ 0 & HP + D_uL & D_w & \gamma I \end{bmatrix} > 0$$

D. STATIC DECENTRALIZED AND DISTRIBUTED STATE-FEEDBACK CONTROL

As discussed in the previous chapters, decentralized and distributed static state-feedback control laws have the following general form, if written in a collective fashion.

- for continuous-time systems: $u(t) = K_x x(t)$;
- for discrete-time systems (digital control): $u_k = K_x x_k$.

The main issue, for decentralized and distributed control, is the fact that the gain matrix K_x is not totally free, but has a specific structure (i.e., some of its blocks are constrained to be zero due to the information structure constraints). For example, considering $N = 3$ subsystems

- for decentralized controllers
$$K_x = \mathrm{diag}(K_x^1, \dots, K_x^3)$$
i.e., the gain is block-diagonal;

- for distributed controllers with a cyclic communication graph
$$K_x = \begin{bmatrix} K_x^{11} & 0 & K_x^{13} \\ K_x^{21} & K_x^{22} & 0 \\ 0 & K_x^{32} & K_x^{33} \end{bmatrix}$$
- for distributed controllers with a star-like unidirectional communication graph of type 1,
$$K_x = \begin{bmatrix} K_x^{11} & 0 & 0 \\ K_x^{21} & K_x^{22} & 0 \\ K_x^{31} & 0 & K_x^{33} \end{bmatrix}$$
- for distributed controllers with a star-like unidirectional communication graph of type 2,
$$K_x = \begin{bmatrix} K_x^{11} & K_x^{12} & K_x^{13} \\ 0 & K_x^{22} & 0 \\ 0 & 0 & K_x^{33} \end{bmatrix}$$
- for distributed controllers with a star-like bidirectional communication graph,
$$K_x = \begin{bmatrix} K_x^{11} & K_x^{12} & K_x^{13} \\ K_x^{21} & K_x^{22} & 0 \\ K_x^{31} & 0 & K_x^{33} \end{bmatrix}$$

The design with LMIs consists of defining the gain matrix as a structured one. However, K_x is not a decision variable of the LMI problem, while matrices Y and L (P and L for control of discrete-time systems) are. So, the problem translates into the following one: how to impose a structure to K_x by imposing a structure to matrices Y (or P) and L? This can be done by

- imposing to L the same structure required to K_x.
- imposing a block-diagonal structure to Y (and therefore to P).

It is worth remarking that YALMIP allows to define a proper structure to its free variables without any restriction.

LMI PROBLEM COMPUTATIONAL EFFORT

Considering the fact that the symmetric matrix Y – or P – is now block diagonal (where each of the N symmetric blocks has dimension $n_i \times n_i$) and matrix L is structured (it is composed by non-zero blocks of dimensions $m_i \times n_j$, i.e., for all $j \in \mathcal{N}_i$, i.e., where communication is allowed). We obtain that

$$\eta_{DIST} = \sum_{i=1}^{N} \left\{ \frac{n_i(n_i+1)}{2} + m_i \sum_{j \in \mathcal{N}_i} n_j \right\}$$

In the particular case of decentralized control, we basically have that $\mathcal{N}_i = \{i\}$. In this case we obtain that

$$\eta_{DEC} = \sum_{i=1}^{N} \left\{ \frac{n_i(n_i+1)}{2} + n_i m_i \right\}$$

It is apparent that this is a significant advantage of decentralized and distributed control design with respect to the centralized one.

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PART III NETWORKED CONTROL SYSTEMS

III.1: DATA RATE LIMITATIONS AND QUANTIZATION

A. MAIN ASSUMPTIONS

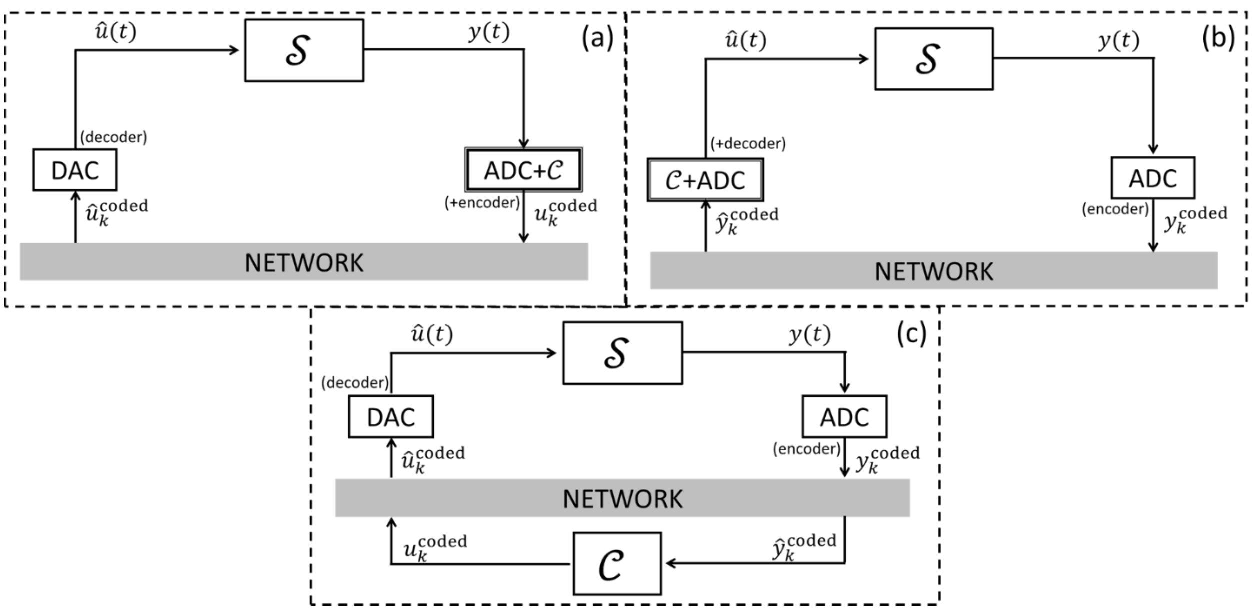
In this chapter the following main assumptions will be done.

A1. **Configuration**.

We consider a NCS (*networked control system*, i.e., a feedback-controlled system over a serial communication channel). In this chapter, the system configuration requires that the controller is integrated/coordinated with the encoder (i.e., the ADC, analog-to-digital converter). In particular, the control law and the quantization rule must be defined concurrently and consistent with each other. As depicted in the following figures (a)-(c), the controller may be either

- embedded into the encoder (a)
- embedded in the decoder (b)
- remote (c).

In this chapter we focus on scheme (a). Cases (b) and (c) are discarded here for simplicity of presentation, but the extension to these cases is straightforward.



A2. **Ideal communication channel**.

In this chapter we will neglect transmission noise, delays, or possible packet dropouts. This implies that (with reference, for example, to Figure (a) above) $\hat{u}_k^{\text{coded}} = u_k^{\text{coded}}$ for all $k \geq 0$. The channel has a capacity $R_{\max}$. Recall that, based on the Shannon's theorem,

$$R_{\max} = B\log_2(1 + S/N)$$

where

- B is the channel bandwidth [Hz];
- S/N is the channel signal-to-noise ratio (in linear scale);
- $R_{\max}$ is the maximum rate at which information (i.e., bits) can be transmitted over the communication channel [b/s].

Of particular interest, in this chapter, is the case where rate limitations are severe. This may be the case in which the loop is closed using wireless links, e.g., Bluetooth or IEEE 802.11(b).

A3. **Packet network**.

In this chapter, we consider the case where information is transmitted in packets. Each packet is composed of N_B bits. However, some bits in the packet are devoted to header, address, access code, etc., while just $N_B^I \leq N_B$ are dedicated to conveying information (i.e., the coded/quantized control signal $\hat{u}_k^{\text{coded}}$ in this case). Note that:

- Under the assumption that $N_B^I < N_B$ we can define the effective capacity as the maximum rate at which effective information (i.e., control signals) is transmitted:
$$R_{\max}^{\text{eff}} = R_{\max} \frac{N_B^I}{N_B} < R_{\max}$$
From now on, to simplify the presentation, we will assume that $N_B^I = N_B$, which implies that $R_{\max}^{\text{eff}} = R_{\max}$. The results discussed in this chapter can be easily extended to the case $N_B^I < N_B$.
- In case u_k is scalar, it can take only $N = 2^{N_B}$ values, i.e., only 2^{N_B} different control actions can be taken at most. For example, when $N_B = 1$, only 2 input values are possible: indeed $\hat{u}_k^{\text{coded}}$ can take values 0 and 1, and the corresponding "decoded" input can take values $u_{\min}$ and $u_{\max}$.

B. PROBLEM STATEMENT

In this chapter we assume that the **transmission rate** R is given, while the **number of bits per packet** N_B and the **sampling time** h are to be considered as degrees of freedom, i.e., design choices.

In principle, the ideal case would be to

- reduce the sampling time h for more reactive control;
- increase N_B, for obtaining a finer quantization.

Note that the time required to transmit the N_B-bits packet is $T_{\text{frame}} = \frac{N_B}{R}$ and that each packet is required to arrive at destination in less than h seconds (before the sampling time expires), i.e., $\frac{N_B}{R} \leq h$, i.e.,

$$\frac{N_B}{h} \leq R$$

This fundamental inequality establishes an often serious trade-off between sampling and quantization. More specifically, if we opt to increase N_B to enhance the quantization quality, we have to accept large values of h. On the other hand, to reduce h a coarser quantization quality is obtained.

It is important to note that, while $T_{\text{frame}} \leq h$ is a fundamental inequality to be respected to guarantee the feasibility of the NCS, it is often convenient to set $T_{\text{frame}} \ll h$, resulting in the inequality $\frac{N_B}{h} \ll R$.

In this chapter we will give an answer (although in a simplified setting) to the following main questions.

- Given a linear system $\mathcal{S}$, what is the minimum value of $\frac{N_B}{h}$ that allows to make the NCS "stable"?
- In the case the rate $\frac{N_B}{h}$ is greater than the lower bound required, how to design a suitable stabilizing – quantized – control law?

C. EXAMPLES

We consider now the following scalar linear system.

$$\dot{x}(t) = ax(t) + bu(t) \qquad (i)$$

We set $N_B = 1$ bit/packet and $h = 1$ s, under the assumption that $R \geq 1$. Assuming sample-and-hold actuation, the corresponding discrete-time system is

$$x_{k+1} = fx_k + gu_k \qquad (ii)$$

where $f = e^{ah}$ and $g = -\frac{b}{a}(1 - e^{ah})$ if $a \neq 0$.

Also we set $b = 1$. Since $N_B = 1$, the control input can take $2^{N_B} = 2$ values, i.e., $u_k \in \mathcal{U}, \mathcal{U} := \{-1,1\}$. More specifically, we use the control law

$$u_k = -sign(x_k)$$

More specifically,

$$u_k = \begin{cases} -1 & \text{if } x_k \geq 0 \\ 1 & \text{if } x_k < 0 \end{cases}$$

In closed-loop, the discrete-time dynamical system is

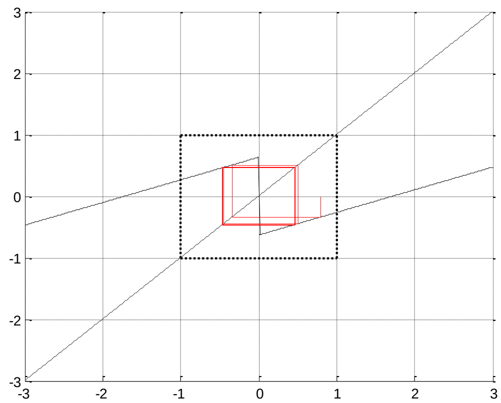
$$x_{k+1} = \begin{cases} fx_k - g & \text{if } x_k \geq 0 \\ fx_k + g & \text{if } x_k < 0 \end{cases} \qquad (iii)$$

We now consider three different cases, i.e., $a = -1, 0.5, 1$.

Case 1: $\boldsymbol{a = -1}$.

Since $a < 0$, the continuous-time system (i) is open-loop asymptotically stable. We compute that $f \cong 0.37$, $g \cong 0.63$. Therefore, as expected, the corresponding discrete-time system (ii) is also open-loop asymptotically stable, i.e., $|f| < 1$.

We now analyze the closed-loop behaviour of the NCS by studying the evolution of the trajectories using the graphical method. We obtain the following graph.

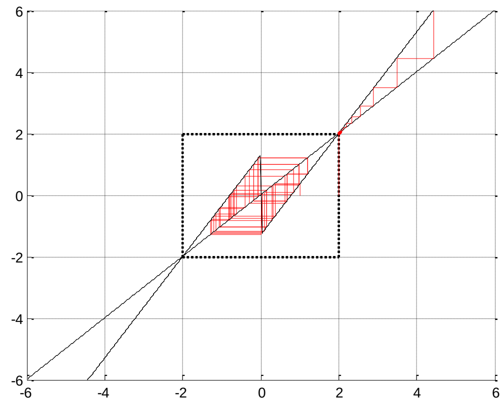


As a result, for all possible initial conditions $x_0 \in \mathbb{R}$, the trajectories tend to enter the set $[-0.63,0.63]$, and are eventually confined there. For this reason, the set $[-0.63,0.63]$ is denoted positively invariant. Note that, however, the trajectories do not converge to zero!

Case 2: $\boldsymbol{a = 0.5}$.

Since $a > 0$, the continuous-time system (i) is open-loop unstable. We compute that $f \cong 1.65$, $g \cong 1.3$. Therefore, as expected, the corresponding discrete-time system (ii) is also unstable, i.e., $|f| > 1$.

We now analyze the closed-loop behaviour of the NCS by studying the evolution of the trajectories using the graphical method. We obtain the following graph.



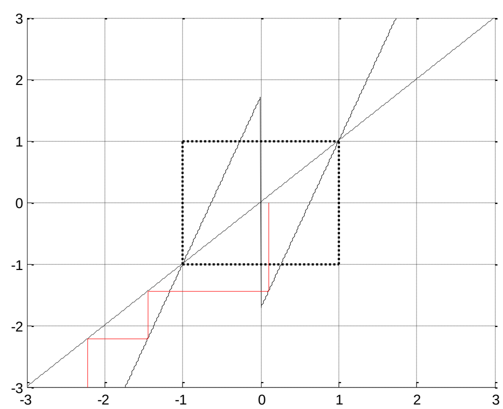
As a result,

- if $x_0 \in (-2,2)$, the trajectories tend to enter the set $[-1.3,1.3]$, and are eventually confined there. Note that, however, the trajectories do not converge to zero!
- if $|x_0| = 2$, if $x_k = x_0$ for all if $k \geq 0$. Indeed ± 2 are equilibria.
- if $|x_0| > 2$, $|x_k| \to +\infty$ as $k \to +\infty$.

Case 3: $\boldsymbol{a = 1}$.

Since $a > 0$, the continuous-time system (i) is open-loop unstable. We compute that $f \cong 2.7$, $g \cong 1.7$. Therefore, as expected, the corresponding discrete-time system (ii) is also unstable, i.e., $|f| > 1$.

We now analyze the closed-loop behaviour of the NCS by studying the evolution of the trajectories using the graphical method. We obtain the following graph.



As a result, for <u>almost</u> all initial conditions (e.g., $|x_0| \neq 1$), $|x_k| \to +\infty$ as $k \to +\infty$. The system has, in general, an unstable behavior.

Remarks from the examples.

The following conclusions can be drawn from the examples in this section.

- If $a < 0$ (open-loop asymptotically stable system), using the proposed control law, the NCS displays a “stable behaviour”.
- If $a > 0$ (open-loop unstable system), the NCS may display a locally “stable behaviour” (i.e., in case a is “not too large”) or an unstable behavior (if a is “large”).
- When the system displays a “stable” behavior, it is a non-standard stability property: it is *weaker* than asymptotic stability, but *stronger* that simple (i.e., non-asymptotic) stability.

In the following we will give an answer to the following questions:

- What type of stability properties are enjoyed by NCSs subject to quantization and what is the stabilizability-type property that allows to confer it?
- What kind of open-loop instabilities can be compensated at most (i.e., how “small” $a > 0$ should be)?

D. BOUNDABILITY

As clear from the examples, the classic notions of (closed-loop) stability and asymptotic stability fail to be applicable to the NCS subject to quantization. Even more so, even the concepts of controllabilty and stabilizability (i.e., the controllability of at least the unstable modes) cannot be used, due to the fact that the set of admissible inputs is discrete (finite), while the latter are defined in case of a continuous input set. The notion of *boundability* is indeed introduced to denote systems that *can be controlled* using a control sequences drawn from a specified set $\mathcal{U}$.

Definition (Boundability)

A linear time-invariant system

$$\dot{x}(t) = Ax(t) + Bu(t)$$

with an admissible control set $\mathcal{U}$ is *boundable* if there exists a bounded set $\mathbb{S}$ and an open set $\mathbb{M} \subseteq \mathbb{S}$ such that, for all $x_0 \in \mathbb{M}$ there exists a control sequence $U = u_0, u_1, u_2, \ldots$, where $u_k \in \mathcal{U}$ for all k, such that the continuous-time state trajectory defined by (i) considering $x(0) = x_0$ and the control sequence U (applied in a sample-and-hold fashion to the system with sampling time h) remains in $\mathbb{S}$ for all time.

Two comments are in order.

- It is easy to prove that, if the set $\mathcal{U}$ is continuous, then boundability is equivalent to (local) *stabilizability*, i.e., controllability of all unstable modes. In this sense, boundability can be regarded as a generalization of the concept of stabilizability to general input sets, and where inuts are applied in a sample-and-hold way.
- If the system is open-loop *asymptotically stable*, it is also *boundable*. In fact, we can define a constant control sequence $u_k = \bar{u} \in \mathcal{U}$ for all $k \geq 0$. For all $x_0 \in \mathbb{R}$, the trajectory converges to the equilibrium point $\bar{x} = (I - A)^{-1}B\bar{u}$. We can define arbitrarily large *positive invariant sets* centered in $\bar{x}$ and readily verify *boundability*.
- The case where *boundability* is not automatically verified is the case where A is not Hurwitz stable. Recalling the examples: in case 2 ($a = 0.5 > 0$) boundability is verified, while in case 3 ($a = 1 > 0$) boundability is not verified, at least using the specified control law.

E. THE CASE OF SCALAR SYSTEMS

System set-up.

In this section we consider the following set-up.

- We consider a scalar system, described by the equation
$$\dot{x}(t) = ax(t) + bu(t) \qquad (i)$$
- For simplicity of presentation (but without compromising the generality of the results, since the generalization is straightforward) we assume that $b > 0$.
- We assume sample-and-hold actuation, i.e., $u(t) = u_k$ for all $t \in [kh, (k+1)h)$, for all $k \geq 0$.
- The control input u_k can take a finite number of values in the discrete set $\mathcal{U}$, whose cardinality is $|\mathcal{U}| = N = 2^{N_B}$. The input variable u_k can take values in the set $[u_{\min}, u_{\max}]$.
- The transmission rate of the control network supporting the control system is R.
- The sampling time is h, such that $\frac{N_B}{h} \leq R$ (if possible $\frac{N_B}{h} \ll R$).
- The (discrete-time) dynamics of $x_k = x(kh)$ ($k \geq 0$) is described by the discrete-time system
$$x_{k+1} = fx_k + gu_k \qquad (ii)$$
where $f = e^{ah}$ and $g = b\int_0^h e^{a\tau} d\tau$.
- The control law is a static state-feedback one, i.e.,
$$u_k = s(x_k)$$
to be later specified, where $s(\cdot)$ is denoted *selection function*.

Main result.

The following theorem can be proved, the proof of which can be found in the appendix.

Theorem

Consider the system (i), with sampled realization (ii). There exists a finite set of admissible input values $\mathcal{U}$ such that the system (i) is *boundable* if and only if the rate $\frac{N_B}{h}$ satisfies the following inequality:

$$\frac{N_B}{h} \geq a \log_2 e$$

Also, in case the inequality holds, a suitable selection function $s(\cdot)$ exists that allows to steer the state to lie in a bounded set.

Remark that, in the examples of the previous section, $\frac{N_B}{h} = 1$. In case 1, $a < 0$, and therefore $\frac{N_B}{h} > a \log_2 e$. In case 2, $a = 0.5$, and therefore $a \log_2 e \cong 0.72 < \frac{N_B}{h}$, verifying the necessary and sufficient condition for boundability. Finally, in case 3, $a = 1$, and therefore $a \log_2 e \cong 1.44 > \frac{N_B}{h}$: in this case the Theorem allows us to conclude that there does not exist a control law (i.e., a selection function $s(\cdot)$) that makes the system trajectory bounded.

Operating at the rate limit: definition of a suitable control law.

Here we consider the case where the system is open-loop unstable and where N_B and h are selected in such a way that $\frac{N_B}{h} = a\log_2 e$ (where $\frac{N_B}{h} \leq R$ as required), i.e., the minimal value required to make the system boundable. The following results can be proved.

- To make the system boundable, the $N = 2^{N_B}$ control levels must be equally spaced. Letting $u_{\min}$ and $u_{\max}$ be the minimal and maximal, respectively, values that u_k can take, the control levels must be, for all $i = 0, \dots, N-1$
$$\bar{u}_i = u_{\max} + \frac{i}{N-1}(u_{\min} - u_{\max})$$
- The selection function $s(\cdot)$ must be the following
$$s(x) = \begin{cases} \bar{u}_0 = u_{\max} & \text{if } x \leq \bar{x}_1 \\ \bar{u}_i & \text{if } \bar{x}_i < x \leq \bar{x}_{i+1}, i = 1, \dots, N-2 \\ \bar{u}_{N-1} = u_{\min} & \text{if } x > \bar{x}_{N-1} \end{cases}$$
where, for all $i = 0, \dots, N$
$$\bar{x}_i = -\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max} - u_{\min})\frac{i}{N}$$
- It can be proved that the corresponding invariant set is
$$\mathbb{I} = [\bar{x}_0, \bar{x}_N]$$
In particular
    - If $x_0 \in \mathbb{I}$, then $x_k \in \mathbb{I}$ for all $k \geq 1$.
    - If $x_0 \notin \mathbb{I}$, then $|x_k| \to +\infty$ as $k \to +\infty$.

It is important to remark that the control law defined in this section is effective also when we do not operate at the rate limit, i.e., when $\frac{N_B}{h} > a\log_2 e$. However, in this case, the control law is not unique, and more performing control laws can be designed (e.g., by taking not equally-spaced quantization level – e.g., logarithmically-spaced).

F. HIGH-ORDER SYSTEMS

In this section we consider very briefly the case of high-order systems with scalar input, i.e., of type

$$\dot{x}(t) = Ax(t) + Bu(t)$$

where $x \in \mathbb{R}^n$ while $u \in \mathbb{R}$, and where the pair (A, B) is *controllable*. In this case, it can be proved that the *data rate inequality* stated before generalizes in

$$R \geq \frac{N_B}{h} \geq (Re(\lambda_1(A)) + \dots + Re(\lambda_k(A)))\log_2 e$$

where $\lambda_1(A)$, …, $\lambda_k(A)$ are the eigenvalues of A with positive real part.

APPENDIX. PROOF OF THE THEOREM

Note that the following proof is derived in the following in case $a > 0$. The case $a \leq 0$ can be proved trivially. Indeed, if $a \leq 0$, $\frac{N_B}{h} \geq 0$ is always verified, and can be proved to be sufficient for deriving a suitable control law that confines the state to lie in a given invariant set.

Necessity

In this part we prove that boundability can be obtained only if $\frac{N_B}{h} \geq a \log_2 e$, i.e., that boundability implies that $\frac{N_B}{h} \geq a \log_2 e$.

First of all, boundability implies that the sampling time h and N are compatible with the existence of a positively-invariant set $\mathbb{I}$ with respect to the closed-loop system resulting from

- the discrete-time system (ii) and
- the control law $u_k = s(x_k)$.

Such invariant set $\mathbb{I}$ is bounded since $\mathcal{U}$ is discrete. This is clear by the following facts.

- The equilibrium point, corresponding with $u_k = u_{\max}$ is $\underline{x} = -\frac{g}{f-1} u_{\max}$; when $x_k < \underline{x}$ then, irrespectively of the control law,
  $$x_{k+1} = f x_k + g u_k < f x_k + g u_{\max} = f x_k - (f-1)\underline{x} \pm x_k = x_k - (f-1)(\underline{x} - x_k) < x_k$$
  In practice, since the control input cannot take sufficiently large values, the state x_k diverges to $-\infty$.
- The equilibrium point, corresponding with $u_k = u_{\min}$ is denoted $\bar{x} = -\frac{g}{f-1} u_{\min} > \underline{x}$; when $x_k > \bar{x}$ then, irrespectively of the control law,
  $$x_{k+1} = f x_k + g u_k > f x_k + g u_{\min} = f x_k - (f-1)\bar{x} \pm x_k = x_k + (f-1)(x_k - \bar{x}) > x_k$$
  In practice, since the control input cannot take sufficiently small values, the state x_k diverges to $+\infty$.

This implies that the positively-invariant set $\mathbb{I}$ is bounded: it must have a lower bound (denoted $\hat{x}_0$) and an upper bound (denoted $\hat{x}_N$) and that, necessarily, $\bar{x}_0 \geq \underline{x}$ and $\bar{x}_N \leq \bar{x}$.

Secondly, we consider the discrete-time system (ii) and we recall that u_k can take a finite number $N = 2^{N_B}$ of values. We denote these values $\bar{u}_0, \dots, \bar{u}_{N-1}$. To define the proper selection function, we need to partition the interval $\mathbb{I} = [\bar{x}_0, \bar{x}_N]$ in N sub-intervals in each of which u takes a different value:

$$[\bar{x}_0, \bar{x}_1], (\bar{x}_1, \bar{x}_2], \dots, (\bar{x}_{N-2}, \bar{x}_{N-1}], (\bar{x}_{N-1}, \bar{x}_N]$$

for selected values of $\bar{x}_1$, …, $\bar{x}_{N-1}$. In this way we can introduce the general control law

$$u = s(x)$$

that must be defined in the following form

$$s(x) = \begin{cases} \bar{u}_0 \text{ if } x \leq \bar{x}_1 \\ \bar{u}_i \text{ if } \bar{x}_i < x \leq \bar{x}_{i+1} \quad \text{for } i = 1, \dots, N-2 \\ \bar{u}_{N-1} \text{ if } x > \bar{x}_{N-1} \end{cases}$$

In the following we will find a necessary condition for boundability. Considering the closed-loop NCS dynamics

$$x_{k+1}=\begin{cases} fx_k+g\bar{u}_0 & \text{if } x_k\le \bar{x}_1\\ fx_k+g\bar{u}_1 & \text{if } \bar{x}_1<x_k\le \bar{x}_2\\ \vdots & \\ fx_k+g\bar{u}_{N-2} & \text{if } \bar{x}_{N-2}<x_k\le \bar{x}_{N-1}\\ fx_k+g\,\bar{u}_{N-1} & \text{if } x>\bar{x}_{N-1}\end{cases}$$

boundability is obtained only if

$$\boxed{\text{for all } x_k\in\mathbb{I}=[\bar{x}_0,\bar{x}_N],\, x_{k+1}=fx_k+gs(x_k)\in\mathbb{I}}$$

This condition can be simply achieved provided that $\bar{x}_i$ (for $i=1,\dots,N-1$) and $\bar{u}_i$ (for $i=1,\dots,N-2$) are defined compatibly with the fact that, for all intervals

$$[\bar{x}_0,\bar{x}_1],(\bar{x}_1,\bar{x}_2],\dots,(\bar{x}_{N-2},\bar{x}_{N-1}],(\bar{x}_{N-1},\bar{x}_N]$$

the function $fx+gs(x)$ takes values between $\bar{x}_0$ and $\bar{x}_N$. Indeed, we must verify

$$fx+g\bar{u}_i\in[\bar{x}_0,\bar{x}_N] \text{ for all } x\in(\bar{x}_i,\bar{x}_{i+1}]$$

This is possible only if $f\bar{x}_i+g\bar{u}_i\ge\bar{x}_0$ and $f\bar{x}_{i+1}+g\bar{u}_i\le\bar{x}_N$, which is possible only if $(f\bar{x}_{i+1}+g\bar{u}_i)-(f\bar{x}_i+g\bar{u}_i)\le\bar{x}_N-\bar{x}_0$ for all $i=0,\dots,N-1$. Therefore, the necessary condition is $fl_i\le\bar{x}_N-\bar{x}_0$, where $l_i=\bar{x}_{i+1}-\bar{x}_i$. This condition must hold for all $i=0,\dots,N-1$: the least conservative case is the one where $l_i=l$ for all $i=0,\dots,N-1$, i.e., the quantization levels are equally spaced, i.e., where

$$l=\frac{\bar{x}_N-\bar{x}_0}{N}$$

The necessary condition is therefore

$$fl=f\frac{\bar{x}_N-\bar{x}_0}{N}\le\bar{x}_N-\bar{x}_0$$

This is equivalent to require that

$$\frac{f}{N}\le 1$$

i.e., that $f=e^{ah}\le N=2^{N_B}$, which is equivalent to

$$\log_2 e^{ah}=ah\log_2 e\ \le\log_2 2^{N_B}=N_B$$

and eventually to

$$\frac{N_B}{h}\ge a\log_2 e$$

Sufficiency

Sufficiency is proved by simply showing that, if $\frac{N_B}{h}\ge a\log_2 e$, there exists a control law that allows to make the system trajectories bounded. Such a control law is:

$$
s(x)=\begin{cases} u_{\max} & \text{if } x\le \bar{x}_1 \\ \bar{u}_i = u_{\max}-(u_{\max}-u_{\min})\dfrac{i}{N-1} & \text{if } \bar{x}_i < x \le \bar{x}_{i+1},\ i=1,\dots,N-2 \\ u_{\min} & \text{if } x > \bar{x}_{N-1} \end{cases}
$$

where, for all $i = 0, \dots, N$

$$
\bar{x}_i = -\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max}-u_{\min})\frac{i}{N}
$$

where, for consistency with the NECESSITY proof, note that $\frac{g}{f-1} = \frac{\frac{b}{a}(e^{ah}-1)}{e^{ah}-1} = \frac{b}{a}$. It can be proved that the set $[\bar{x}_0, \bar{x}_N]$ is invariant for the closed-loop system. For all $i = 0, \dots, N-1$, if $x_k \in (\bar{x}_i, \bar{x}_{i+1}]$, then

$$
x_{k+1} = f x_k + g s(x_k) \in (f\bar{x}_i + g\bar{u}_i, f\bar{x}_{i+1} + g\bar{u}_i]
$$

We compute that

$$
\begin{aligned}
f\bar{x}_i + g\bar{u}_i &= f\left(-\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max}-u_{\min})\frac{i}{N}\right) + g\left(u_{\max}-(u_{\max}-u_{\min})\frac{i}{N-1}\right) \\
&= e^{ah}\left(-\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max}-u_{\min})\frac{i}{N}\right) + \frac{b}{a}\left(e^{ah}-1\right)\left(u_{\max}-(u_{\max}-u_{\min})\frac{i}{N-1}\right) \\
&= \frac{b}{a}\left\{-u_{\max} + \frac{N-e^{ah}}{N(N-1)}(u_{\max}-u_{\min})i\right\}
\end{aligned}
$$

The term $\frac{N-e^{ah}}{N(N-1)}(u_{\max}-u_{\min})i \ge 0$ since $\frac{N_B}{h} \ge a\log_2 e$, and therefore $f\bar{x}_i + g\bar{u}_i \ge -\frac{b}{a}u_{\max} = \bar{x}_0$.

Secondly

$$
\begin{aligned}
f\bar{x}_{i+1} + g\bar{u}_i &= f\left(-\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max}-u_{\min})\frac{(i+1)}{N}\right) + g\left(u_{\max}-(u_{\max}-u_{\min})\frac{i}{N-1}\right) \\
&= e^{ah}\left(-\frac{b}{a}u_{\max} + \frac{b}{a}(u_{\max}-u_{\min})\frac{(i+1)}{N}\right) \\
&\quad + \frac{b}{a}\left(e^{ah}-1\right)\left(u_{\max}-(u_{\max}-u_{\min})\frac{i}{N-1}\right) \\
&= \frac{b}{a}\left\{-u_{\max} + (u_{\max}-u_{\min})\left[\frac{e^{ah}(i+1)}{N} - \frac{\left(e^{ah}-1\right)i}{N-1}\right]\right\}
\end{aligned}
$$

Note that

$$
\frac{e^{ah}(i+1)}{N} - \frac{\left(e^{ah}-1\right)i}{N-1} = \frac{i}{(N-1)}\frac{N-e^{ah}}{N} + \frac{e^{ah}(N-1)}{N(N-1)} \le \frac{N-e^{ah}}{N} + \frac{e^{ah}}{N} = 1
$$

From this it follows that

$$
f\bar{x}_{i+1} + g\bar{u}_i \le \frac{b}{a}\{-u_{\max} + (u_{\max}-u_{\min})\} = -\frac{b}{a}u_{\min} = \bar{x}_N
$$

Eventually, we obtain that

$$
x_{k+1} = f x_k + g\bar{u}_i \in [\bar{x}_0, \bar{x}_N]
$$

RELATED REFERENCE

J. Baillieul, Feedback coding for information-based control: operating near the data-rate limit. *Proceedings of the IEEE Conference on Decision and Control*, 2002, pp. 3229-3236.

EXERCISES: CONTROL NETWORKS AND QUANTIZATION

EXERCISE 1

Consider the following data, regarding some widespread control networks.

	ETHERNET	**DEVICENET**	**CONTROLNET**	**WIFI**
Link length l	2.5 km	100 m	1000 m	150 m
Propagation speed v	$2 \cdot 10^8$ m/s	$2 \cdot 10^8$ m/s	$2 \cdot 10^8$ m/s	$3 \cdot 10^8$ m/s
Bitrate R	1 Gb/s	500 kb/s	5 Mb/s	54 Mb/s
Packet size	1526 bytes	8 bytes (message) + 47 bits (overhead)	510 bytes	2346 bytes

A) For each control network, compute the time delay T_{tx} on the network channel, including the frame (transmission) time T_{frame} and the propagation time T_{prop}.
B) Assuming for simplicity that the network-related delay corresponds with T_{tx}, what is the minimum allowable sampling time $h_{\min}$?
C) Compute, for each control network, the maximum eigenvalue of the unstable (continuous-time) scalar system that can be controlled (ideally) if the minimum sampling time $h_{\min}$ were used.

SOLUTION

The parameters required in A) are computed according to the following equations.

$$T_{\text{tx}} = T_{\text{frame}} + T_{\text{prop}}, T_{\text{frame}} = \frac{N_B}{R}, T_{\text{prop}} = \frac{l}{v}$$

where N_B is the number of bits per packet.

Since a packet is assumed to arrive at destination before the sampling time expires, the minimum allowable sampling time required in B) is equal to

$$h_{\min} = T_{\text{delay}} = T_{\text{tx}}$$

Finally, the maximum eigenvalue of the unstable scalar system that can be controlled (ideally) if the minimum sampling time $h_{\min}$ were taken is computed based on the following equation

$$a_{\max} = \frac{N_B/h_{\min}}{\log_2(e)}$$

More precisely, in the previous equation, N_B should be replaced by the number of *informative* bits N_B^I transmitted in each packet (note that, however, the overhead is limited for all the considered networks). The following table summarizes the numerical results

	ETHERNET	**DEVICENET**	**CONTROLNET**	**WIFI**
$\boldsymbol{N_B}$	12208	111	4080	18768
$\boldsymbol{T_{\text{frame}}}$	$12.2\ \mu s$	$222\ \mu s$	$816\ \mu s$	$348\ \mu s$
$\boldsymbol{T_{\text{prop}}}$ (ideal)	$12.5\ \mu s$	$0.5\ \mu s$	$5\ \mu s$	$0.5\ \mu s$
$\boldsymbol{h_{\min} = T_{\text{tx}}}$	$24.7\ \mu s$	$222.5\ \mu s$	$821\ \mu s$	$348.5\ \mu s$
$\boldsymbol{a_{\max}}$	$3.42 \cdot 10^8$	$3.46 \cdot 10^5$	$3.44 \cdot 10^6$	$37.38 \cdot 10^6$

EXERCISE 2

Consider the continuous-time system

$$\dot{x} = x + u$$

We aim to control the system using a digital controller,

A) According to the theory, what is the minimum rate $\frac{N_B}{h}$ required to make the system boundable?
B) If we set the sampling time $h = \log_e(2.5) = 0.9163$ s, what is the minimum number of bits N_B required to be transmitted at each sampling instant in order to make the system boundable?
C) Set $h = \log_e(2.5) = 0.9163$ s and assume that $N_B = 1$. Using the control law $u_k = -\text{sign}(x_k)$, show the properties of the state trajectories using the graphical method.
D) Set $h = \log_e(2.5) = 0.9163$ s and assume that $N_B = 2$. Using the control law defined in case we operate at the rate limits (with $u_{\min} = -1, u_{\max} = 1$), show the properties of the state trajectories using the graphical method. In case of boundability, define the corresponding positively-invariant set.

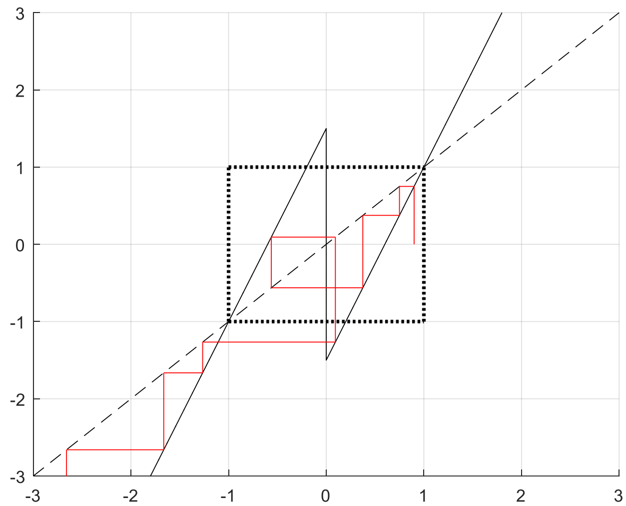
SOLUTION

A) In order to make the system boundable, the following inequality must be satisfied
$$\frac{N_B}{h} \geq a \cdot \log_2 e = 1.442 \text{ b/s}$$
B) Setting $h = \log_e(2.5) = 0.9163$ s, the minimum number of bits N_B required to be transmitted at each sampling instant in order to make the system boundable is
$$N_B \geq 1.442 \cdot 0.9163 = 1.32$$
Therefore, being N_B an integer number, the minimum number of bits required to make the system boundable is $N_B = 2$.

C) Setting $h = \log_e(2.5) = 0.9163$ s, $N_B = 1$, and using the control law $u_k = -\text{sign}(x_k)$, the dynamics of the system under control is

$$\begin{cases} x_{k+1} = 2.5x_k + 1.5 & x_k \le 0 \\ x_{k+1} = 2.5x_k - 1.5 & x_k > 0 \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, for almost any possible initial conditions, the state trajectories diverge.

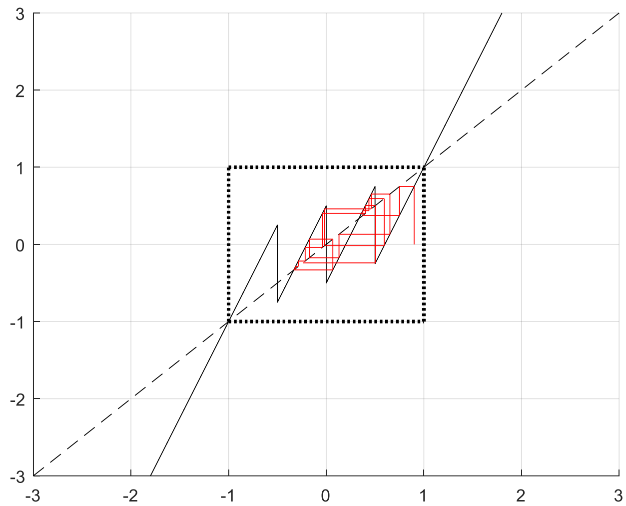
D) We set $h = \log_e(2.5) = 0.9163$ s, $N_B = 2$. The control law defined in case we operate at the rate limits is

$$u_k = \begin{cases} u_{\max} = 1 & x_k \le \frac{b}{a}\left(-u_{\max} + \frac{u_{\max} - u_{\min}}{N}\right) = -\frac{1}{2} \\ u_{\max} - \frac{u_{\max} - u_{\min}}{N-1} = \frac{1}{3} & -\frac{1}{2} < x_k \le \frac{b}{a}\left(-u_{\max} + 2\frac{u_{\max} - u_{\min}}{N}\right) = 0 \\ u_{\max} - 2\frac{u_{\max} - u_{\min}}{N-1} = -\frac{1}{3} & 0 < x_k \le \frac{b}{a}\left(-u_{\max} + 3\frac{u_{\max} - u_{\min}}{N}\right) = \frac{1}{2} \\ u_{\min} = -1 & x_k > \frac{1}{2} \end{cases}$$

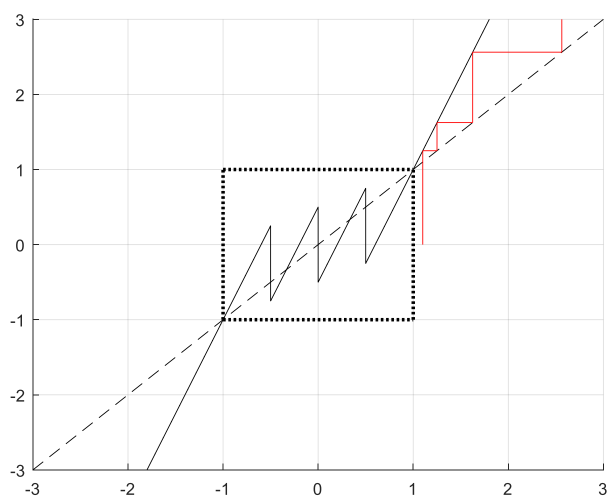
The dynamics of the system under control is

$$\begin{cases} x_{k+1} = 2.5x_k + 1.5 & x_k \le -\frac{1}{2} \\ x_{k+1} = 2.5x_k + 0.5 & -\frac{1}{2} < x_k \le 0 \\ x_{k+1} = 2.5x_k - 0.5 & 0 < x_k \le \frac{1}{2} \\ x_{k+1} = 2.5x_k - 1.5 & x_k > \frac{1}{2} \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, if $-1 \leq x_0 \leq 1$, the state trajectories remain bounded in the set $[-1,1]$. On the other hand, if $|x_0| > 1$, the trajectories diverge. This apparent in the following plot



EXERCISE 3

Consider the continuous-time system

$$\dot{x} = -x + u$$

A) According to the theory, what is the minimum rate $\frac{N_B}{h}$ required to make the system boundable?

B) Set $h = \log_e(2.5) = 0.9163$ s and assume that $N_B = 1$. Using the control law $u_k = -\text{sign}(x_k)$, show the properties of the state trajectories using the graphical method.

C) Set $h = \log_e(2.5) = 0.9163$ s and assume that $N_B = 2$. Using the control law defined in case we operate at the rate limits (with $u_{\min} = -1, u_{\max} = 1$), show the properties of the state trajectories using the graphical method. In case of boundability, define the corresponding positively-invariant set.

D) Consider the previous control law: is it possible to define an alternative control law that guarantees attractivity of the origin?

SOLUTION

A) In order to make the system boundable, the following inequality must be satisfied:

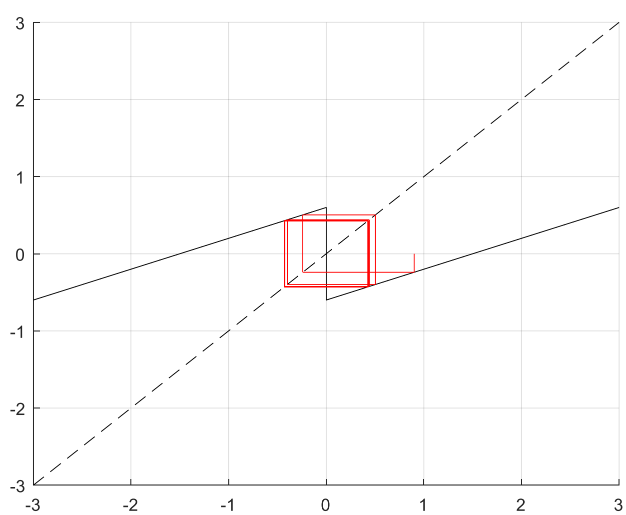
$$\frac{N_B}{h} \geq a \cdot \log_2 e$$

Therefore, since $a = -1 < 0$, the system is "stabilizable" with any rate.

B) Setting $h = \log_e(2.5) = 0.9163$ s, $N_B = 1$, and using the control law $u_k = -\text{sign}(x_k)$, the dynamics of the system under control is

$$\begin{cases} x_{k+1} = 0.4x_k + 0.6 & x_k \leq 0 \\ x_{k+1} = 0.4x_k - 0.6 & x_k > 0 \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, for any possible initial conditions, the state trajectories converge to a set, where the trajectory is bounded to lie.

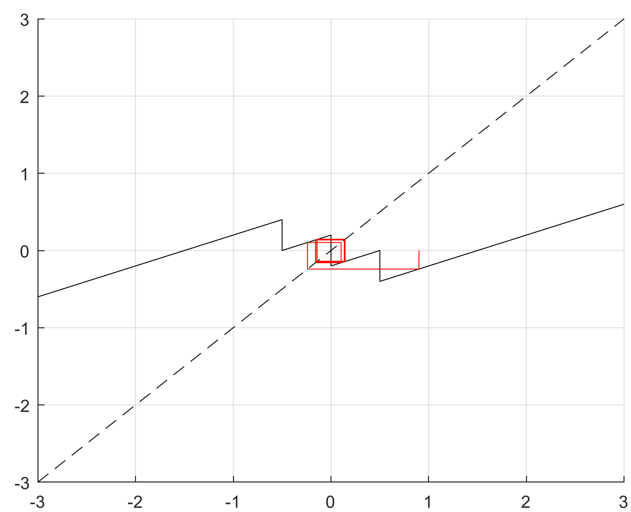
C) We set $h = \log_e(2.5) = 0.9163$ s, $N_B = 2$. The control law defined in case we operate at the rate limits (where, in the definition of the threshold values, we replace a with $|a|$) is

$$u_k = \begin{cases} u_{\max} = 1 & x_k \leq \dfrac{b}{|a|}\left(-u_{\max} + \dfrac{u_{\max} - u_{\min}}{N}\right) = -\dfrac{1}{2} \\ u_{\max} - \dfrac{u_{\max} - u_{\min}}{N-1} = \dfrac{1}{3} & -\dfrac{1}{2} < x_k \leq \dfrac{b}{|a|}\left(-u_{\max} + 2\dfrac{u_{\max} - u_{\min}}{N}\right) = 0 \\ u_{\max} - 2\dfrac{u_{\max} - u_{\min}}{N-1} = -\dfrac{1}{3} & 0 < x_k \leq \dfrac{b}{|a|}\left(-u_{\max} + 3\dfrac{u_{\max} - u_{\min}}{N}\right) = \dfrac{1}{2} \\ u_{\min} = -1 & x_k > \dfrac{1}{2} \end{cases}$$

The dynamics of the system under control is

$$\begin{cases} x_{k+1} = 0.4x_k + 0.6 & x_k \leq -\dfrac{1}{2} \\ x_{k+1} = 0.4x_k + 0.2 & -\dfrac{1}{2} < x_k \leq 0 \\ x_{k+1} = 0.4x_k - 0.2 & 0 < x_k \leq \dfrac{1}{2} \\ x_{k+1} = 0.4x_k - 0.6 & x_k > \dfrac{1}{2} \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, for any possible initial conditions, the state trajectories converge to a set, where the trajectory is bounded to lie.

D) Considering the previous control law, it is possible to notice that, despite all possible trajectories converge to a neighborhood of the origin, the point $x = 0$ is not attractive, i.e., it does not hold that $x_k \to 0$ as $k \to +\infty$. Since the system is open-loop stable (i.e.,

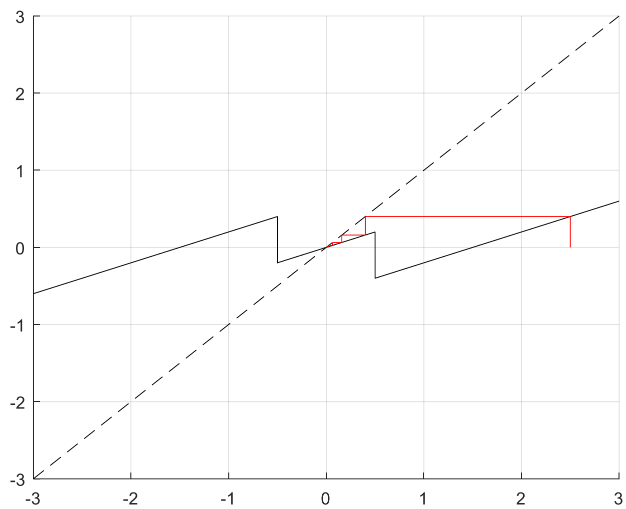
$a = -1 < 0$), it is possible to define an alternative control law that guarantees attractivity of the origin. For example, we can modify the previous control law as follows

$$u_k = \begin{cases} 1 & x_k \leq -\dfrac{1}{2} \\ 0 & -\dfrac{1}{2} < x_k \leq 0 \\ 0 & 0 < x_k \leq \dfrac{1}{2} \\ -1 & x_k > \dfrac{1}{2} \end{cases}$$

The dynamics of the system under control is

$$\begin{cases} x_{k+1} = 0.4x_k + 0.6 & x_k \leq -\dfrac{1}{2} \\ x_{k+1} = 0.4x_k & -\dfrac{1}{2} < x_k \leq \dfrac{1}{2} \\ x_{k+1} = 0.4x_k - 0.6 & x_k > \dfrac{1}{2} \end{cases}$$

We apply the graphical method to analyze the properties of the state trajectories, starting from random initial conditions.



It is possible to verify that, for any possible initial conditions, the state trajectories converge to a zero.

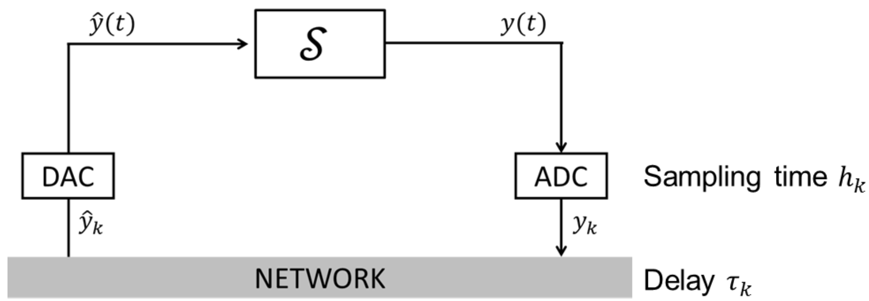
III.2: SAMPLING AND DELAYS

As discussed, delays are generated from the process of sampling and encoding of data in digital format, access time, transmission over the network (frame time + propagation time), and decodification. The overall delay may be highly variable, e.g., due to network access delay variability.

A. MAIN ASSUMPTIONS

In this chapter the following main assumptions will be done.

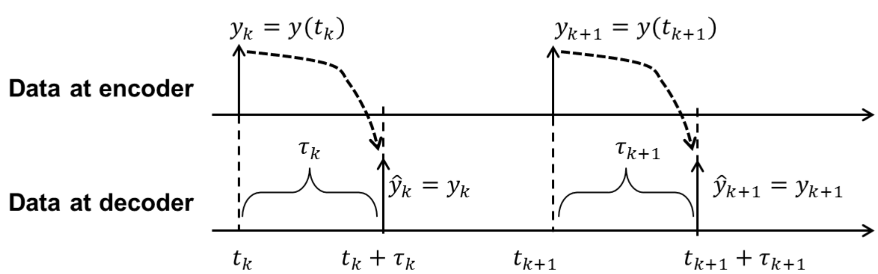
A1. **Configuration**.
We consider the NCS (*networked control system*, i.e., a feedback-controlled system over a serial communication channel) depicted below.



The dynamics of $\mathcal{S}$ is

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

The corresponding timing diagram is depicted below.



More specifically, it is here assumed that the encoder has a clock and releases data according to a possibly time-varying sampling period h_k. As soon as the data is conveyed to the decoder (with a possibly time-varying delay τ_k), it is converted to an analog signal (i.e., kept constant until new data are available) and applied to the system.

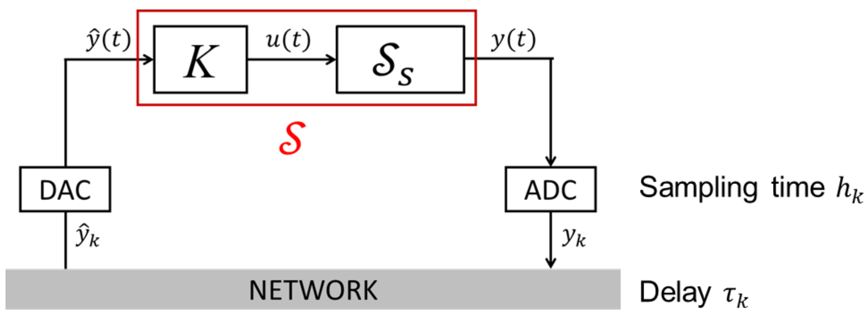
It is important to remark that this scheme is very general. In fact, we can describe, through the above scheme and with a suitable model reformulation, a number of different notable configurations (control systems), e.g.,

- Static and dynamic output-feedback controller co-located with the actuator.
- Static and dynamic output-feedback controller co-located with the sensor.
- Remote state-feedback controller.
- …

In the following we will describe how simple analytic reformulations can allow to cast the above-mentioned schemes as the reference configuration and model.

a. Static output-feedback control co-located with the actuator

The model configuration that includes a static output-feedback controller co-located with the actuator is the following.



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

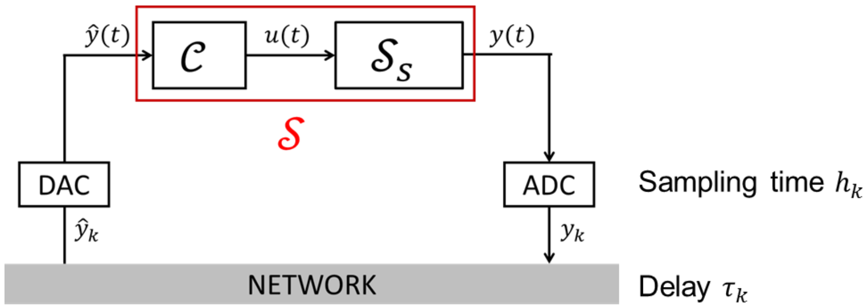
and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. Renaming $x(t) = x_S(t), A = A_S, B = B_S K, C = C_S$, the system dynamics is

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

i.e., the one of the reference configuration.

b. Dynamic output-feedback control co-located with the actuator

The model configuration that includes a dynamic output-feedback controller co-located with the actuator is the following.



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

and the dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) & = A_C x_C(t) + B_C \hat{y}(t) \\ u(t) & = C_C x_C(t) + D_C \hat{y}(t) \end{cases}$$

Overall, we write

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S(C_C x_C(t) + D_C \hat{y}(t)) \\ \dot{x}_C(t) & = A_C x_C(t) + B_C \hat{y}(t) \end{cases}$$

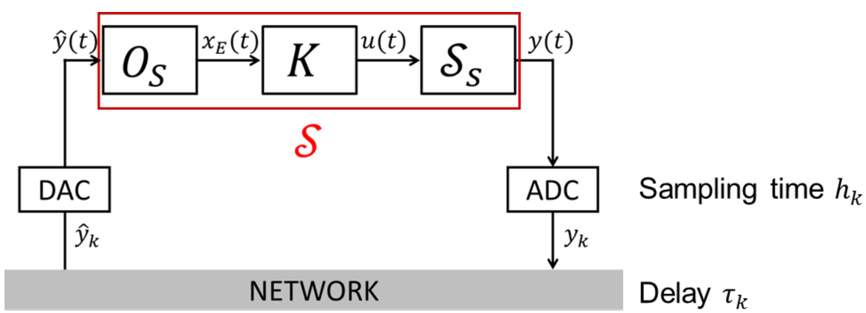
Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}^T$, $A = \begin{bmatrix} A_S & B_S C_C \\ 0 & A_C \end{bmatrix}$, $B = \begin{bmatrix} B_S D_C \\ B_C \end{bmatrix}$, $C = [C_S \quad 0]$, the system dynamics is

$$\begin{cases} \dot{x}(t) & = A x(t) + B \hat{y}(t) \\ y(t) & = C x(t) \end{cases}$$

i.e., the one of the reference configuration.

c. **Observer + static state-feedback control co-located with the actuator**

The model configuration that includes an observer and a static state-feedback controller co-located with the actuator is the following.



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

the dynamics of the obsever O_S is, e.g., $\dot{x}_E(t) = A_S x_E(t) + B_S\, u(t) + L_E\big(\hat{y}(t) - C_S x_E(t)\big)$, and the control action is set to $u(t) = K x_E(t)$. Overall, we write

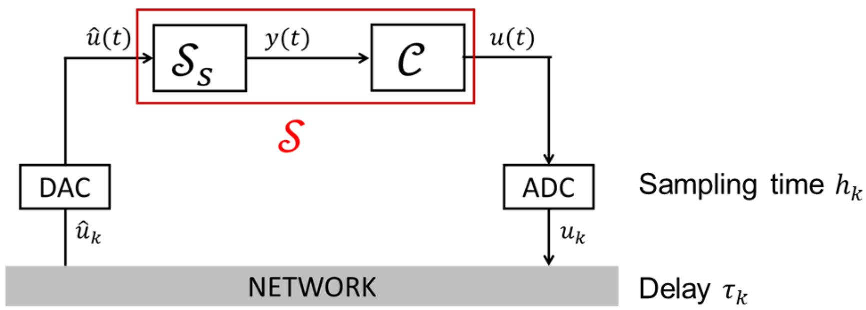
$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S K x_E(t) \\ \dot{x}_E(t) & = A_S x_E(t) + B_S K x_E(t) + L_E\big(\hat{y}(t) - C_S x_E(t)\big) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_E(t) \end{bmatrix}^T, A = \begin{bmatrix} A_S & B_S K \\ 0 & A_S + B_S K - L_E C_S \end{bmatrix}, B = \begin{bmatrix} 0 \\ L_E \end{bmatrix}, C = [C_S \quad 0]$, we obtain

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

d. Output-feedback control co-located with the sensor

The model configuration that includes an output-feedback controller co-located with the sensor is the following.



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

The dynamics of the dynamic output-feedback controller $\mathcal{C}$ is

$$\begin{cases} \dot{x}_C(t) & = A_C x_C(t) + B_C\, y(t) \\ u(t) & = C_C x_C(t) + D_C y(t) \end{cases}$$

Overall, we write

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S \hat{u}(t) \\ \dot{x}_C(t) & = A_C x_C(t) + B_C C_S x_S(t) \end{cases}$$

Renaming $x(t) = \begin{bmatrix} x_S(t) \\ x_C(t) \end{bmatrix}^T, A = \begin{bmatrix} A_S & 0 \\ B_C C_S & A_C \end{bmatrix}, B = \begin{bmatrix} B_S \\ 0 \end{bmatrix}, C = [D_C C_S \quad C_C]$, we obtain

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{u}(t) \\ u(t) & = Cx(t) \end{cases}$$

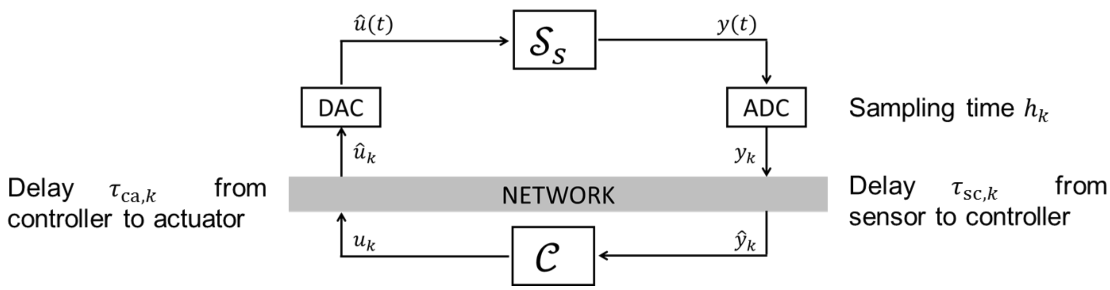
The original scheme is recovered by simply replacing the signal names as follows $y(t) \to u(t)$ and $\hat{y}(t) \to \hat{u}(t)$

Note that

- Setting $C_C = 0$, $D_C = K$ this is the static output-feedback scheme.
- With suitable straightforward definition of C_C and D_C also the scheme with the observer is included.

e. Output-feedback remote control

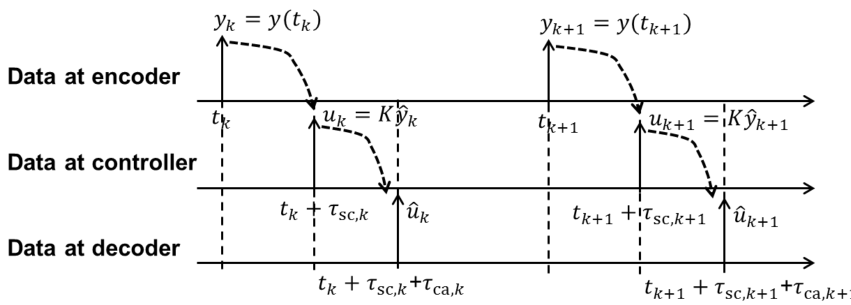
The model configuration that includes an output-feedback remote controller is the following.



The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

An output-feedback static controller is used, i.e., $u_k = K\hat{y}_k$. The corresponding timing diagram is shown below.



More specifically, it is here assumed that the encoder has a clock and releases data according to a possibly time-varying sampling period h_k. As soon as the data is conveyed to the controller (with a possibly time-varying delay $\tau_{\mathrm{sc},k}$), the control action $u_k = K\hat{y}_k$ is computed and instantaneously transmitted. As soon as the data $\hat{u}_k$ is conveyed to the decoder (with a possibly additional time-varying delay $\tau_{\mathrm{ca},k}$), it is converted to an analog signal (i.e., kept constant until new data are available) and applied to the system.

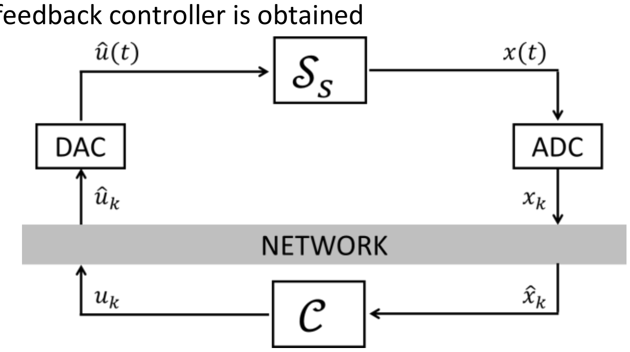
Renaming $x(t) = x_S(t), A = A_S, B = B_S K, C = C_S$, this scheme is equivalent to the one of the general configuration with dynamics

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

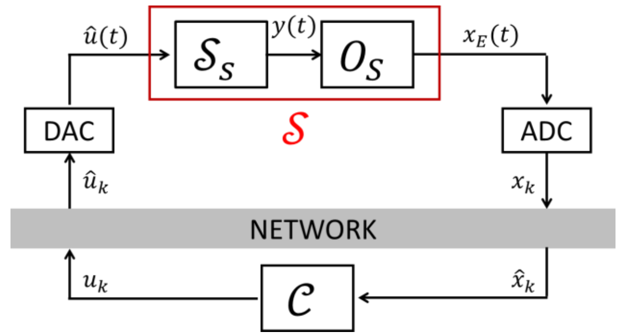
provided that the (unique) delay is set to $\tau_k = \tau_{\mathrm{sc},k} + \tau_{\mathrm{ca},k}$.

Note also that

- If $C_S = I$ a static state-feedback controller is obtained



- If $\mathcal{S}$ includes both system and state observer, also the case of dynamic output-feedback control can be considered



A2. **Assumptions on the communication channel**.
The number of bits contained in each packet is sufficiently large, so that quantization errors can be ignored. Thanks to this assumption, the control network can be viewed as a channel that can carry real numbers without distortion, but with possible delays and message dropouts (see also the next Chapter II.3).

A3. **Sampling time and delay**.
Let h_k be the possibly time varying sampling time. The sampling instants are here denoted t_k, being $k \in \mathbb{N}$ the time step. Therefore, the following relationship is verified: $h_k = t_{k+1} - t_k$.
Letting τ_k be the possibly time-varying delay, for each time step k, it holds that $\tau_k < h_k$, i.e., the network-induced delay is smaller than the sampling interval.

B. NETWORKED CONTROL SYSTEM DYNAMICS

In this section we describe how the dynamics of the NCS can be reformulated as the one of a discrete-time system. We start by considering the dynamics of system $\mathcal{S}$

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

where, in view of the sampling process and of the presence of delays,

$$\hat{y}(t) = \begin{cases} \hat{y}_{k-1} = y_{k-1} = y(t_{k-1}) & t \in [t_k, t_k + \tau_k) \\ \hat{y}_k = y_k = y(t_k) & t \in [t_k + \tau_k, t_{k+1}) \end{cases}$$

Letting $x_k = x(t_k)$ and using the Lagrange equation

$$x_{k+1} = x(t_k + h_k) = e^{Ah_k}x(t_k) + \int_{t_k}^{t_k+h_k} e^{A(t_k+h_k-\nu)}B\hat{y}(\nu)d\nu$$

We compute that

$$\begin{aligned} \int_{t_k}^{t_k+h_k} e^{A(t_k+h_k-\nu)}B\hat{y}(\nu)d\nu &= \int_{t_k}^{t_k+\tau_k} e^{A(t_k+h_k-\nu)}B\hat{y}_{k-1}d\nu + \int_{t_k+\tau_k}^{t_k+h_k} e^{A(t_k+h_k-\nu)}B\hat{y}_k d\nu \\ &= e^{A(h_k-\tau_k)}\int_{t_k}^{t_k+\tau_k} e^{A(t_k+\tau_k-\nu)}d\nu\, B\hat{y}_{k-1} + \int_{t_k+\tau_k}^{t_k+h_k} e^{A(t_k+h_k-\nu)}d\nu\, B\hat{y}_k \\ &= e^{A(h_k-\tau_k)}\int_0^{\tau_k} e^{A\nu}d\nu\, B\hat{y}_{k-1} + \int_0^{h_k-\tau_k} e^{A\nu}d\nu\, B\hat{y}_k \end{aligned}$$

Defining, for a generic time interval ξ, $\Gamma(\xi) = \int_0^{\xi} e^{A\nu}d\nu$, we compute that

$$x_{k+1} = e^{Ah_k}x_k + e^{A(h_k-\tau_k)}\Gamma(\tau_k)B\hat{y}_{k-1} + \Gamma(h_k-\tau_k)B\hat{y}_k$$

Recalling that $\hat{y}_k = Cx_k$ we write

$$z_{k+1} = \begin{bmatrix} x_{k+1} \\ \hat{y}_k \end{bmatrix} = \begin{bmatrix} e^{Ah_k} + \Gamma(h_k-\tau_k)BC & e^{A(h_k-\tau_k)}\Gamma(\tau_k)B \\ C & 0 \end{bmatrix}\begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}$$

Defining the augmented state vector

$$z_k = \begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}$$

and the matrix

$$\Phi(h_k, \tau_k) = \begin{bmatrix} e^{Ah_k} + \Gamma(h_k-\tau_k)BC & e^{A(h_k-\tau_k)}\Gamma(\tau_k)B \\ C & 0 \end{bmatrix}$$

the NCS dynamics is

$$z_{k+1} = \Phi(h_k, \tau_k)z_k$$

Remark

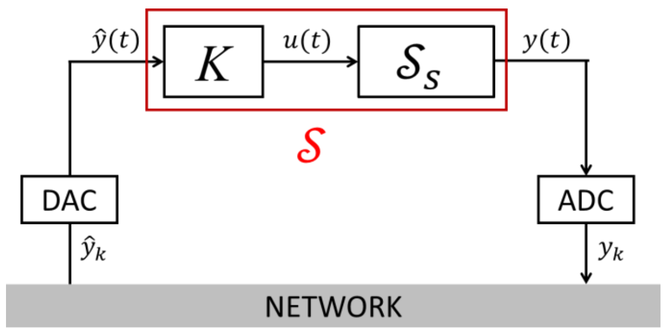
In absence of delays and in case the sampling time is constant we have $\tau_k = 0$ and $h_k = h$ for all $k \geq 0$. In this case $\Gamma(\tau_k) = 0$ and

$$\Phi(h_k, \tau_k) = \Phi(h, 0) = \begin{bmatrix} e^{Ah} + \Gamma(h)BC & 0 \\ C & 0 \end{bmatrix}$$

is block-triangular. Therefore, the NCS is asymptotically stable if and only if $e^{Ah} + \Gamma(h)BC = F + GH$ is asymptotically stable, where F and G are the matrices of the discretized system. As expected, in absence of delays, only the effect of sampling matters.

Examples

In case system $\mathcal{S}$ consists of system+output feedback controller co-located with the actuator, the scheme is the following



where the dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. As discussed previously, the control scheme in the above figure can be cast in the class of the reference configuration provided that we rename $x(t) = x_S(t), A = A_S, B = B_S K, C = C_S$. So, we rewrite

$$\Phi(h, \tau) = \begin{bmatrix} e^{A_S h} + \Gamma(h-\tau)B_S K C_S & e^{A_S(h-\tau)}\Gamma(\tau)B_S K \\ C_S & 0 \end{bmatrix}, \Gamma(\xi) = \int_0^{\xi} e^{A_S v} dv$$

Case 1: scalar system with $A_S = 0$ (integrator).

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = u(t) \\ y(t) & = x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h, \tau) = \begin{bmatrix} 1 + K(h-\tau) & K\tau \\ 1 & 0 \end{bmatrix}$$

Case 2: scalar system with $A_S = a \neq 0$ ($B_S = 1$ for simplicity).

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = a x_S(t) + u(t) \\ y(t) & = x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h,\tau) = \begin{bmatrix} e^{ah} + \frac{K}{a}\left(e^{a(h-\tau)} - 1\right) & \frac{K}{a} e^{a(h-\tau)}\left(e^{a\tau} - 1\right) \\ 1 & 0 \end{bmatrix}$$

Case 3: high-order system with $\det(A_S) \neq 0$.

The dynamics of the system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\Phi(h,\tau) = \begin{bmatrix} e^{A_S h} + A_S^{-1}\left(e^{A_S(h-\tau)} - I\right) B_S K C_S & e^{A_S(h-\tau)} A_S^{-1}\left(e^{A_S \tau} - I\right) B_S K \\ C_S & 0 \end{bmatrix}$$

C. STABILITY OF THE NCS

1) The case of constant sampling period and delay

In this case the following formal assumption will be assumed to hold.

Assumption 1

For all $k \in \mathbb{N}$, $h_k = h$ and $\tau_k = \tau$, where $0 < \tau < h$.

Notably, some network protocols allow to verify Assumption 1, for example

- CAN, where the value of τ_k is constant, at least for high-priority messages.
- Token ring or token bus, and in general protocols based on time-division multiplexing.

In general, time delays may be equalized by introducing a buffer (endowed with a clock) at the receiver. In the buffer, packets can be held so that all packets appear to have the same delay from the perspective of the NCS. The downside of this idea is that the *artificial* common delay must be defined as the worst-case delay that the network can introduce, and results conservative.

Under Assumption 1, the discrete-time model describing the NCS dynamics is

$$z_{k+1} = \Phi(h,\tau) z_k$$

The model is linear time invariant, and therefore its stability properties can be established in a straightforward way by the following theorem.

Theorem 1

Under Assumption 1, the NCS is (exponentially) asymptotically stable if and only if $\Phi(h,\tau)$ is Schur stable.

Examples

Let us now consider the examples introduced in the previous section.

Case 1: scalar system with $A_S = 0$ (integrator).

In case

$$\Phi(h,\tau) = \begin{bmatrix} 1 + K(h-\tau) & K\tau \\ 1 & 0 \end{bmatrix}$$

the characteristic polynomial is

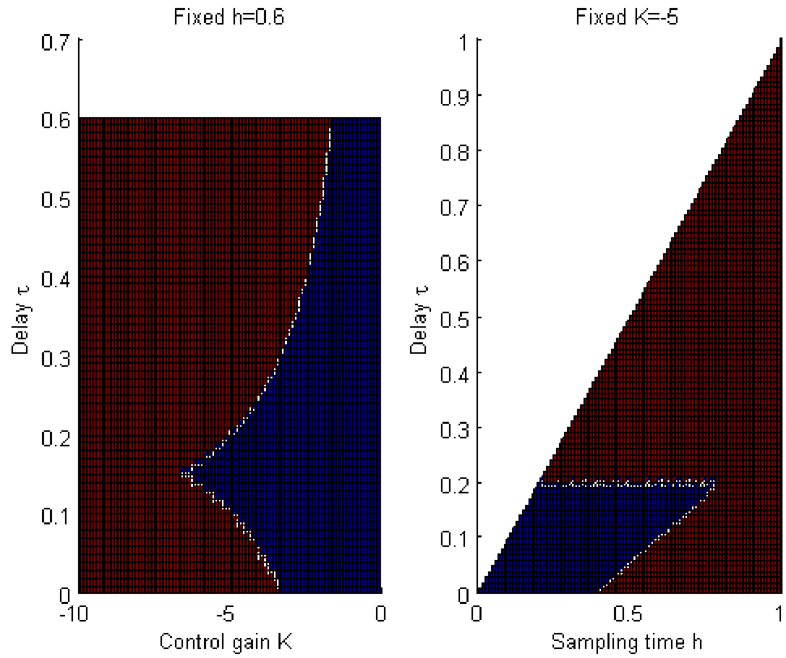
$$p_\Phi(\lambda) = \lambda^2 - (1 + (h-\tau)K)\lambda - K\tau$$

Applying the conditions derived from the application of the Jury criterion to systems of order 2, the networked control system is asymptotically stable if and only if

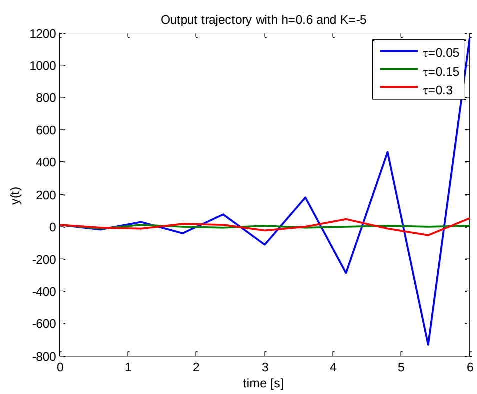
$$\begin{cases} -1 < K\tau < 1 \\ Kh < 0 \\ \left(\tau - \dfrac{h}{2}\right)K < 1 \end{cases}$$

- In case $\tau > \frac{h}{2}$ (large delay), this is possible if and only if $-1/\tau < K < 0$.
  Note that this condition is more conservative than the one obtained in case only sampling is present, i.e., $-2 < Kh < 0$.
- In case $\tau < \frac{h}{2}$ ("small" delay), this is possible if and only if $-1/\tau < K < 0$ and $-1 < K(\frac{h}{2} - \tau) < 0$. Note that this condition is less conservative than the one obtained in case only sampling is present, i.e., $-2 < Kh < 0$.

In the following figures we show the result of the stability analysis (red region: instability, blue region: asymptotic stability) on $\Phi(h,\tau)$ when parameters h,τ and gain K vary. In particular, in the left panel we set $h = 0.6$ and we vary τ and K, while on the right panel we set $K = -5$ and we vary h and τ.

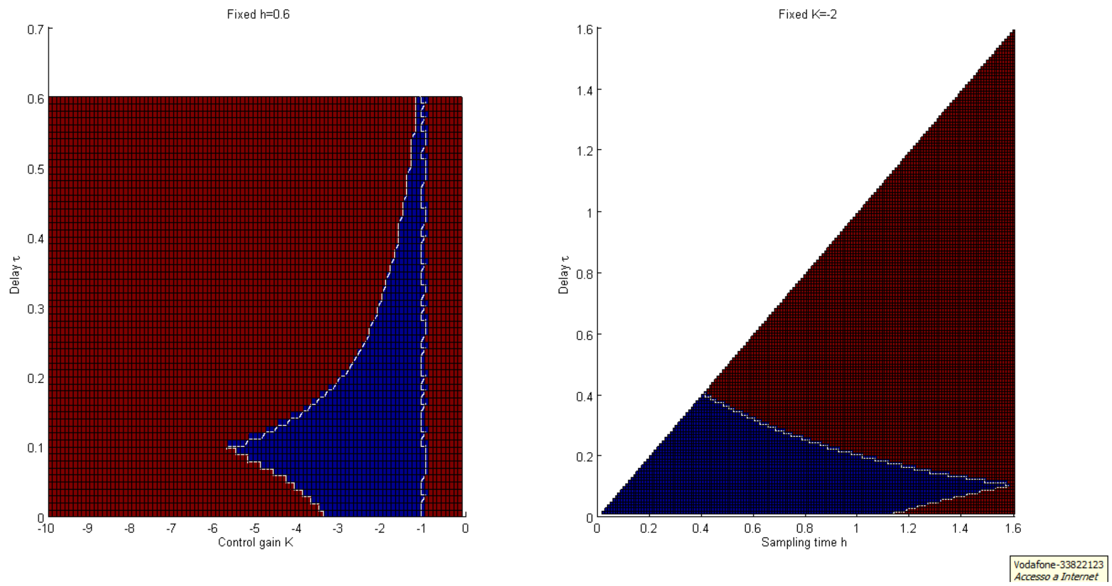


In the following figure we show some simulations validating the previous analysis. Indeed, we set $h = 0.6$ s and $K = -5$ and we depict three trajectories, obtained setting $\tau = 0.05, 0.15, 0.2$.

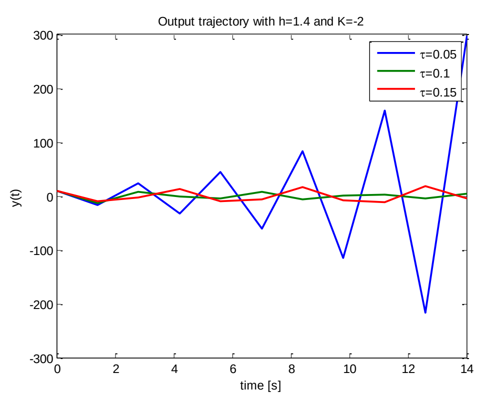


Case 2a: scalar system with $A_S = a = 1$ ($B_S = 1$).

In the following figures we show the result of the stability analysis (red region: instability, blue region: asymptotic stability) on $\Phi(h, \tau)$ when parameters h, τ and gain K vary. In particular, in the left panel we set $h = 0.6$ and we vary τ and K, while on the right panel we set $K = -2$ and we vary h and τ.

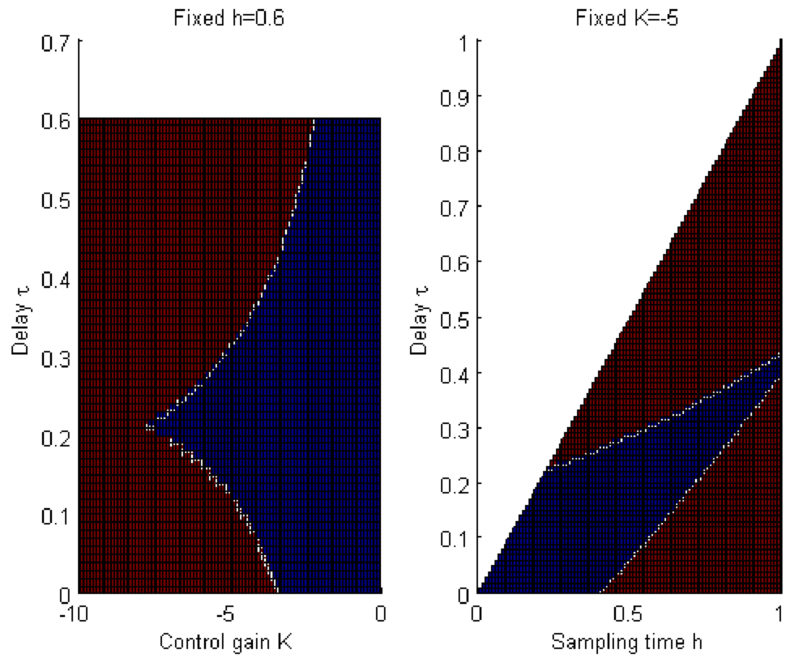


In the following figure we show some simulations validating the previous analysis. Indeed, we set $h = 1.4$ s and $K = -2$ and we depict three trajectories, obtained setting $\tau = 0.05, 0.1, 0.15$.

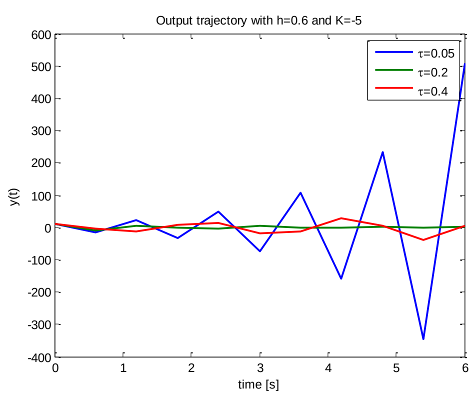


Case 2b: scalar system with $A_S = a = -1$ ($B_S = 1$).

In the following figures we show the result of the stability analysis (red region: instability, blue region: asymptotic stability) on $\Phi(h, \tau)$ when parameters h, τ and gain K vary. In particular, in the left panel we set $h = 0.6$ and we vary τ and K, while on the right panel we set $K = -5$ and we vary h and τ.



In the following figure we show some simulations validating the previous analysis. Indeed, we set $h = 0.6$ s and $K = -5$ and we depict three trajectories, obtained setting $\tau = 0.05, 0.2, 0.4$.

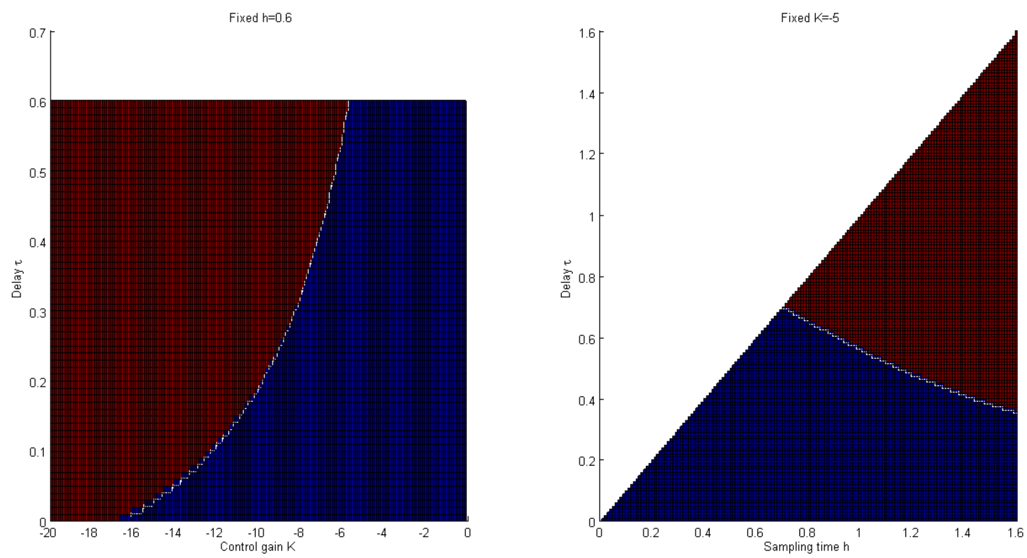


Case 3: high-order system with $\det(A_S) \neq 0$.

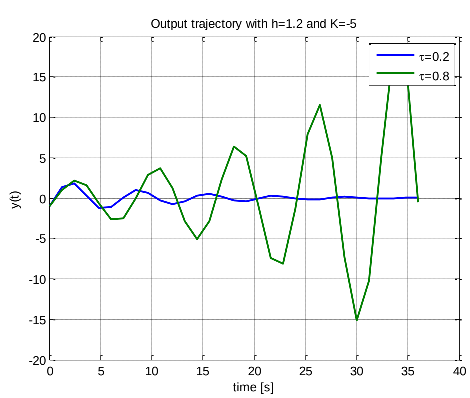
We consider the case

$$A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1, 1]$$

In the following figures we show the result of the stability analysis (red region: instability, blue region: asymptotic stability) on $\Phi(h, \tau)$ when parameters h, τ and gain K vary. In particular, in the left panel we set $h = 0.6$ and we vary τ and K, while on the right panel we set $K = -5$ and we vary h and τ.



In the following figure we show some simulations validating the previous analysis. Indeed, we set $h = 1.2$ s and $K = -5$ and we depict three trajectories, obtained setting $\tau = 0.2, 0.8$.



2) The case of time-varying sampling period and delay

In the more general case in which the sampling time h_k and the delay τ_k are time-varying, the system can be regarded as a parameter-varying linear one and the following result holds.

Theorem 2

Assume that there exist constants $h_{\min}, h_{\max}, \tau_{\min}, \tau_{\max}$ such that, for all $k \in \mathbb{N}$,

$$0 \leq h_{\min} \leq h_k \leq h_{\max}$$

$$0 \leq \tau_{\min} \leq \tau_k \leq \tau_{\max}$$

Then, the NCS is exponentially stable **if** there exists a symmetric matrix $P > 0$ such that, for all $h \in [h_{\min}, h_{\max}]$ and for all $\tau \in [\tau_{\min}, \tau_{\max}]$

$$\Phi(h,\tau)^T P \Phi(h,\tau) - P < 0 \qquad (i)$$

Remark that

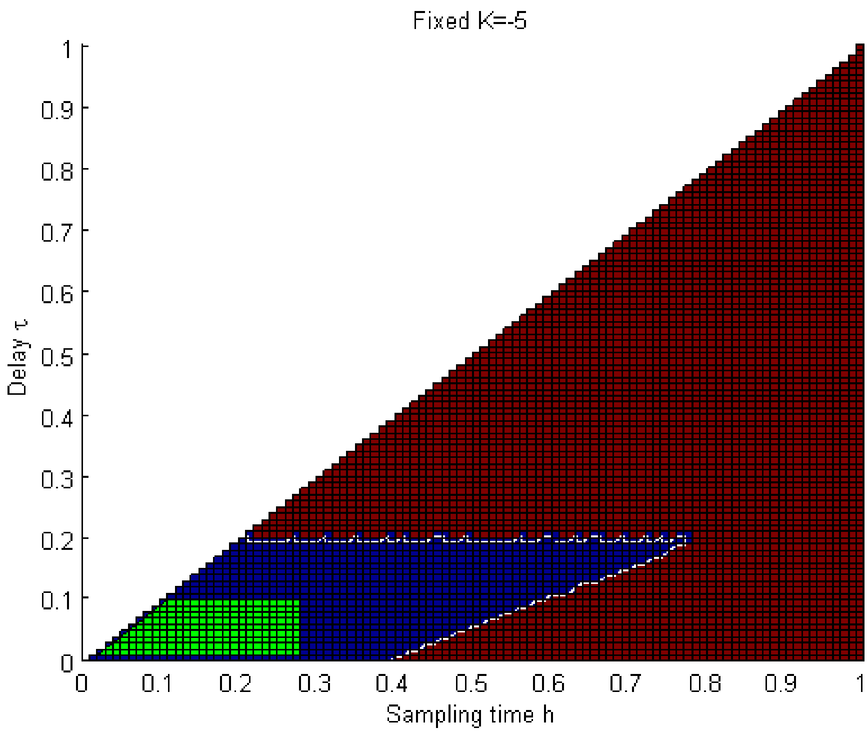
- It is in general not simple to find – numerically - a matrix P that satisfies the LMI (i) for all admissible values of h and τ. Testing the existence of such a matrix can be done for a finite number of h and τ (*gridding* method), leading to the solution to an LMI feasibility problem that includes a finite set of linear matrix inequalities (i.e., one for each pair of tested values of h and τ).
- Note that the set of LMIs (i) guarantees the existence of a *common Lyapunov function* (i.e., that, for all admissible values of h and τ, the system has the same Lyapunov function), which is however only *sufficient* for guaranteeing the asymptotic stability of the time-varying system, but it is *not necessary*.

Examples

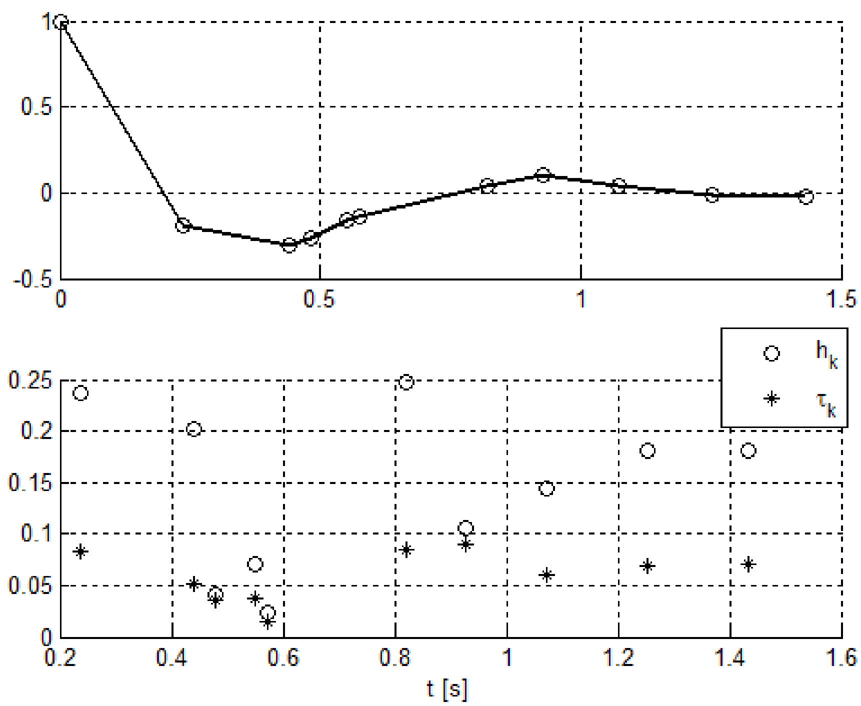
Let us now consider the examples introduced in the previous section.

Case 1: scalar system with $A_S = 0$ (integrator).

In the following figure we show the result of the stability analysis of the NCS when $K = -5$. In particular, in green the region of values (h,τ) where the set of LMIs (i) is verified is indicated. As a comparison, we show (in blue) the region of values (h,τ) where $\Phi(h,\tau)$ is Schur stable (i.e., the NCS is stable under Assumption 1).

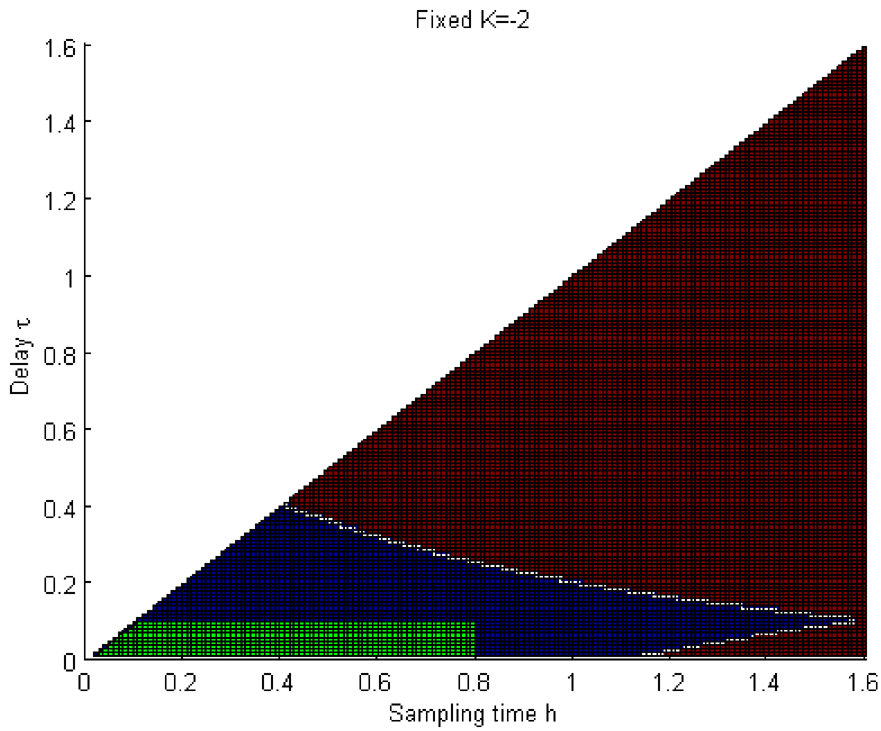


In the following figure (upper panel) we show a simulation result, taking $K = -5$ and with (h_k, τ_k) randomly-varying in the range indicated in green in the previous figure. In the lower panel we show the corresponding randomly-varying values of h_k and τ_k.

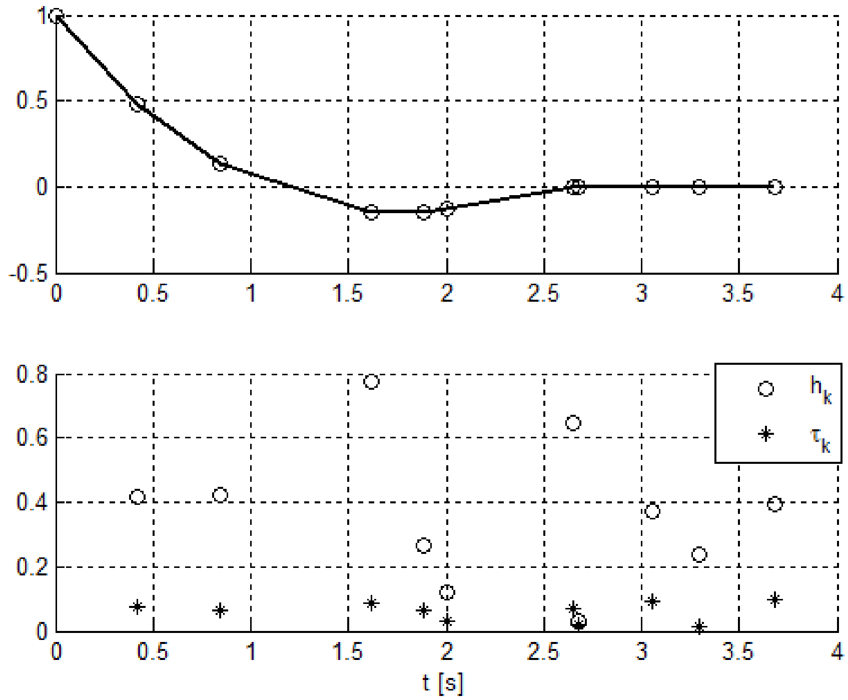


Case 2a: scalar system with $A_S = a = 1$ ($B_S = 1$).

In the following figure we show the result of the stability analysis of the NCS when $K = -2$. In particular, in green the region of values (h, τ) where the set of LMIs (i) is verified is indicated. As a comparison, we show (in blue) the region of values (h, τ) where $\Phi(h, \tau)$ is Schur stable (i.e., the NCS is stable under Assumption 1).

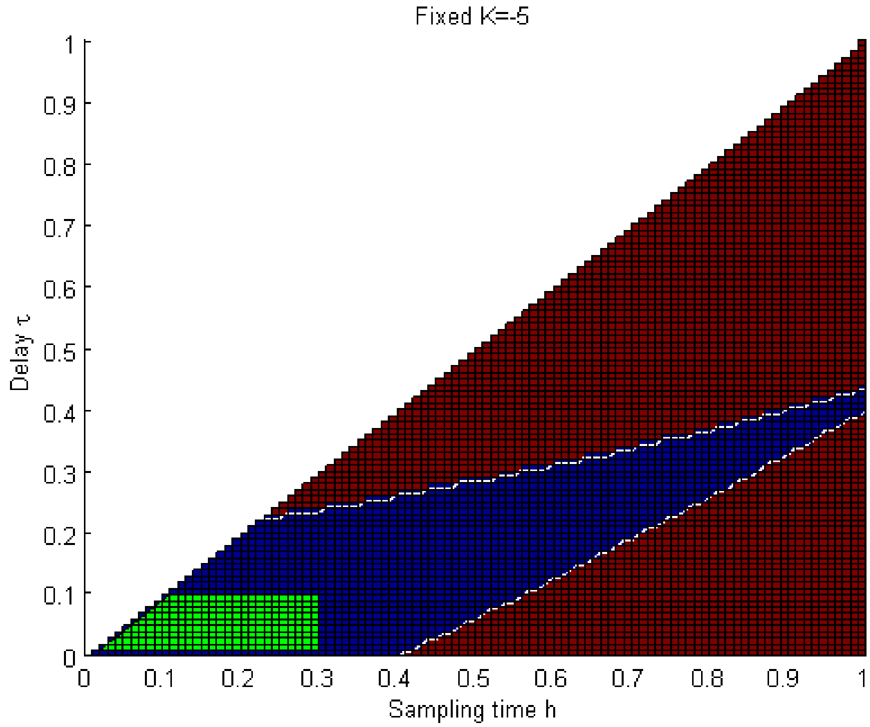


In the following figure (upper panel) we show a simulation result, taking $K = -2$ and with (h_k, τ_k) randomly-varying in the range indicated in green in the previous figure. In the lower panel we show the corresponding randomly-varying values of h_k and τ_k.

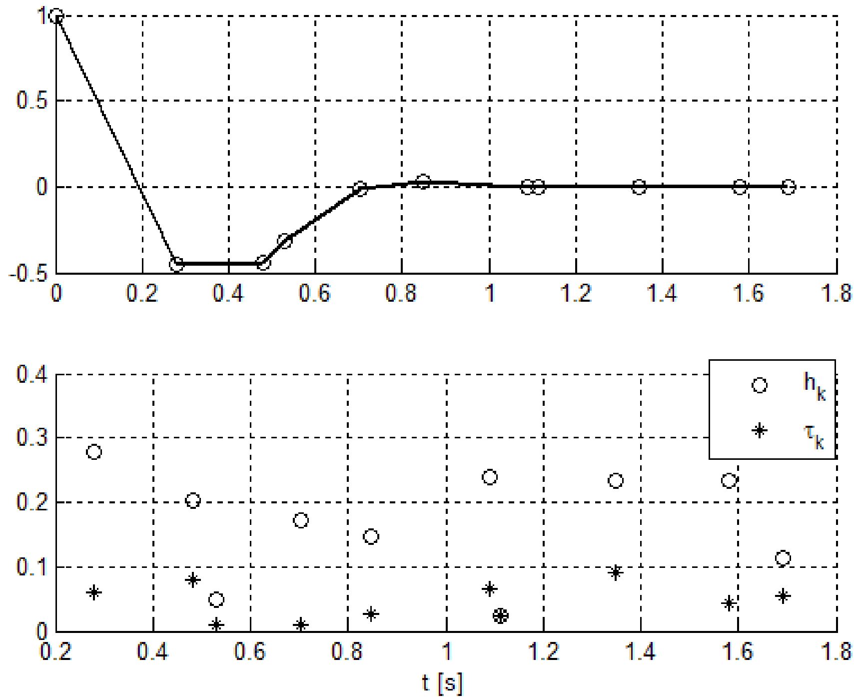


Case 2b: scalar system with $A_S = a = -1$ ($B_S = 1$).

In the following figure we show the result of the stability analysis of the NCS when $K = -5$. In particular, in green the region of values (h, τ) where the set of LMIs (i) is verified is indicated. As a comparison, we show (in blue) the region of values (h, τ) where $\Phi(h, \tau)$ is Schur stable (i.e., the NCS is stable under Assumption 1).

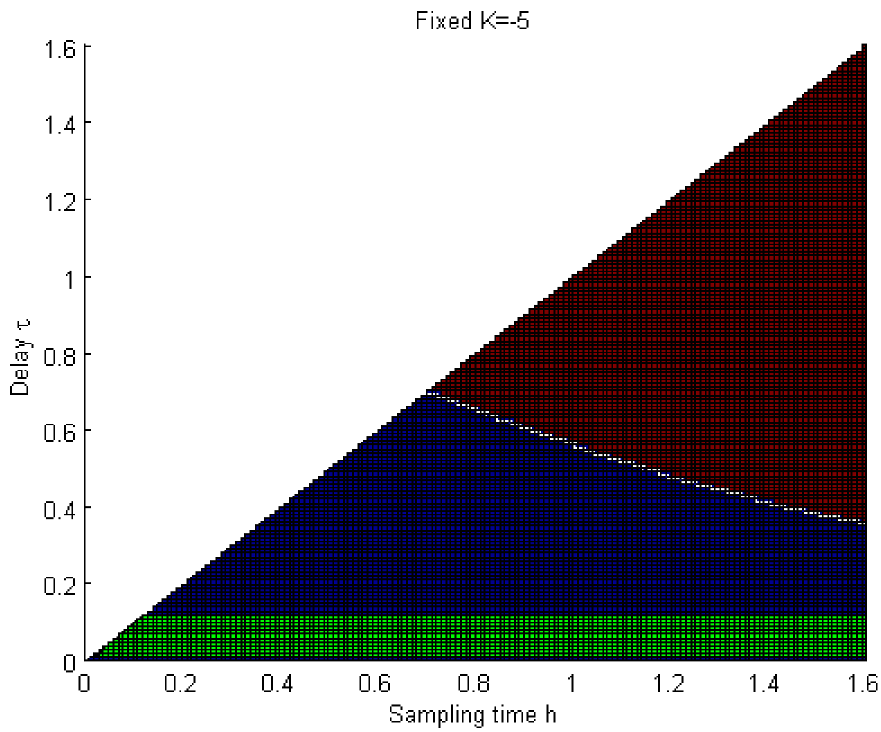


In the following figure (upper panel) we show a simulation result, taking $K = -5$ and with (h_k, τ_k) randomly-varying in the range indicated in green in the previous figure. In the lower panel we show the corresponding randomly-varying values of h_k and τ_k.

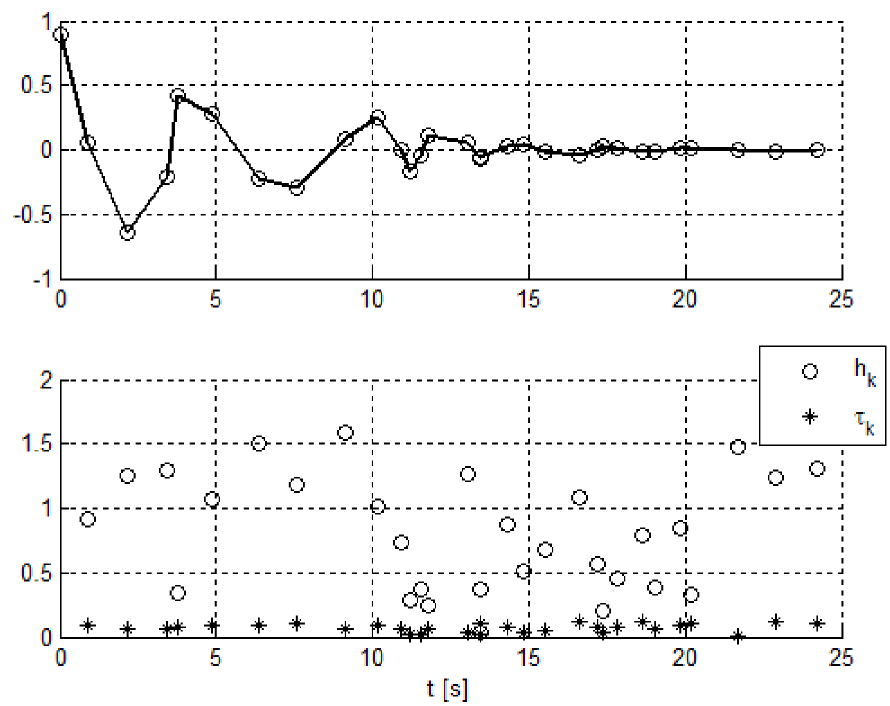


Case 3: high-order system with $\det(A_S) \neq 0$**.**

In the following figure we show the result of the stability analysis of the NCS when $K = -5$. In particular, in green the region of values (h, τ) where the set of LMIs (i) is verified is indicated. As a comparison, we show (in blue) the region of values (h, τ) where $\Phi(h, \tau)$ is Schur stable (i.e., the NCS is stable under Assumption 1).



In the following figure (upper panel) we show a simulation result, taking $K = -5$ and with (h_k, τ_k) randomly-varying in the range indicated in green in the previous figure. In the lower panel we show the corresponding randomly-varying values of h_k and τ_k.



D. MAXIMAL ALLOWABLE TRANSFER INTERVAL (MATI)

For given values of $\tau_{\min}$ and $\tau_{\max}$, we define the *maximal allowable transfer interval* (MATI) $\bar{h}$ as the largest value of $h \in \mathbb{R}$ such that, if $h \le \bar{h}$, then the NCS is asymptotically stable for all $\tau_k \in [\tau_{\min}, \tau_{\max}]$.

In other words, the successive sensor messages must be separated by at most $\bar{h}$ s.

The knowledge of the MATI allows to set the sampler in order to

- Preserve stability.
- Avoid too small sampling intervals, which would increase the network load.

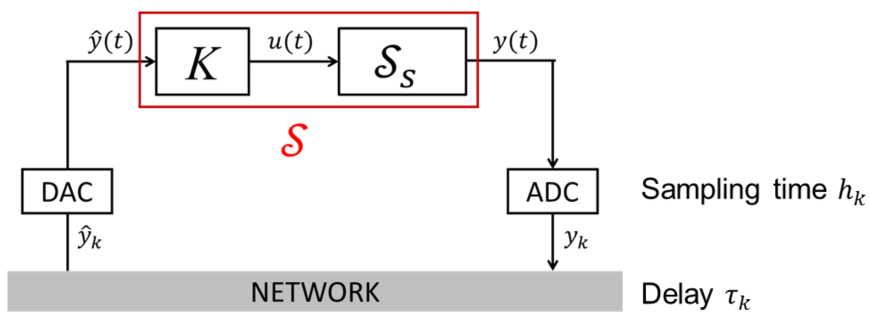
Under Assumption 1 (constant sampling time and delay), the MATI can be easily computed, using a numerical approach similar to the one used in the examples introduced in Section C.1. If Assumption 1 does not hold, we can find an estimate of $\bar{h}$ by finding the maximal value of $h_{\max}$ that allows to solve the LMI set (i) for all $h \in [h_{\min}, h_{\max}]$ and for all $\tau \in [\tau_{\min}, \tau_{\max}]$.

E. COMPENSATION OF DELAYS

In case the measurements are time-stamped, we can adopt some strategies to compensate for the transmission delays. Here we consider two schemes considered so far: the case where controller and actuator are co-located and the case where the controller is remote.

1) Controller co-located with the actuator

In this case we consider the general scheme, discussed previously:



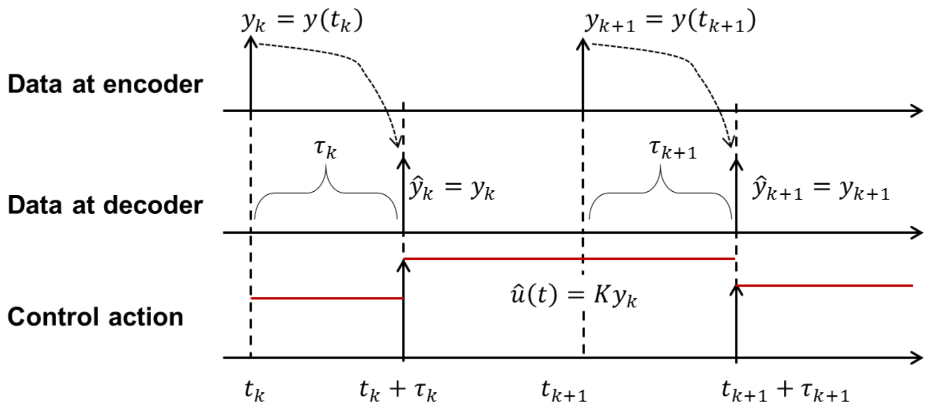
As already introduced, the dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$.

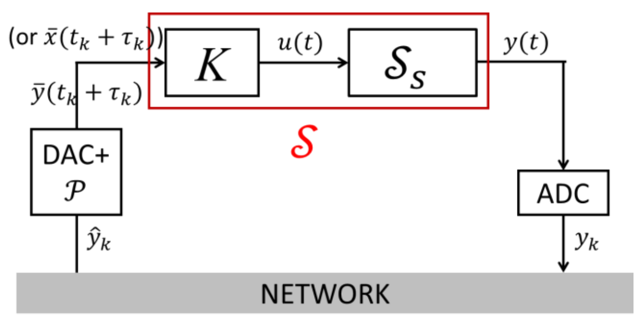
In case measurements are time-stamped, it is possible to precisely estimate, at the receiver/decoder level, the network-induced delay τ_k at each time instant. We can use this information to devise a control scheme that is able to compensate for these delays using suitable state/output predictors.

- In case time stamps are not used, the following time diagram describes the implementation of the control scheme.

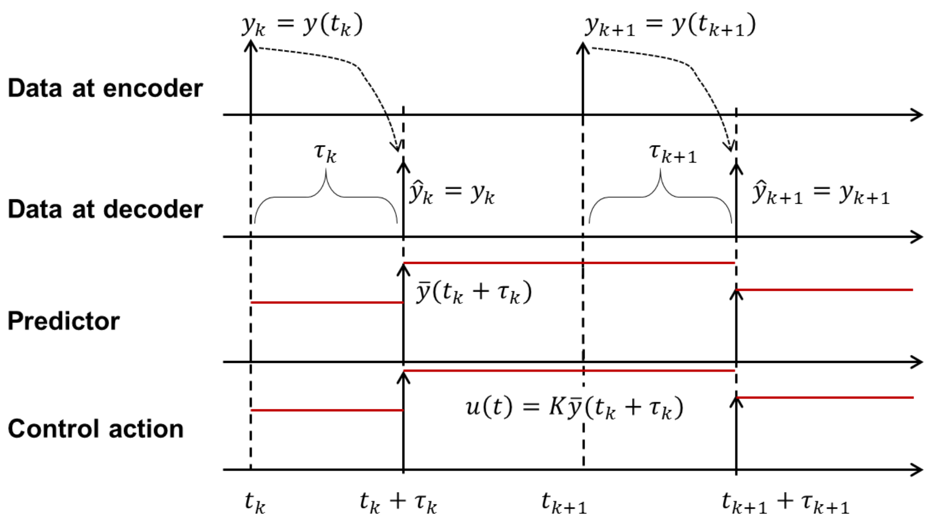


In particular, the control action $u(t) = Ky(t_k)$ is applied to the system in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$ – it is kept constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s -, while in "nominal" digital control the control action $u(t) = Ky(t_k)$ should be applied in the interval $[t_k, t_{k+1})$ and kept constant for h_k s.

- In case time stamps are used and the time delay is known, we can introduce an output/state predictor (embedded, for example, in the decoder), as in the following figure.



The idea is sketched in the time diagram below.



In particular, the control action $u(t) = K\bar{y}(t_k + \tau_k) \cong Ky(t_k + \tau_k)$ is applied to the system in the interval $[t_k + \tau_k, t_{k+1} + \tau_{k+1})$ - it is kept constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s. Essentially, this is equivalent to applying the **proper digital control action** at time instants $t_k^* = t_k + \tau_k$ and with sampling time h_k^*. Note also that, if the delay is constant, $h_k^* = h_k$, i.e., the actual sampling time is equal to the nominal one.

The *predictor* can be devised by implementing a digital observer of the system state $x_k = x(t_k)$ of the type

$\bar{x}_{k|k-1} = e^{A_S h_{k-1}}\bar{x}_{k-1} + e^{A_S(h_{k-1}-\tau_{k-1})}\Gamma(\tau_{k-1})B_S\hat{u}_{k-2} + \Gamma(h_{k-1}-\tau_{k-1})B_S\hat{u}_{k-1}$

$\bar{x}_k = \bar{x}_{k|k-1} + L(y_k - C_S\bar{x}_{k|k-1})$

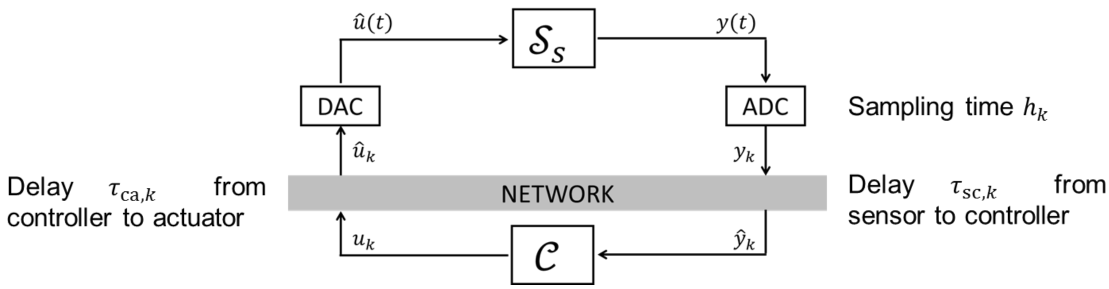
and a predictor of the type

$\bar{x}(t_k+\tau_k) = e^{A_S\tau_k}\bar{x}_k + \Gamma(\tau_k)B_S\hat{u}_{k-1}$

Then, if an output feedback control law is applied, it is possible to compute $\bar{y}(t_k+\tau_k) = C_S\bar{x}(t_k+\tau_k)$. Note that this approach allows also to apply state-feedback control laws using $\bar{x}(t_k+\tau_k)$.

2) Remote controller

In this case we consider the general scheme, discussed previously:

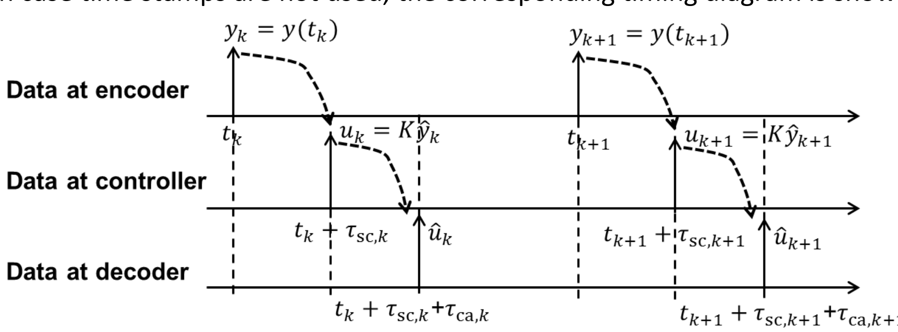


The dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

An output-feedback static controller is used, i.e., $u_k = K\hat{y}_k$.

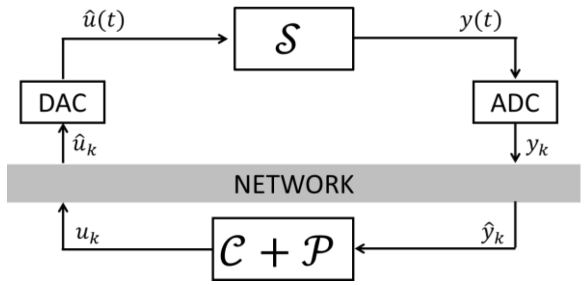
- In case time stamps are not used, the corresponding timing diagram is shown below.



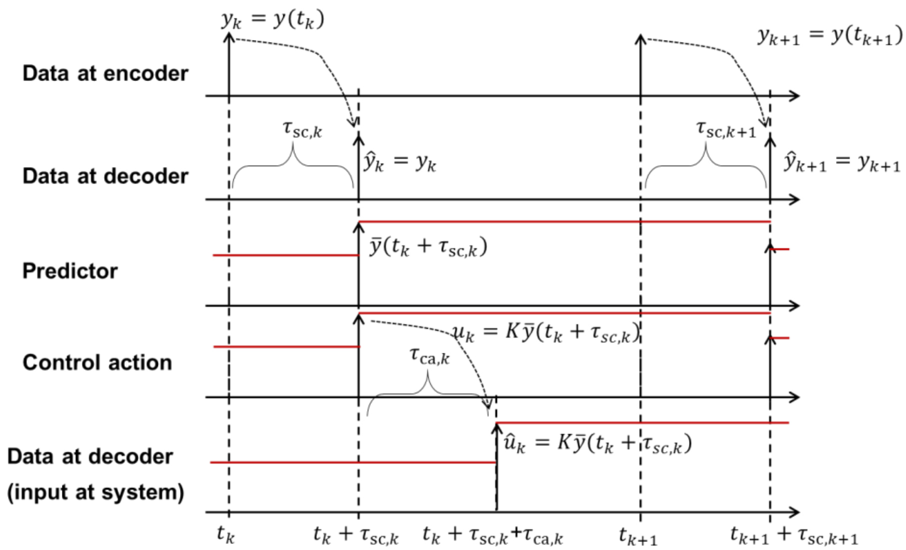
As soon as the data is conveyed to the controller (with a possibly time-varying delay $\tau_{sc,k}$), the control action $u_k = K\hat{y}_k$ is computed and instantaneously retransmitted. Also, as soon as $\hat{u}_k$ is conveyed to the decoder (with a possibly additional time-varying delay $\tau_{ca,k}$), it is converted to an analog signal (i.e., kept constant until new data are available) and applied to the system.

Therefore, being $\tau_k = \tau_{sc,k} + \tau_{ca,k}$, the control action $u(t) = Ky(t_k)$ is applied to the system in the interval $[t_k+\tau_k, t_{k+1}+\tau_{k+1})$ – it is kept constant for $h_k^* = h_k + \tau_{k+1} - \tau_k$ s.

- In case time stamps are used, the time delay between sensor and controller $\tau_{\text{sc},k}$ is known and can be used for compensation. In fact, we can introduce an output/state predictor (embedded, in the controller), as in the following figure.



However, it is not possible to use similarly the possibly-available information about the delay $\tau_{\text{ca},k}$. The main idea is sketched in the time diagram below.



In particular, the control action $\hat{u}(t) = K\bar{y}\big(t_k + \tau_{\text{sc},k}\big) \cong Ky(t_k + \tau_{\text{sc},k})$ is computed by the controller at time $t_k + \tau_{\text{sc},k}$ and instantaneously transmitted. This information is received by the decoder at time instant $t_k + \tau_{\text{sc},k} + \tau_{\text{ca},k}$ and applied to the system until new data is available (i.e., at time instant $t_{k+1} + \tau_{\text{sc},k+1} + \tau_{\text{ca},k+1}$). Therefore, $\hat{u}(t) = K\bar{y}\big(t_k + \tau_{\text{sc},k}\big)$ is applied to the system in the interval $\big[t_k + \tau_{\text{sc},k} + \tau_{\text{ca},k}, t_{k+1} + \tau_{\text{sc},k+1} + \tau_{\text{ca},k+1}\big)$ - it is kept constant for $h_k^* = h_k + \tau_{\text{sc},k+1} + \tau_{\text{ca},k+1} - \tau_{\text{sc},k} - \tau_{\text{ca},k}$. In in way the delay $\tau_{\text{sc},k}$ is in practice well compensated, while the delay $\tau_{\text{ca},k}$ cannot be considered.

RELATED REFERENCES

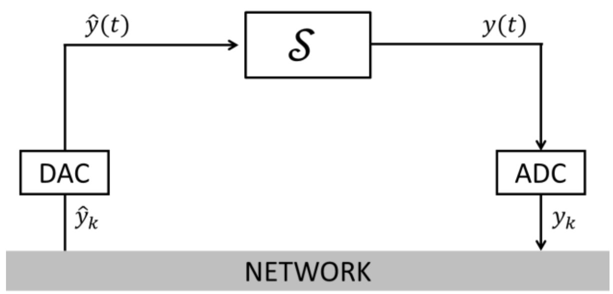
- W. Zhang, M. S. Branicky, S. M. Phillips, Stability of networked control systems, *IEEE Control Systems Magazine*, 2001, pp. 84-99.
- J. P. Hespanha, P. Naghshtabrizi, Y. Xu, A survey of recent results in networked control systems, *Proceedings of the IEEE*, 95 (1), 2007, pp. 138-162.

III.3: PACKET DROPOUTS

Control networks can be considered as unreliable data channels, where packet collisions and network node failures occasionally occur. In this cases, some packets are lost. In this chapter we discuss some results concerning the stability properties of a NCS subject to packet losses.

A. NETWORKED CONTROL SYSTEM CONFIGURATION

The main working assumptions are those specified in Chapter III.2. In particular, we consider again the NCS



where the dynamics of system $\mathcal{S}$ is described by

$$\begin{cases} \dot{x}(t) & = Ax(t) + B\hat{y}(t) \\ y(t) & = Cx(t) \end{cases}$$

Also, the signal $\hat{y}(t)$, due to the system delay, is

$$\hat{y}(t) = \begin{cases} \hat{y}_{k-1} & t \in [t_k, t_k + \tau_k) \\ \hat{y}_k & t \in [t_k + \tau_k, t_{k+1}) \end{cases}$$

B. NETWORKED CONTROL SYSTEM DYNAMICS

In this section we consider, in addition to the framework described in the Chapter III.2, that some of the information to be transmitted by the network channel may be lost.

- When the package is not lost, as in Chapter III.2, it is set $\hat{y}_k = y_k = y(t_k)$.

- When the package is lost, no useful information is available at the decoder level at time $t_k + \tau_k$, i.e., $\hat{y}_k$ is actually not available. A number of different approaches can be taken: the most usual one consists of setting $\hat{y}_k = \hat{y}_{k-1}$, i.e., re-transmitting the previously-available significant value of y_k.

The availability of the information is described introducing the variable θ_k. More specifically, when transmission occurs we set $\theta_k = 1$ while, when the information is lost, we set $\theta_k = 0$. We write

$$\hat{y}_k = \theta_k y_k + (1 - \theta_k)\hat{y}_{k-1}$$

The NCS dynamics is therefore described by the following model

$$z_{k+1} = \begin{bmatrix} x_{k+1} \\ \hat{y}_k \end{bmatrix} = \begin{bmatrix} e^{Ah_k} + \theta_k \Gamma(h_k - \tau_k)BC & e^{A(h_k - \tau_k)}\Gamma(\tau_k)B + (1 - \theta_k)\Gamma(h_k - \tau_k)B \\ \theta_k C & (1 - \theta_k)I \end{bmatrix} \begin{bmatrix} x_k \\ \hat{y}_{k-1} \end{bmatrix}$$

For simplicity of presentation, in the following we adopt **Assumption 1**, i.e., the delay and the sampling time are constant. Therefore we set $\tau_k = \tau$ and $h_k = h$ for all $k \in \mathbb{N}$. On the other hand, θ_k is obviously time-varying. We denote

$$\widetilde{\Phi}_{\theta_k} = \begin{bmatrix} e^{Ah} + \theta_k \Gamma(h - \tau)BC & e^{A(h-\tau)}\Gamma(\tau)B + (1 - \theta_k)\Gamma(h - \tau)B \\ \theta_k C & (1 - \theta_k)I \end{bmatrix}$$

C. DETERMINISTIC DROPOUTS

In this section we consider the case where we can define an *asymptotic rate* r at which dropouts occur, defined by the following time average

$$r = \lim_{T \to +\infty} \frac{1}{T} \sum_{k=k_0}^{k_0+T-1} (1 - \theta_k)$$

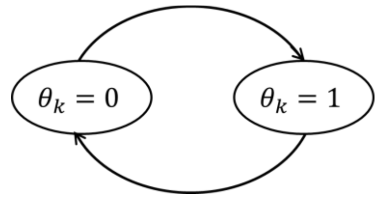
for all $k_0 \in \mathbb{N}$. Under the assumption that r is well defined, the network model

$$z_{k+1} = \widetilde{\Phi}_{\theta_k} z_k$$

is a so-called *asynchronous dynamical system* (ADS), i.e., a *hybrid system*, with

- **continuous dynamics**
  $$z_{k+1} = \widetilde{\Phi}_0 z_k, \text{or}$$
  $$z_{k+1} = \widetilde{\Phi}_1 z_k$$
  depending upon the value taken by variable θ_k;

- **discrete dynamics** (i.e., describing the switch from the case $\theta_k = 0$ to $\theta_k = 1$ and vice-versa), i.e., a *finite automaton*



driven by external events occurring at a predefined rate.

1) Stability of the NCS with deterministic dropouts

The stability properties of the NCS subject to deterministic packet dropouts can be established thanks to the following theorem.

Theorem 1

Assume that there exists a symmetric matrix $P > 0$ and scalars α_0, α_1, α such that

$$\alpha_0^r \alpha_1^{1-r} > \alpha > 1 \quad (i)$$

$$\widetilde{\Phi}_0^T P \widetilde{\Phi}_0 \leq \alpha_0^{-2} P \quad (ii)$$

$$\widetilde{\Phi}_1^T P \widetilde{\Phi}_1 \leq \alpha_1^{-2} P \quad (iii)$$

then, the NCS is exponentially stable, in the sense that

$$\lim_{k \to +\infty} \alpha^k \left\| z_k \right\| = 0 \,, \text{i.e.,} \left\| z_k \right\| \leq \mu \alpha^{-k} \text{ for some } \mu > 0$$

Remarks

- Inequalities (ii) and (iii) are LMIs for fixed values of scalars α_0, α_1.
- If α, α_0, α_1 are free variables, the set of inequalities $(i) - (iii)$ is a system of bilinear (nonlinear) matrix inequalities, which is more difficult to solve than a set of LMIs.
- To solve the system $(i) - (iii)$ we can select different values of scalars α_0, α_1 (*gridding method*) and check the feasibility of (ii)-(iii) for at least of feasible pair (α_0, α_1).
- The function $V(z) = z^T P z$ results to be the Lyapunov function for the hybrid system, as shown in the proof of Theorem 1.

Proof of Theorem 1

We chose $V(z) = z^T P z$ as a candidate Lyapunov function for the hybrid system ADS. We compute

$$\begin{aligned} V(z_k) = z_k^T P z_k &= z_{k-1}^T \widetilde{\Phi}_{\theta_{k-1}}^T P \widetilde{\Phi}_{\theta_{k-1}} z_{k-1} \leq \alpha_{\theta_{k-1}}^{-2} z_{k-1}^T P z_{k-1} \leq \cdots \leq \alpha_{\theta_{k-1}}^{-2} \alpha_{\theta_{k-2}}^{-2} \ldots \alpha_{\theta_0}^{-2} z_0^T P z_0 \\ &= \alpha_{\theta_{k-1}}^{-2} \alpha_{\theta_{k-2}}^{-2} \ldots \alpha_{\theta_0}^{-2} V(z_0) \end{aligned}$$

From the definition of asymptotic rate r, packet dropouts (corresponding to $\theta = 0$,. i.e., to α_0) occur rk times in k steps (for $k \to +\infty$, i.e., asymptotically), while packet transmissions (corresponding to $\theta = 1$,. i.e., to α_1) occur $(1 - r)k$ times in k steps. This means that, asymptotically,

$$V(z_k) \leq \alpha_0^{-2rk} \alpha_1^{-2(1-r)k} V(z_0) = \left(\alpha_0^r \alpha_1^{(1-r)} \right)^{-2k} V(z_0) \leq \alpha^{-2k} V(z_0)$$

Recall that, since P is symmetric definite positive, $\lambda_{\min}(P)\|z\|^2 \le V(z) \le \lambda_{\max}(P)\|z\|^2$. Therefore

$$\lambda_{\min}(P)\|z_k\|^2 \le V(z_k) \le \alpha^{-2k}V(z_0) \le \alpha^{-2k}\lambda_{\max}(P)\|z_0\|^2$$

and so

$$\|z_k\| \le \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}}\alpha^{-k}\|z_0\|$$

2) Computation of the maximal dropout rate

In the previous section the *dropout rate* r was regarded as a network property, and given as a constraint. However, it can also be regarded as a free NCS design choice. In fact, the possibility of having packet losses may be an opportunity to make the network traffic less intense.

The question we address in this section is: at which maximal rate can we drop packets without compromising the system stability?

Before to answer this question, let us consider the NCS dynamics in the two limit cases:

- **Case 1**: $\theta_k = 1$ for all $k \in \mathbb{N}$, i.e., no packet losses occur.
  In this case the network dynamics is
  $$z_{k+1} = \widetilde{\Phi}_1 z_k$$
  where
  $$\widetilde{\Phi}_1 = \begin{bmatrix} e^{Ah} + \Gamma(h-\tau)BC & e^{A(h-\tau)}\Gamma(\tau)B \\ C & 0 \end{bmatrix} = \Phi(h,\tau)$$
  This case has been thoroughly studied in Chapter III.2. In the current chapter we assume that $\widetilde{\Phi}_1 = \Phi(h,\tau)$ is Schur stable, i.e., that stability is guaranteed in case losses are absent, but only a time delay τ. Under this assumption, the spectral radius of $\widetilde{\Phi}_1$ is
  $$\rho(\widetilde{\Phi}_1) < 1$$
- **Case 2**: $\theta_k = 0$ for all $k \in \mathbb{N}$, i.e., packets are lost all the time. In this case the network dynamics is
  $$z_{k+1} = \widetilde{\Phi}_0 z_k$$
  where
  $$\widetilde{\Phi}_0 = \begin{bmatrix} e^{Ah} & e^{A(h-\tau)}\Gamma(\tau)B + \Gamma(h-\tau)B \\ 0 & I \end{bmatrix} = \begin{bmatrix} e^{Ah} & \Gamma(h)B \\ 0 & I \end{bmatrix}$$
  Since $\widetilde{\Phi}_0$ is block-triangular, its spectral radius is
  - $\rho(\widetilde{\Phi}_0) = 1$ if $F = e^{Ah}$ is Schur stable, i.e., if the system $\mathcal{S}$ is open-loop stable;
  - $\rho(\widetilde{\Phi}_0) = \rho(F)$ if the system $\mathcal{S}$ is open-loop not asymptotically stable.

  As a result, we can infer that, if dropouts always occur, the NCS cannot enjoy asymptotic stability properties.

As a result of this preliminary considerations, we have that

- If $\theta_k = 1$ for all $k \in \mathbb{N}$ (i.e., no packet losses occur, $r = 0$), the NCS is asymptotically stable.

- If $\theta_k = 0$ for all $k \in \mathbb{N}$ (i.e., packets are lost all the time, $r = 1$), the NCS is not asymptotically stable.

Therefore, there certainly exists a maximum allowed asymptotic dropout rate $\bar{r} \in [0,1)$. In this section we provide a procedure that allows to quantify it (although in a possibly conservative way).

We can prove the following result, derived by **Theorem 1** where, for simplicity, we redefine $\beta_0 = \alpha_0^{-2}$ and $\beta_1 = \alpha_1^{-2}$.

Theorem 2

Assume that there exists a symmetric matrix $P > 0$ and scalars $\beta_0 \geq 1$, $0 < \beta_1 < 1$ such that

$$\widetilde{\Phi}_0^T P \widetilde{\Phi}_0 \leq \beta_0 P \quad (ii^*)$$

$$\widetilde{\Phi}_1^T P \widetilde{\Phi}_1 \leq \beta_1 P \quad (iii^*)$$

then, the NCS is exponentially stable for all $r < \bar{r}$, where

$$\bar{r} = \frac{1}{1 - \frac{\log \beta_0}{\log \beta_1}}$$

Proof of Theorem 2.

First note that, under $\beta_0 = \alpha_0^{-2}$ and $\beta_1 = \alpha_1^{-2}$, (ii^*) and (iii^*) are equivalent to (ii) and (iii) in Theorem 1. If $\beta_0 \geq 1$, $0 < \beta_1 < 1$, to verify (i) we have to chose r such that $\alpha_0^r \alpha_1^{1-r} > 1$. Since under $\beta_0 = \alpha_0^{-2}$ and $\beta_1 = \alpha_1^{-2}$, this is equivalent to enforce $\beta_0^r \beta_1^{(1-r)} < 1$. Taking the logarithm of both sides, this is equivalent to set

$$r \log \beta_0 \; + (1 - r) \log \beta_1 \; < 0$$

This is equivalent to

$$r \underbrace{(\log \beta_0 \; - \log \beta_1 \;)}_{>0} < - \underbrace{\log \beta_1}_{<0}$$

which is possible if and only if

$$r < \frac{1}{1 - \frac{\log \beta_0}{\log \beta_1}}$$

All the assumptions (ii)-(iii) of Theorem 1 are therefore verified. We resort to Theorem 1 to prove the asymptotic stability of the NCS.

Remarks

- To guarantee the existence of $P > 0$ such that $\widetilde{\Phi}_0^T P \widetilde{\Phi}_0 \leq \beta_0 P$ (proof not shown) it must hold that
$$\beta_0 \geq \rho^2\left(\widetilde{\Phi}_0\right) \geq 1$$

which is compatible with the assumptions of Theorem 2.

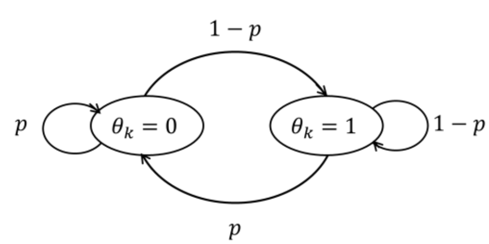
- To guarantee the existence of $P > 0$ such that $\widetilde{\Phi}_1^T P \widetilde{\Phi}_1 \leq \beta_1 P$ (proof not shown) it must hold that
$$\beta_1 \geq \rho^2(\widetilde{\Phi}_1)$$
Since $\rho(\widetilde{\Phi}_1) < 1$ it may be possible to find $\beta_1 < 1$, i.e., compatible with the requirements in Theorem 2.
- Recall that
$$\bar{r} = \frac{1}{1 - \frac{\log \beta_0 \geq 0}{\log \beta_1 < 0}}$$
Inspecting the previous equation, we infer that
    - $\bar{r} < 1$
    - To maximize $\bar{r}$ (i.e., to obtain the least conservative approximation of the maximum allowable dropout rate) it would be helpful to choose (if possible), at the same time
        - β_0 as small as possible (i.e., as close as possible to 1 or, more precisely, to $\rho^2(\widetilde{\Phi}_0)$).
        - β_1 as small as possible (i.e., as close as possible to 0 or, more precisely, to $\rho^2(\widetilde{\Phi}_1)$).

D. STOCHASTIC DROPOUTS

In the *stochastic* framework, the variable θ_k is described by a *Bernoulli process* with dropout *probability* $p \in [0,1]$. More specifically

$$\mathcal{Pr}\{\theta_k = 0\} = p, \quad \mathcal{Pr}\{\theta_k = 1\} = 1 - p$$

This makes the NCS a special case of a *discrete-time Markov jump linear system* (MJLS). In this framework, θ_k is the state of a discrete-time *Markov chain* with a finite (i.e., 2) number of discrete states and with given transition probability, as in figure.



Stability of MJLSs has been thorough studied in dedicated works. The following result can be derived.

Theorem 3

The NCS, with dropout probability p, is exponentially mean-square stable **if** there exists symmetric matrix $Z > 0$ such that

$$\begin{bmatrix} Z & \sqrt{p}(\widetilde{\Phi}_0 Z)^T & \sqrt{1-p}(\widetilde{\Phi}_1 Z)^T \\ \sqrt{p}(\widetilde{\Phi}_0 Z) & Z & 0 \\ \sqrt{1-p}(\widetilde{\Phi}_1 Z) & 0 & Z \end{bmatrix} > 0$$

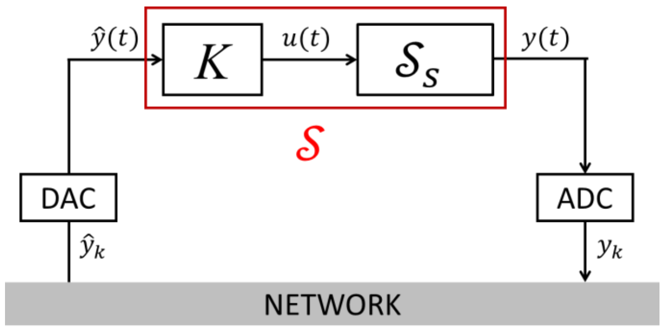
The system is mean-square stable if, for all initial conditions z_0,

$$\lim_{k\to+\infty} \mathbb{E}[z_k] = 0$$

$$\lim_{k\to+\infty} \mathbb{E}\left[z_k z_k^T\right] = \lim_{k\to+\infty} \left(\mathrm{var}(z_k) + \mathbb{E}[z_k]\mathbb{E}\left[z_k^T\right]\right) = 0$$

E. EXAMPLES

In this section we consider the examples used in Chapter III.2. In particular, system $\mathcal{S}$ consists of system+output feedback controller co-located with the actuator. The scheme is the following



where the dynamics of $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = A_S x_S(t) + B_S u(t) \\ y(t) & = C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. As discussed previously, the control scheme in the above figure can be cast in the class of the reference configuration provided that we rename $x(t) = x_S(t), A = A_S, B = B_S K, C = C_S$. So, we rewrite

$$\tilde{\Phi}_1 = \Phi(h,\tau) = \begin{bmatrix} e^{A_S h} + \Gamma(h-\tau)B_S K C_S & e^{A_S(h-\tau)}\Gamma(\tau)B_S K \\ C_S & 0 \end{bmatrix}$$

$$\tilde{\Phi}_0 = \begin{bmatrix} e^{A_S h} & \Gamma(h)B_S K \\ 0 & I \end{bmatrix}$$

where , $\Gamma(\xi) = \int_0^\xi e^{A_S v} dv$.

Case 1: scalar system with $A_S = 0$ (integrator).

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = u(t) \\ y(t) & = x_S(t) \end{cases}$$

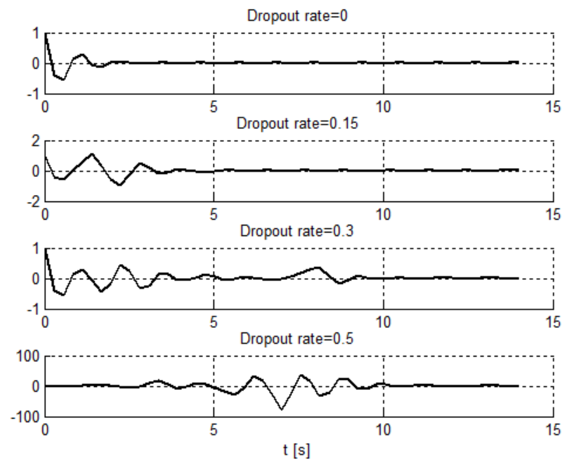
and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} 1 + K(h-\tau) & K\tau \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} 1 & Kh \\ 0 & 1 \end{bmatrix}$$

Taking $K = -5$, $h = 0.28$ s, and $\tau = 0.1$. we evaluate (in a possibly conservative way) the maximal admissible dropout rate $\bar{r}$ allowed, in the deterministic setting, according to Theorem 2. Also, we compare such rate with the maximal dropout probability allowed by Theorem 3. We compute that

$$\bar{r} = 0.2802, p = 0.23$$

In the following figure we show simulation results considering different cases, i.e., $r = 0{,}0.15{,}0.3{,}0.5$.



Case 2: scalar system with $A_S = a \neq 0$ ($B_S = 1$ for simplicity).

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) & = a x_S(t) + u(t) \\ y(t) & = x_S(t) \end{cases}$$

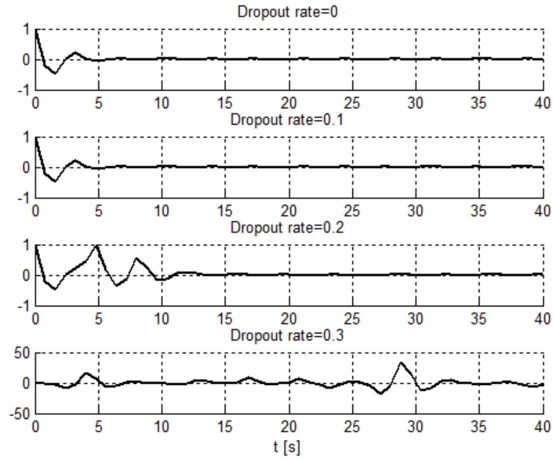
and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\tilde{\Phi}_1 = \begin{bmatrix} e^{ah} + \frac{K}{a}(e^{a(h-\tau)} - 1) & \frac{K}{a} e^{a(h-\tau)}(e^{a\tau} - 1) \\ 1 & 0 \end{bmatrix}, \tilde{\Phi}_0 = \begin{bmatrix} e^{ah} & \frac{K}{a}(e^{ah} - 1) \\ 0 & 1 \end{bmatrix}$$

- In case $a = 1$ and taking $K = -2$, $h = 0.8$ s, and $\tau = 0.1$ we evaluate (in a possibly conservative way) the maximal admissible dropout rate $\bar{r}$ allowed, in the deterministic setting, according to Theorem 2. Also, we compare such rate with the maximal dropout probability allowed by Theorem 3. We compute that

$$\bar{r} = 0.2224, p = 0.08$$

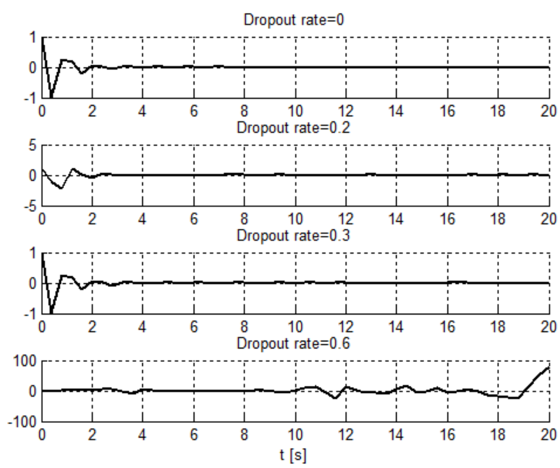
In the following figure we show simulation results considering different cases: $r = 0{,}0.1{,}0.2{,}0.3$.



- In case $a = -1$ and taking $K = -5$, $h = 0.4$ s, and $\tau = 0.1$, we evaluate (in a possibly conservative way) the maximal admissible dropout rate $\bar{r}$ allowed, in the deterministic setting, according to Theorem 2. Also, we compare such rate with the maximal dropout probability allowed by Theorem 3. We compute that

$$\bar{r} = 0.2975, p = 0.19$$

In the following figure we show simulation results considering different cases: $r = 0,0.2,0.3,0.6$.



Case 3: high-order system with $\det(A_S) \neq 0$.

The dynamics of the scalar system $\mathcal{S}_S$ is

$$\begin{cases} \dot{x}_S(t) &= A_S x_S(t) + B_S u(t) \\ y(t) &= C_S x_S(t) \end{cases}$$

and an output-feedback static controller is applied, i.e., $u(t) = K\hat{y}(t)$. We obtain that

$$\widetilde{\Phi}_1 = \begin{bmatrix} e^{A_S h} + A_S^{-1}\big(e^{A_S(h-\tau)} - I\big)B_S K C_S & e^{A_S(h-\tau)}A_S^{-1}(e^{A_S\tau} - I)B_S K \\ C_S & 0 \end{bmatrix}$$

$$\widetilde{\Phi}_0 = \begin{bmatrix} e^{A_S h} & A_S^{-1}\big(e^{A_S h} - I\big)B_S K \\ 0 & 1 \end{bmatrix}$$

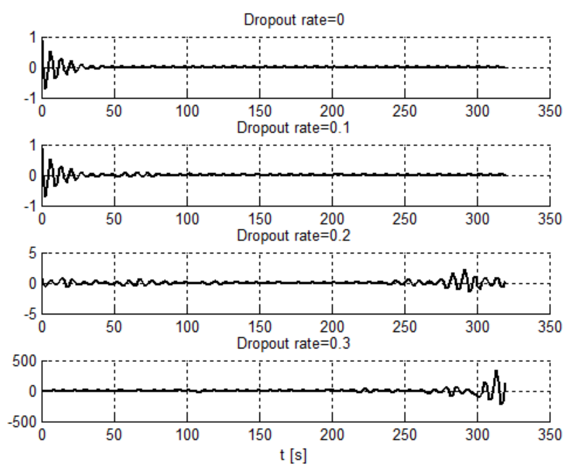
We consider the system

$$A_S = \begin{bmatrix} -1 & 0.9 \\ 0.2 & -0.2 \end{bmatrix}, B_S = \begin{bmatrix} 1 \\ 0.1 \end{bmatrix}, C_S = [-0.1,1]$$

Taking $h = 1.6$ s, and $\tau = 0.12$ and the gain $K = -5$, we evaluate (in a possibly conservative way) the maximal admissible dropout rate $\bar{r}$ allowed, in the deterministic setting, according to Theorem 2. Also, we compare such rate with the maximal dropout probability allowed by Theorem 3. We compute that

$$\bar{r} = 0.1119, p = 0.11$$

In the following figure we show simulation results considering different cases: $r = 0{,}0.15{,}0.3{,}0.5$.



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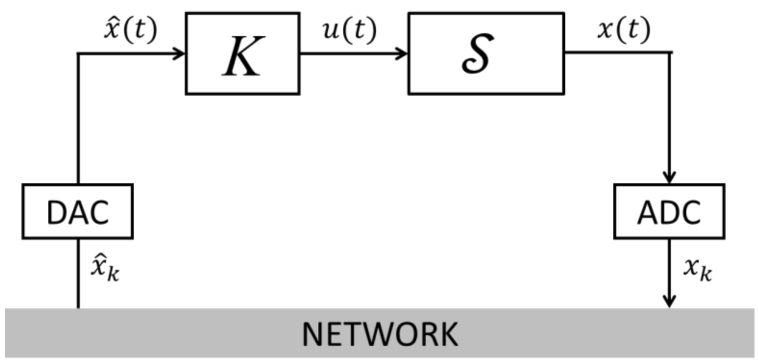
EXERCISES: DELAY AND PACKET DROPOUTS

EXERCISE 1

Consider the continuous-time system

$$\mathcal{S}: \dot{x} = -x + u$$

A) Assuming that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.
B) Assuming that the system is controlled using a digital control law (i.e., with a sample-and-hold actuator) of the type $u_k = Kx_k$ and with sampling time $h = \log_e 2.5$ s=0.9163 s, compute the range of values of K that make the closed-loop system asymptotically stable.
C) Assume that the system is controlled using the networked control system below.



The sampling time is constant and equal to $h = \log_e 2.5$ s=0.9163 s. Define the range of values of K that make the closed-loop system asymptotically stable in case the network delay is constant and equal to

   a. $\tau = \log_e 1.5$ s=0.4055 s
   b. $\tau = \log_e 2$ s=0.6931 s

D) Assume again that the system is controlled using the networked control system sketched above, with constant sampling time, gain $K = -1$, and in presence of a constant delay $\tau = \log_e 1.5$. Compute the maximal allowable transmission interval (MATI), i.e., the largest value of h that guarantees asymptotic stability of the NCS.
E) Assume that $h = \log_e 2.5$ s, $\tau = 0$, $K = -1$, and that one in two information packets drops. More specifically, variable θ_k has the following behavior:

k	1	2	3	4	$\cdots$
θ_k	1	0	1	0	$\cdots$

Is the corresponding NCS asymptotically stable?

SOLUTION

A) Setting $u(t) = Kx(t)$ for all $t \in \mathbb{R}$, the closed loop system equation is $\dot{x} = (-1 + K)x$, which is asymptotically stable if and only if $K < 1$.

B) The corresponding discrete-time system (with sampling time $h = \log_e 2.5$ s) is $x_{k+1} = 0.4x_k + 0.6u_k$. Setting $u_k = Kx_k$, the closed-loop system equation is $x_{k+1} = (0.4 + 0.6K)x_k$, which is asymptotically stable if and only if $-\frac{7}{3} = -2.33 < K < 1$.

C) When the system is controlled using the network control system in figure, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} e^{-h} + \Gamma(h-\tau)K & e^{-(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{-\nu} d\nu = (1 - e^{-\tau})$.

a. In case $h = \log_e 2.5$ s and $\tau = \log_e 1.5$ s,

$$\Phi = \begin{bmatrix} 0.4 + 0.4K & 0.2K \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - 0.4(1+K)\lambda - 0.2K$$

According to the rule derived in the Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -0.2K < 1 \\ -(1 - 0.2K) < -0.4(1+K) < (1 - 0.2K) \end{cases}$$

This is verified for $-5 < K < 1$. Note that this condition is less conservative than the one obtained in case of “pure” digital control.

b. In case $h = \log_e 2.5$ s and $\tau = \log_e 2$ s,

$$\Phi = \begin{bmatrix} 0.4 + 0.2K & 0.4K \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - (0.4 + 0.2K)\lambda - 0.4K$$

According to the rule derived in the Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -0.4K < 1 \\ -(1 - 0.4K) < -(0.4 + 0.2K) < (1 - 0.4K) \end{cases}$$

This is verified for $-2.5 < K < 1$. Note that also this condition is less conservative than the one obtained in case of “pure” digital control.

D) When the system is controlled using the networked control system in figure, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} e^{-h} + \Gamma(h-\tau)K & e^{-(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{-\nu} d\nu = (1 - e^{-\tau})$. We denote $f = e^{-h}$ and we rewrite

$$\Phi(h,\tau) = \begin{bmatrix} \frac{5}{2}f - 1 & -\frac{1}{2}f \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - \left(\frac{5}{2}f - 1\right)\lambda + \frac{1}{2}f$$

According to the rule derived in the Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < \frac{1}{2}f < 1 \\ -\left(1 + \frac{1}{2}f\right) < -\left(\frac{5}{2}f - 1\right) < \left(1 + \frac{1}{2}f\right) \end{cases}$$

This is verified for $0 < f = e^{-h} < 1$. Therefore, all values of $h > \tau$ are valid.

E) To analyze the stability of the system under packet dropouts defined by the sequence of values of variable θ_k, we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\widetilde{\Phi}_1 = \Phi(h,\tau) = \begin{bmatrix} e^{-h} + \Gamma(h-\tau)K & e^{-(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -0.2 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\widetilde{\Phi}_0 = \begin{bmatrix} e^{-h} & \Gamma(h)K \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.4 & -0.6 \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\widetilde{\Phi}_1$ has eigenvalues in -0.2 and 0 while $\widetilde{\Phi}_0$ has eigenvalues in 0.4 and 1.

Since the sequence of values of θ_k is 1,0,1,0,1,0,1,0:

- if k is odd, $z_{k+1} = \widetilde{\Phi}_1 z_k$, $z_{k+2} = \widetilde{\Phi}_0\widetilde{\Phi}_1 z_k$.
- if k is even, $z_{k+1} = \widetilde{\Phi}_0 z_k$, $z_{k+2} = \widetilde{\Phi}_1\widetilde{\Phi}_0 z_k$.

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{2k+2} = \widetilde{\Phi}_1\widetilde{\Phi}_0 z_{2k}$$

$$z_{2k+3} = \widetilde{\Phi}_0\widetilde{\Phi}_1 z_{2k+1}$$

Therefore, $\widetilde{\Phi}_1\widetilde{\Phi}_0$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1$ can be regarded as the two-steps transition matrices for the NCS state. We resort to the fact that $\widetilde{\Phi}_1\widetilde{\Phi}_0$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\widetilde{\Phi}_1\widetilde{\Phi}_0 = \begin{bmatrix} -0.2 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0.4 & -0.6 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.08 & -0.12 \\ 0.4 & -0.6 \end{bmatrix}$$

whose eigenvalues are 0 and -0.68. The NCS under packet dropout is therefore asymptotically stable.

EXERCISE 2

Consider the continuous-time system

$$\dot{x} = x + u$$

A) Assuming that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.
B) Assuming that the system is controlled using a digital control law (i.e., with a sample-and-hold actuator) of the type $u_k = Kx_k$ and with sampling time $h = \log_e 2.5$ s=0.9163 s, compute the range of values of K that make the closed-loop system asymptotically stable.
C) Assume that the system is controlled using the networked control system sketched in Exercise 1, where the sampling time is constant and equal to $h = \log_e 2.5$ s. Define the range of values of K that make the closed-loop system asymptotically stable in case the network delay is constant and equal to
   a. $\tau = \log_e 1.5$ s=0.4055 s
   b. $\tau = \log_e 2$ s=0.6931 s
D) Assume again that the system is controlled using the networked control system sketched above, with constant sampling time, gain $K = -1.1$, and in presence of a constant delay $\tau = \log_e 1.5$. Compute the maximal allowable transmission interval (MATI), i.e., the largest value of h that guarantees asymptotic stability of the NCS.
E) Assume that $h = \log_e 2.5$ s, $\tau = 0$, $K = -2$, and that one in two information packets drops. More specifically, variable θ_k has the following behavior:

k	1	2	3	4	$\cdots$
θ_k	1	0	1	0	$\cdots$

Is the corresponding NCS asymptotically stable?

SOLUTION

A) Setting $u(t) = Kx(t)$ for all $t \in \mathbb{R}$, the closed loop system equation is $\dot{x} = (1 + K)x$, which is asymptotically stable if and only if $K < -1$.
B) The corresponding discrete-time system (with sampling time $h = \log_e 2.5$ s) is $x_{k+1} = 2.5x_k + 1.5u_k$. Setting $u_k = Kx_k$, the closed-loop system equation is $x_{k+1} = (2.5 + 1.5K)x_k$, which is asymptotically stable if and only if $-\frac{7}{3} = -2.33 < K < -1$.
C) When the system is controlled using the networked control system in figure, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} e^h + \Gamma(h-\tau)K & e^{(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^\nu d\nu = (e^\tau - 1)$.

   a. In case $h = \log_e 2.5$ s and $\tau = \log_e 1.5$ s,

$$\Phi = \begin{bmatrix} \frac{5}{2} + \frac{2}{3}K & \frac{5}{6}K \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - \left(\frac{5}{2} + \frac{2}{3}K\right)\lambda - \frac{5}{6}K$$

According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -\frac{5}{6}K < 1 \\ -\left(1 - \frac{5}{6}k\right) < -\left(\frac{5}{2} + \frac{2}{3}K\right) < (1 - \frac{5}{6}K) \end{cases}$$

This is verified for $-\frac{6}{5} = -1.2 < K < -1$. Note that this condition is more conservative than the one obtained in case of "pure" digital control.

b. In case $h = \log_e 2.5$ s and $\tau = \log_\mathrm{e} 2$ s,

$$\Phi = \begin{bmatrix} \frac{5}{2} + \frac{1}{4}K & \frac{5}{4}K \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - \left(\frac{5}{2} + \frac{1}{4}K\right)\lambda - \frac{5}{4}K$$

According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -\frac{5}{4}K < 1 \\ -\left(1 - \frac{5}{4}K\right) < -\left(\frac{5}{2} + \frac{1}{4}K\right) < \left(1 - \frac{5}{4}K\right) \end{cases}$$

This is never verified.

D) When the system is controlled using the networked control system in figure, the transition matrix $\Phi(h,\tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} e^h + \Gamma(h-\tau)K & e^{(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix}$$

where $\Gamma(\tau) = \int_0^\tau e^{-\nu}d\nu = (e^\tau - 1)$. We denote $f = e^h$ and we rewrite

$$\Phi(h,\tau) = \begin{bmatrix} 0.2667f + 1.1 & -0.3667f \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - (0.2667f + 1.1)\lambda + 0.3667f$$

According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < 0.3667f < 1 \\ -(1 + 0.3667f) < -(0.2667f + 1.1) < (1 + 0.3667f) \end{cases}$$

This is verified for $1 < f = e^h < 2.727 = e$. Therefore, all values of $\tau < h < 1$ are valid.

E) To analyze the stability of the system under packet dropouts defined by the sequence of values of variable θ_k, we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\widetilde{\Phi}_1 = \Phi(h,\tau) = \begin{bmatrix} e^h + \Gamma(h-\tau)K & e^{(h-\tau)}\Gamma(\tau)K \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -0.5 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\widetilde{\Phi}_0 = \begin{bmatrix} e^h & \Gamma(h)K \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2.5 & -3 \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\widetilde{\Phi}_1$ has eigenvalues in -0.5 and 0, while $\widetilde{\Phi}_0$ has eigenvalues in 2.5 and 1.

Since the sequence of values of θ_k is 1,0,1,0,1,0,1,0:

- if k is odd, $z_{k+1} = \widetilde{\Phi}_1 z_k$, $z_{k+2} = \widetilde{\Phi}_0\widetilde{\Phi}_1 z_k$.
- if k is even, $z_{k+1} = \widetilde{\Phi}_0 z_k$, $z_{k+2} = \widetilde{\Phi}_1\widetilde{\Phi}_0 z_k$.

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{2k+2} = \widetilde{\Phi}_1\widetilde{\Phi}_0 z_{2k}$$

$$z_{2k+3} = \widetilde{\Phi}_0\widetilde{\Phi}_1 z_{2k+1}$$

Therefore, $\widetilde{\Phi}_1\widetilde{\Phi}_0$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1$ can be regarded as the two-steps transition matrices for the NCS state. We resort to the fact that $\widetilde{\Phi}_1\widetilde{\Phi}_0$ and $\widetilde{\Phi}_0\widetilde{\Phi}_1$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\widetilde{\Phi}_1\widetilde{\Phi}_0 = \begin{bmatrix} -0.5 & 0 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 2.5 & -3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1.25 & 1.5 \\ 2.5 & -3 \end{bmatrix}$$

whose eigenvalues are 0 and -4.25. The NCS under packet dropout is therefore unstable.

EXERCISE 3

Consider the continuous-time system

$$\dot{x} = u$$

A) Assuming that the system is controlled using a continuous-time control law of the type $u(t) = Kx(t)$, compute the range of values of K that make the closed-loop system asymptotically stable.

B) Assuming that the system is controlled using a digital control law (i.e., with a sample-and-hold actuator) of the type $u_k = Kx_k$ and with sampling time $h = 1$ s, compute the range of values of K that make the closed-loop system asymptotically stable.

C) Assume that the system is controlled using the networked control system sketched in Exercise 1, where the sampling time is constant and equal to $h = 1$ s. Define the range of values of K that make the closed-loop system asymptotically stable in case the network delay is constant and equal to
   a. $\tau = 0.25$ s
   b. $\tau = 0.5$ s
   c. $\tau = 0.75$ s

D) Assume again that the system is controlled using the networked control system sketched above, with constant sampling time, gain $K = -1$, and in presence of a constant delay $\tau = 0.75$. Compute the maximal allowable transmission interval (MATI), i.e., the largest value of h that guarantees asymptotic stability of the NCS.

E) Assume that $h = 1$ s, $\tau = 0.75$, $K = -1$, and that one in two information packets drops. More specifically, variable θ_k has the following behavior:

k	1	2	3	4	$\cdots$
θ_k	1	0	1	0	$\cdots$

Is the corresponding NCS asymptotically stable?

SOLUTION

A) Setting $u(t) = Kx(t)$ for all $t \in \mathbb{R}$, the closed loop system equation is $\dot{x} = Kx$, which is asymptotically stable if and only if $K < 0$.

B) The corresponding discrete-time system (with sampling time $h = 1$ s) is $x_{k+1} = x_k + u_k$. Setting $u_k = Kx_k$, the closed-loop system equation is $x_{k+1} = (1 + K)x_k$, which is asymptotically stable if and only if $-2 < K < 0$.

C) When the system is controlled using the network control system in figure, the transition matrix $\Phi(h, \tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} 1 + (h - \tau)K & K\tau \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - (1 + (h - \tau)K)\lambda - K\tau$$

   a. In case $\tau = 0.25$ s, $p_\Phi(\lambda) = \lambda^2 - (1 + 0.75K)\lambda - 0.25K$

According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -0.25K < 1 \\ -(1-0.25K) < -(1+0.75K) < (1-0.25K) \end{cases}$$

This is verified for $-4 < K < 0$. Note that this condition is less conservative than the one obtained in case of "pure" digital control.

b. In case $\tau = 0.5$ s, $p_\Phi(\lambda) = \lambda^2 - (1+0.5K)\lambda - 0.5K$
According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -0.5K < 1 \\ -(1-0.5K) < -(1+0.5K) < (1-0.5K) \end{cases}$$

This is verified for $-2 < K < 0$. Note that this condition is the one obtained in case of "pure" digital control.

c. In case $\tau = 0.75$ s, $p_\Phi(\lambda) = \lambda^2 - (1+0.25K)\lambda - 0.75K$
According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < -0.75K < 1 \\ -(1-0.75K) < -(1+0.25K) < (1-0.75K) \end{cases}$$

This is verified for $-\frac{4}{3} < K < 0$. Note that this condition is more conservative than the one obtained in case of "pure" digital control.

D) When the system is controlled using the networked control system in figure, the transition matrix $\Phi(h,\tau)$ is

$$\Phi(h,\tau) = \begin{bmatrix} 1+(h-\tau)K & K\tau \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1.75-h & -0.75 \\ 1 & 0 \end{bmatrix}$$

The corresponding characteristic polynomial is

$$p_\Phi(\lambda) = \lambda^2 - (1.75-h)\lambda + 0.75$$

According to the rule derived in Part I (see Section B.6) for 2x2 matrices, the networked control system is asymptotically stable if and only if

$$\begin{cases} -1 < 0.75 < 1 \\ -(1+0.75) < -(1.75-h) < (1+0.75) \end{cases}$$

Therefore, all values $\tau < h < 3.5$ are valid.

E) To analyze the stability of the system under packet dropouts defined by the sequence of values of variable θ_k, we first recall that the transition matrices in case transmission occurs and in case of dropout are, respectively

$$\widetilde{\Phi}_1 = \Phi(h,\tau) = \begin{bmatrix} 1+K(h-\tau) & K\tau \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0.75 & -0.75 \\ 1 & 0 \end{bmatrix}$$

$$\widetilde{\Phi}_0 = \begin{bmatrix} 1 & Kh \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

Note, in passing, that $\widetilde{\Phi}_1$ has eigenvalues in 0.375 ± 0.7086j and 0, while $\widetilde{\Phi}_0$ has eigenvalues in 1.

Since the sequence of values of θ_k is 1,0,1,0,1,0,1,0:

- if k is odd, $z_{k+1} = \widetilde{\Phi}_1 z_k$, $z_{k+2} = \widetilde{\Phi}_0 \widetilde{\Phi}_1 z_k$.
- if k is even, $z_{k+1} = \widetilde{\Phi}_0 z_k$, $z_{k+2} = \widetilde{\Phi}_1 \widetilde{\Phi}_0 z_k$.

The update rule that can be derived is, for all $k \in \mathbb{N}$,

$$z_{2k+2} = \widetilde{\Phi}_1 \widetilde{\Phi}_0 z_{2k}$$

$$z_{2k+3} = \widetilde{\Phi}_0 \widetilde{\Phi}_1 z_{2k+1}$$

Therefore, $\widetilde{\Phi}_1 \widetilde{\Phi}_0$ and $\widetilde{\Phi}_0 \widetilde{\Phi}_1$ can be regarded as the two-steps transition matrices for the NCS state. We resort to the fact that $\widetilde{\Phi}_1 \widetilde{\Phi}_0$ and $\widetilde{\Phi}_0 \widetilde{\Phi}_1$ have the same eigenvalues to study the convergence properties of the NCS. In particular

$$\widetilde{\Phi}_1 \widetilde{\Phi}_0 = \begin{bmatrix} 0.75 & -0.75 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.75 & -1.5 \\ 1 & -1 \end{bmatrix}$$

Whose characteristic polynomial is $p(\lambda) = \lambda^2 + 0.25\lambda + 0.75$. The eigenvalues are eigenvalues are -0.125 ± 0.857j, that have absolute value <1. The NCS under packet dropout is therefore asymptotically stable.